

Towards Generalist Game Players : A Survey of Foundation Models in the Multiverse of Games

The real world unfolds along a single set of physics laws, yet human intelligence demonstrates a remarkable capacity to generalize experiences from this singular physical existence into a multiverse of games, each governed by entirely different rules, aesthetics, physics, and objectives. This omni-reality adaptability is a hallmark of general intelligence. As Artificial Intelligence (AI) progresses towards Artificial General Intelligence (AGI), the multiverse of games has evolved from mere entertainment into the ultimate ground for training and evaluating AGI. In this survey, we present a comprehensive review of the paradigm shift in game-playing AI: from environment-specific Symbolic and Reinforcement Learning (RL) agents in to Large Foundation Models (LFMs) acting as generalist game players. We propose a unified pipeline that includes four core pillars: Datasets, Models, Harness, and Benchmarks. By systematically reviewing recent advancements, we highlight the trajectory toward this omni-reality adaptability. Finally, we discuss the open challenges and future directions for creating the omnipotent generalist agent capable of seamlessly mastering any challenge within the multiverse of games, thereby paving the way for AGI.

1. Introduction: From a Single Worldline to the Multiverse of Games

“Man only plays when in the full meaning of the word he is a man, and he is only completely a man when he plays.”

— Friedrich Schiller

Games are far from mere pastimes; they are profound abstractions of reality—originating from life, yet transcending it. Collectively, they constitute a vast *multiverse*, where each universe is governed by entirely distinct rules, aesthetics, physics and objectives, different from the single reality.

From a cognitive science perspective, human intelligence was developed to navigate the survival constraints of a single physical reality—a worldline characterized by consistent gravity, linear time, shared environment and fixed material properties (Cosmides and Tooby, 1994; Gigerenzer and Goldstein, 1996). However, armed with the singular set of real-world experiences, humans can seamlessly adapt to the infinite *multiverse of games* (Dubey et al., 2018; Lake et al., 2017), from managing block-based ecosystems in Minecraft (Fan et al., 2022) to orchestrating grand strategies in StarCraft (Ma et al., 2024b) and Civilization (Qi et al., 2024). Within minutes, a human can adapt to a digital universe where the rules are different, the aesthetics surreal, the physics inverted, and the objectives entirely abstracted from physical survival. The *omni-reality adaptability* to utilize localized experience to rapidly master a multiverse of entirely novel environments is the ultimate symbol of generalized intelligence (Lake et al., 2017; Legg and Hutter, 2007; Turing, 1953).

Driven by the pursuit of this generalized intelligence, researchers naturally turned to these digital games as the ultimate testbed for AI since its inception (Campbell et al., 2002; Laird and Lent, 2001; Turing, 1953). Digital worlds offer high quality, dynamic, visually rich, and long-horizon environments that demand a synthesis of perception, planning, reasoning, and motor control, while AI pushes the boundaries of what can be automated and solved within these worlds. For decades,

the AI community has utilized games as a crucible to train and measure capability (Ahn et al., 2025; Bellemare et al., 2013b; Berner et al., 2019; Campbell et al., 2002; Fan et al., 2022; Hu et al., 2025a; Johnson et al., 2016; Park et al., 2025; Silver, 2016; Towers et al., 2024; Vinyals, 2019; Wang et al., 2025f; Ying et al., 2026; Zhang et al., 2026a), yet it has historically failed to replicate this omni-reality adaptability. Generally, past paradigms have largely produced *specialists*. Despite achieving superhuman performance through environment-specific Symbolic and Reinforcement Learning (RL) in certain complex games like DeepBlue in Chess (Campbell et al., 2002), AlphaGo in Go (Silver, 2016), AlphaStar in StarCraft II (Vinyals, 2019) and Openai Five in Dota II (Berner et al., 2019), these systems were fundamentally brittle specialists. Typically, these models are optimized from scratch for a singular game worldline with simplified interfaces and interactions. While they achieve superhuman performance within such highly structured environments, they fail completely to transfer their knowledge to another game worldline—rendering them paradoxically powerful yet profoundly fragile. Moreover, these models almost operate as black boxes, heavily obscuring the underlying mechanisms that drive their decision-making processes (Greydanus et al., 2018; Mott et al., 2019). Ultimately, systems confined to this paradigm fall fundamentally short of the true AGI that AI community pursues.

Recently, the advent of Large Foundation Models (LFMs), including Large Language Models (LLMs) (Bai et al., 2023; Brown et al., 2020; Ouyang et al., 2022; Yang et al., 2025), Vision-Language Models (VLMs) (Bai et al., 2025a; Comanici et al., 2025; OpenAI et al., 2024; SIMA-team et al., 2025), Vision-Language-Action Models (VLAs) (Chen et al., 2025d; Li et al., 2025a; Wang et al., 2025f; Zitkovich et al., 2023), Diffusion Models (DMs) (Chi et al., 2023; Ho et al., 2020; Magne et al., 2026a) and World Models (WMs) (Ball et al., 2025; Decart and Julian Quevedo, 2024; Ha and Schmidhuber, 2018; He et al., 2025a; Zhang et al., 2025d), has sparked a transformative paradigm shift. Rather than mastering a single game through billions of trial-and-error episodes from scratch, LFMs were born with vast open-world knowledge and emergent reasoning capabilities. By treating games not as isolated optimization problems, but as diverse instances of interactive environments, these models show the potential of *omni-reality adaptability* as a true *generalist* game player.

To capture this rapid evolution, this survey provides a comprehensive pipeline-oriented perspective towards generalist game player. *To the best of our knowledge, this is the first survey to systematically investigate Large Foundation Models (LFMs) as generalist game players through a comprehensive, end-to-end lifecycle. Our contributions are threefold:*

- **An evolution framework for game-playing AI.** We trace the development of game-playing AI through four eras and unify them under a Goal-Conditioned POMDP formulation. This formulation makes explicit how each transition reshapes the elements of the POMDPs tuple $\mathcal{M}$, providing a principled basis for understanding the ongoing shift from the brittle specialists to generalist game players can be understood.
- **A four-pillar pipeline for generalist game players.** We organize the literature around four interdependent pillars, **datasets, models, harness, and benchmarks**, that together constitute the full lifecycle of a generalist game-playing system. Within this pipeline, datasets fuel model training, trained models are deployed by the harness to interact with game environments, benchmarks measure the resulting capabilities and expose limitations, and these findings motivate the collection of new data and further model improvement.
- **A research roadmap with identified open challenges.** Building on the systematic review, we identify current bottleneck in each pillar. We highlight where current methods cluster and where significant gaps remain, aiming to inform future research priorities in this rapidly evolving field.

2. Preliminary: AI, Games, and the Paradigm Shift

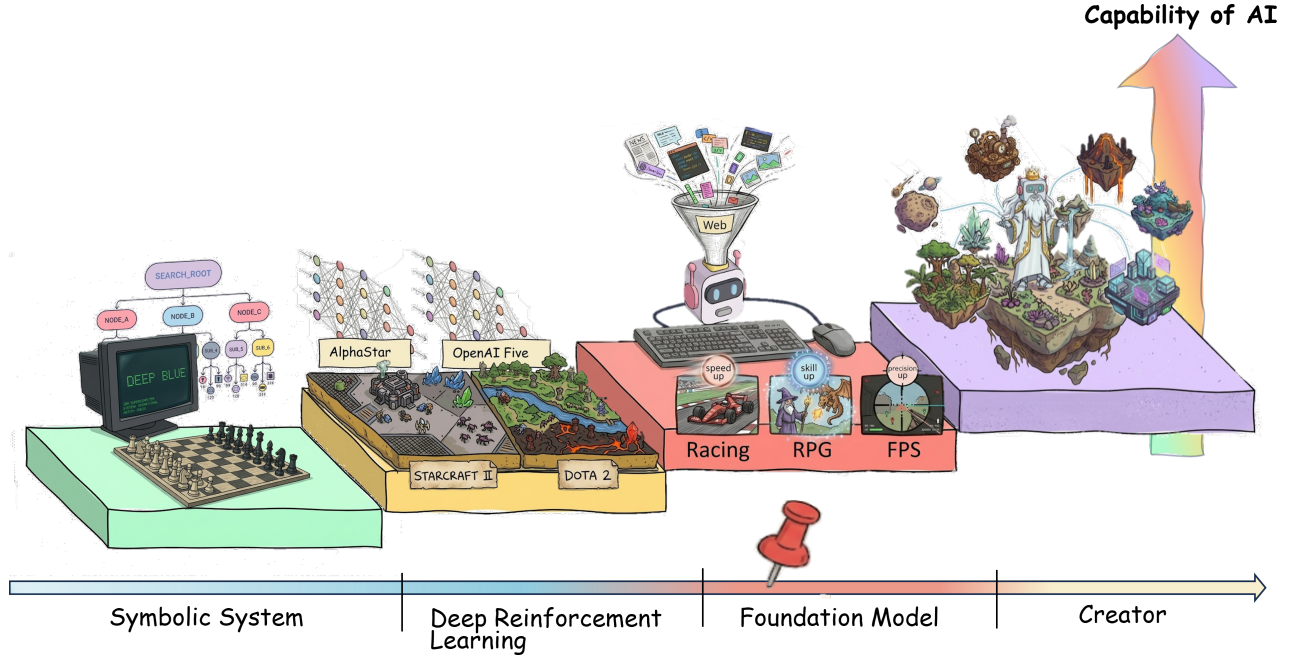


Fig. 1: The evolution of game-playing AI paradigms towards AGI. We categorizes the timeline into four eras: (1) Symbolic Systems, relying on hard-coded heuristics within isolated environments; (2) Deep Reinforcement Learning (DRL), achieving superhuman mastery in specific domains, yet constrained as narrow experts; (3) Large Foundation Models (LFMs), emerging as generalist agents capable of reasoning and adaptation across the human-crafted multiverse of games; and ultimately, (4) The Creator, the future where AI transcends playing to become the simulator itself, autonomously generating and evolving infinite game multiverses.

2.1. The Symbiosis and Formalization of AI in Games

The quest for AI has always been closely linked to games. From the early days of chess programs (Campbell et al., 2002) to modern highly complex strategy games (Ma et al., 2024b), games serve as a standardized, quantifiable proxy for real-world decision-making. They include diverse challenges, such as partial observability, long-horizon planning, and real-time multimodal processing, while without the physical risks and costs of real-world robotics.

To establish a rigorous foundation, we formalize the interaction between an AI agent and a game environment as a Partially Observable Markov Decision Processes (POMDPs) (Smallwood and Sondik, 1971). A game can be described as a tuple $\mathcal{M} = \langle G, S, A, T, R, \Omega, O, \gamma \rangle$, where:

- G is a set of potential goals or objectives. Each $g \in G$ is a task objective (e.g., natural language prompts, target images, or specific sub-tasks) that dictate the agent’s current mission.
- S is a set of states. Each $s \in S$ represents the internal state of the environment.
- A is a set of actions. Each action $a \in A$ can be a combination of textual reasoning, retrieval of external knowledge and memory, tool calls, and execution action.
- $T : S \times A \rightarrow \Delta(S)$ is the state transition probability function which takes a state-action pair (s, a) and outputs the probability distribution $T(s' | s, a)$ of the next state.

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Table 1: The Paradigm Shift in Game-Playing AI Formalization. We frame the evolution through the lens of a Goal-Conditioned POMDP $\mathcal{M} = \langle G, S, A, T, R, \Omega, O, \gamma \rangle$, highlighting the transition from rigid, pre-defined components to agent-generated open-ended dynamics.

Feature	Era 1: Symbolic	Era 2: Deep RL	Era 3: Foundation Models	Era 4: The Creator, Demiurge
Timeline	Pre-2010	2010-2022	2022-Present	Post-Future
Scope	Single Universe	Single Universe	Human-crafted Multiverse	Theoretical Multiverse
Mechanism	Rules $\rightarrow$ Search	State/Pixels $\rightarrow$ Policy	Knowledge $\rightarrow$ Reason $\rightarrow$ Action	Simulate $\rightarrow$ Generate $\rightarrow$ Evolve
Goal (G)	Fixed Condition	Maximize Reward	Dynamic Tasks	Self-assigned & Evolving
Action (A)	Symbolic API	Structured API	Human-like Interface	Unconstrained / Generative
Dynamics (T)	Deterministic	Fixed Black-box	Fixed Black-box	Dynamically Generated
Reward (R)	Terminal / Heuristic	Env-defined Scalar	Env Scalar + Semantic Evaluation	Agent-generated
Obs. (O, Ω)	Bypassed (Direct S)	Pixels / Structured ($\Omega_{px/vec}$)	Multimodal ($\Omega_{text,vision,audio}$)	Omniscient / Bird’s-eye
World Prior	Hand-craft Logic	Tabula Rasa	Pre-trained Knowledge	Self-defined Logic
Target	Specialist	Specialist	Generalist Player	Omnipotent Creator

- $R : S \times A \times G \rightarrow \mathbb{R}$ is the feedback/reward function, conditioned on the specific goal $g \in G$. The feedback $r = R(s, a, g)$ typically takes the form of a scalar score or textual or image feedback.
- Ω is a set of observations accessible to the agent.
- $O : S \times A \rightarrow \Delta(\Omega)$ is the observation probability function which takes a state-action pair (s, a) and outputs the probability distribution $O(o' | s, a)$ of the next observation for the agent.
- $\gamma \in [0, 1)$ is the discount factor.

Under this framework, by explicitly introducing G , the agent aims to find a goal-conditioned policy $\pi(a_t | o_{\leq t}, g)$ that maximizes the expected discounted return $E[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t, g)]$.

2.2. The Evolution of Game-Playing AI Paradigms

The evolution of game-playing AI can be mapped to how agents interact with, and control the elements of the POMDPs tuple $\mathcal{M}$. This progression marks a fundamental shift from rigid rule execution in constrained state spaces to open-ended multiverse simulation. Table 1 visualizes this paradigm shift across four distinct eras:

Era 1: Symbolic Systems - *Know the Rules, Search the Trees*. Prior to 2000, early AI relied on the hard-coded rules and heuristic search algorithms (Campbell et al., 2002; Knuth and Moore, 1975). In this era, the observation function O was often bypassed, giving the agent direct access to a fully observable and simplified state space S (e.g., the exact positions of the chess pieces). The action space A was rigidly defined by game-specific logical rules, and the goal G was a singular, fixed terminal state, such as checkmate in chess. These systems required domain experts to hand-craft features and lacked the ability of true perception and generalization.

Era 2: Deep Reinforcement Learning - *Learn from Scratch via Self-Play*. Since 2010s, the integration of deep neural networks with RL shifted the paradigm toward handling partial observability (Berner et al., 2019; Silver, 2016; Vinyals, 2019). Agents learned policies $\pi(a_t | o_{\leq t})$ from high-dimensional observation spaces Ω (such as raw pixels or structured input). However, the action space A remained a fixed, structured API as pre-defined control vectors in the specific task, and the goal G was implicitly bound to a static, environment-provided scalar reward R . While eliminating hand-crafted rules, these models still learned from scratch for each game $\mathcal{M}$, producing highly capable but narrow specialists with no omni-reality adaptability.

Era 3: Foundation Models - *Understand, Reason, Then Act.* Now, the current era is defined by models pre-trained on internet-scale multimodal data (Magne et al., 2026a; Tan et al., 2025b; Wang et al., 2025c,f; Yue et al., 2026). Here, the observation space Ω expands to include rich multimodal inputs (vision, audio, text). Crucially, the goal space G becomes open-ended, allowing agents to pursue complex, dynamic natural language instructions. Furthermore, the action space A shifts from game-specific APIs to a universal, human-like interface (emulating keyboard and mouse actions). By leveraging pre-trained world knowledge to infer underlying transition dynamics T , these models transition the field toward generalist game player in hand-crafted multiverse of games.

Era 4: The Creator, Demiurge - *Simulate, Generate, Then Evolve.* In the future, the AI transcends the role of a participant constrained by a fixed game $\mathcal{M}$ (Ball et al., 2025; SIMA-team et al., 2025). Instead of merely optimizing a policy within predefined boundaries, the agent becomes the simulator itself. It possesses the capability to actively generate and expand the state space S and action space A dynamically. It invents its own evolutionary goals G , and constructs the transition dynamics T and reward structures R . The interface becomes completely free-form, shifting from existing games to simulating, evolving, and directing the progression of entire theoretical multiverses.

2.3. The Current Frontier: The Nexus of Era 2 and Era 3

Situating the current landscape within the evolutionary trajectory of game-playing AI, we find ourselves at a critical stage: transitioning from the specialized optimization of Era 2 to the generalist paradigms of Era 3. While Deep RL in Era 2 has yielded specialists capable of mastering complex POMDPs, these systems often remain confined by the generalization bottleneck and the cost of learning π from scratch for each game $\mathcal{M}$ is becoming increasingly prohibitive.

Consequently, the field is pivoting toward Era 3: Foundation Model as generalists. By leveraging the world knowledge priors of Large Foundation Models, emerging frameworks demonstrate remarkable multimodal integration, zero-shot adaptation and goal-conditioned reasoning (G). However, significant challenges remain across the board, spanning accurate perception, sparse reasoning, memory retrieval, precise control, and efficiency, affecting virtually every layer of the system.

This survey specifically focuses on this paradigm-shifting frontier. We comprehensively review the ongoing transition, exploring the datasets, models, harness, benchmarks, and persistent challenges that define this pivotal moment in AI evolution, ultimately laying the groundwork for the distant but inevitable realization of Era 4.

3. Datasets: The Foundation Beneath Generalist Game Player

Among the four fundamental pillars of our pipeline, datasets occupy a uniquely structural position. They serve not merely as the starting point of the development loop but rather as the constraint surface that shapes all downstream processes. The model architectures a community can explore, the harness designs that prove viable, and the benchmark dimensions that remain measurable are mostly predetermined, frequently in an implicit manner, by the data available at the onset of training.

In the DeepRL Era 2 paradigm, agents (Berner et al., 2019; Silver, 2016; Vinyals, 2019) generated their own training data through direct environment interaction: each self-play loop was self-contained, the data it produced was inseparable from the game that generated it, and nothing transferred elsewhere. Era 3 breaks this closure. Foundation models require corpora that are collected before training begins, that span multiple games, and that carry value beyond the domain they came

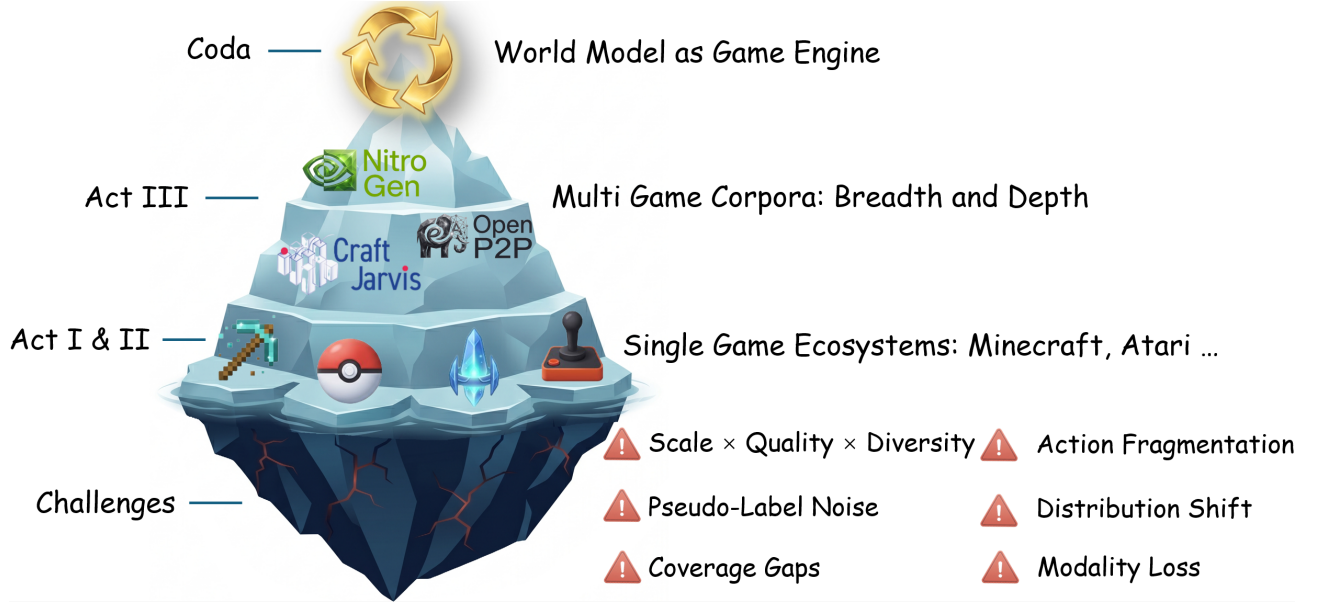


Fig. 2: Overview of Dataset in Game-playing AI. The trajectory of data curation is shifting from isolated, single-game ecosystems (Acts I & II) to comprehensive, multi-game corpora (Act III) in both depth and breadth, tailored for generalist agents. We highlight the future where world model can replace the game engine, and we show the critical bottlenecks in current dataset construction.

from. The question is no longer “*how many environment steps can the agent take?*” but “*what kind of data should we prepare, and where does it come from?*”

This section answers that question through three chapters and a coda.

- **Act I — Requisition: Learning from Human Play.** Act I examines how the field learned to extract training signals from human gameplay, progressing from initial structured demonstration datasets to internet-scale pseudo-labeled video and finally to multimodal time-aligned streams.
- **Act II — Excavation: Mining the Competitive Archive.** Act II investigates a complementary data source encompassing decades of replays archived on gaming platforms, alongside algorithmic synthesis methods that distill strategic knowledge directly from game mechanics.
- **Act III — Scaling: Crossing the Single-Game Boundary.** Act III faces the fundamental challenge of transcending the single-game boundary, documenting the dual pathways of broad automated collection and deep structured curation that the community has pursued to construct multi-game corpora.
- **Coda — Creation: The World Model as Data Engine.** Finally, the Coda looks beyond these three phases to the emerging paradigm of the *world-model-as-data-engine*, effectively connecting the datasets pillar to the Era 4 Demiurge vision that frames our comprehensive survey.

3.1. Act I — Requisition: Learning from Human Play

Human players generate enormous behavioral traces every day, and they constitute the most natural source of training data for game-playing agents. The challenge of learning from human play is a

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Table 2: Summary of representative large datasets in gaming AI

Paper	Modality	Data Source	Label Method	Scale	Game Set
ALE (2013a)	Frame + action	Atari 2600 ROMs	Env-provided reward	57 games	57 Atari games
SC2LE (2017)	State + actions	Ladder replays	Auto-recorded	~800K replays	StarCraft II
LIGHT (2019)	Text dialogues	Crowdsourced roleplay	Human-authored	663 scenes, 11K episodes	Text adventure
MineRL (2019)	RGB + actions	Crowdsourced play	Auto-recorded	60M frame-action pairs	Minecraft
Atari-HEAD (2020)	Frame + gaze + action	Human play + eye-tracker	Sync-recorded	117 hrs, 328M gaze	20 Atari games
NLE (2020)	Symbol + action	Procedural generation	Env-provided reward	Infinite (procedural)	NetHack
RL Unplugged (2020)	Frame + action	RL replay buffers	Auto-recorded	46 games × 200M frames	46 Atari games
CS Deathmatch (2022)	RGB + kbd/mouse	Server spectation	Rule-based IDM	5.77M frames, 100 hrs	CS:GO
VPT (2022)	Video + actions	Web videos	IDM pseudo-label	70K hrs (clean)	Minecraft
MineDojo (2022)	Video + text	YouTube/Wiki/Reddit	None (natural)	730K videos, 300K hrs	Minecraft
CICERO (2022)	Text + board state	webDiplomacy.net	Auto-aligned	40K+ human games	Diplomacy
StarCraftImage (2023)	Rendered images	Match replays	Auto-generated	3.6M imgs / 60K replays	StarCraft II
GROOT (2024b)	Video + actions	Contractor + YouTube	IDM pseudo-label	30 tasks (benchmark)	Minecraft
Werewolf Arena (2024)	Text dialogues	Human play	Auto-recorded	18.8K sessions	Werewolf
Genie (2024a)	Unlabeled video	Web videos	None (unsupervised)	30K hrs (filtered)	100+ platformers
GameGen-X (2024)	Video + captions	Web game videos	GPT-4o captioning	1M+ clips, 150+ games	150+ games
GameNGen (2024)	Frame + action	Video Generation Model	Auto-recorded	20 FPS, multi-minute play	DOOM
PlayGen (2024)	Frame + action	Video Generation Model	Auto-recorded	20 FPS, 1K+ Gen. frames	Mario & DOOM
OpenHA (2025e)	RGB + actions	VPT trajectories	Rule-based pipeline	5.5B tokens, 1K tasks	Minecraft
JARVIS-VLA (2025b)	Image + actions	Scripted agent	Self-supervised tokenizer	3.78M samples, 1K tasks	Minecraft
PLAICraft (2025b)	5-modal aligned	Human play	None (time-aligned)	10K hrs, 10K players	Minecraft
Game-RL (2025)	Image + QA	Code generation	Code-verified	140K QA pairs	30 games
Metamon (2025)	RL trajectories	Platform replays	Auto-parsed	2M human + 18M self-play	Pokémon
PokéChamp (2025)	Text battle logs	Platform replays	Reverse-engineered	3M+ battles	Pokémon
Mastermind (2025b)	Text (encoded)	Algorithm synthesis	Algorithm-labeled	1.7M + 288K samples	2 Chess games
CombatVLA (2025c)	Screenshot + actions	Expert human play	Human AoT labels	25K imgs, 200 hrs	Black Myth
GF-Minecraft (2025)	Video + kbd/mouse	Programmatic capture	Auto-recorded	70 hrs, 2K clips	Minecraft
Orak (2025)	Text/Image/Both	LLM-generated	LLM + human verified	11K fine-tuning samples	12 games
Matrix-Game (2025d)	Video + kbd/mouse	Video Generation Model	Auto-labeled actions	~ 400 frames	Minecraft
Matrix-Game 2.0 (2025a)	Video + kbd/mouse	Video Generation Model	Auto annotations	25 FPS, minute-level gen	GTA5 + UE scenes
WildWorld (2026d)	Video + state	ARPG auto-capture	Auto-extracted	108M frames, 450+ actions	Monster Hunter
NitroGen (2026b)	Video + gamepad	Web videos	CV overlay extraction	40K hrs, 1K+ games	1,000+ games
OpenP2P (2026)	Video + kbd/mouse	Expert human play	Recorded + VLM text	8.3K hrs, 600M pairs	45+ games
Proact-VL (2026)	Video + ASR text	YouTube streams	Multi-stage pipeline	561 hrs, 128K samples	12 games
Solaris (2026)	Video + actions	Video Generation Model	Sync-recorded	Minute-level play	Minecraft
Matrix-Game 3.0 (2026b)	Video + action + pose + text	Video Generation Model	Auto-generated	40 FPS; minute-long play	UE + AAA games

tension between two quantities that move in opposite directions: the scale of available gameplay footage and the cost of attaching action labels to it.

The earliest efforts resolved this tension by choosing precision over scale. MineRL (Guss et al., 2019) compiled over 60 million state-action pairs from human Minecraft demonstrations. A parallel effort collected four million frames of online Counter-Strike gameplay through the same strategy (Pearce and Zhu, 2022). Atari-HEAD (Zhang et al., 2020) captured not only frames and actions but also human gaze data via eye-tracking across 20 Atari games. More recently, CombatVLA (Chen et al., 2025c) invested in collecting 200 hours of Black Myth: Wukong aligned at sub-15-millisecond precision, further annotated with Action-of-Thought labels. These datasets founded imitation learning for game agents, yet their construction cost grew linearly with data volume, leaving the resulting corpora orders of magnitude smaller than the total gameplay available on the internet.

To decouple data scale from human labeling costs, recent works increasingly employ learned models as automated annotators at both perceptual and semantic levels. At the perceptual level, Video PreTraining (VPT) (Baker et al., 2022) utilizes an Inverse Dynamics Model (IDM) to infer low-level actions from large-scale unlabeled internet videos, a paradigm also adopted by GROOT (Cai et al., 2024b) to curate task-oriented benchmarks. Complementing this, semantic-level approaches leverage the reasoning capabilities of foundation models: Orak (Park et al., 2025) deploys strong LLMs to autonomously generate structured expert trajectories across various game genres, while Proact-VL (Yan et al., 2026) and GameGen-X (Che et al., 2024) employs VLMs to autonomously label and caption the video. Despite enabling internet-scale data collection, these automated methods

share a critical limitation: residual noise, from IDM inaccuracies or LLM hallucinations, inevitably propagates into the training objective, degrading downstream policies over long horizons. This accumulation of noise serves as a primary motivation for the reinforcement learning fine-tuning stage discussed in the Models chapter.

While VPT demonstrated that vision and action can be recovered at scale, human gameplay is far richer than frames paired with keyboard inputs. Atari-Head (Zhang et al., 2020) collects 117 hours of Atari-2600 games with human K&B action and gaze. PLAICraft (He et al., 2025b) pushes the boundary by collecting 10,000 hours of Minecraft gameplay with five modalities aligned in time: video, audio, natural language (voice and text), actions, and K&M signals. They capture not only what the player did but also what the player said and aspects of the player’s cognitive state. Training a model requires observations spanning the full perceptual space the agent will operate in. Multimodal temporal alignment is therefore not optional but a prerequisite for agents that perceive and reason about the game world in a human-like manner.

At the opposite end of the cost spectrum, OpenP2P (Yue et al., 2026) invested in over 8,300 hours of meticulous human annotation across more than 45 3D games. Its core contribution is empirical: by varying model scale and data volume, the authors showed that behavior cloning follows a predictable scaling law and that increasing both dimensions leads to emergent causal reasoning. This reframes the quality-versus-quantity debate. Internet-scale pseudo-labeled data and expertly annotated data at moderate scale are not interchangeable. They serve different roles in the training pipeline. The former provides broad behavioral coverage at low per-sample cost, while the latter supplies the high-fidelity signal needed to unlock qualitatively new capabilities. *How to optimally combine these two data regimes remains one of the central open problems in corpus design for game-playing agents.*

Viewed as a whole, the evolution of human-sourced game data follows a clear trajectory: from expensive synchronous annotation (Guss et al., 2019; Pearce and Zhu, 2022), through internet-scale pseudo-labeling (Baker et al., 2022), to multimodal temporal alignment (He et al., 2025b) and quality-centric scaling experiments (Yue et al., 2026). Each advance was driven by a specific limitation of its predecessor, and the limitations that remain collectively, most notably the noise ceiling of pseudo-labels and the prohibitive cost of high-quality annotation at scale, motivate the alternative data sources explored in the following sections.

3.2. Act II — Excavation: Mining the Competitive Archive

Competitive gaming platforms have accumulated decades of match replays, battle logs, and ranked statistics, forming a vast reservoir whose value for training game agents has only recently been recognized. Unlike the human demonstrations of Section 3.1, competitive replays are byproducts of the competitive ecosystem, not produced for teaching AI. Yet they carry a distinctive advantage: match outcomes and player rankings provide implicit quality signals requiring no additional annotation.

The most direct use of this archive is to extract structured training data from historical replays. The practice traces back to SC2LE (Vinyals et al., 2017), which released approximately 800,000 StarCraft II ladder replays as a standardized research resource, establishing the template of leveraging platform recording infrastructure and competitive outcomes as implicit supervision. StarCraftImage (Kulinski et al., 2023) later drew on 60,000 replays from the same ladder to compile 3.6 million images with spatial reasoning labels derived from the game state, using the ladder’s built-in rating system as a natural skill-level filter. Metamon (Grigsby et al., 2025) mines a full decade of Pokémon Showdown competitive replays, capturing the complete evolutionary record of the metagame—from casual play

to tournament-level strategy through repeated cycles of innovation, counter-adaptation, and balance patches. This combination of quality stratification and temporal diversity enables robust policy learning without active environment interaction. The common thread is that competitive replays embed implicit reward signals—rankings and win rates—that directly inform offline RL reward shaping, bypassing both the expensive manual annotation and the noisy pseudo-labeling.

Beyond replays that preserve visual or state-level information, a parallel line of work extracts strategic knowledge from textual battle logs and dialogue records at even larger scale. Pokéchamp (Karten et al., 2025) compiled over three million competitive Pokémon battle logs, one of the largest corpora of structured competitive interactions. CICERO’s Diplomacy corpus (Meta Fundamental AI Research Diplomacy Team (FAIR) et al., 2022) contributes over 40,000 full-press Diplomacy games in which natural language negotiation messages are aligned with board-state actions, encompassing the full spectrum of strategic communication. Werewolf Arena (Bailis et al., 2024) provides 18,800 sessions of nine-player social deduction games with complete dialogue logs and voting records. At this scale, strategic patterns and team-composition preferences emerge from aggregate distributions rather than manual expert identification.

For rule-complete games with tractable state spaces, training data can be algorithmically synthesized or simulated, bypassing the need for human or competitive data collection. Prominent examples include RL Unplugged (Gulcehre et al., 2020), which compiles diverse agent-generated replay buffers across Atari games; NLE (Küttler et al., 2020), which provides procedurally generated NetHack episodes with automated reward annotations; Game-RL (Tong et al., 2025), which utilizes Code2Logic to programmatically generate verifiable QA pairs; and Mastermind (Wang et al., 2025b), which synthesizes Q-value-annotated trajectories and optimal moves for Doudizhu and Go via extensive game-tree search. By enumerating optimal solutions, synthesized data often surpasses human demonstrations in quality, offering verifiable correctness without expensive manual annotation or noisy pseudo-labeling. However, its application is fundamentally bounded by computational tractability. As environments become partially observable, continuous, or open-ended, algorithmic synthesis becomes prohibitive. Consequently, this strategy serves as a robust complement to human- and competition-sourced data, rather than a universal replacement.

Taken together, the data sources examined in this section, including competitive replays with implicit quality gradients (Grigsby et al., 2025; Kulinski et al., 2023; Vinyals et al., 2017), large-scale battle logs and dialogue archives (Bailis et al., 2024; Karten et al., 2025; Meta Fundamental AI Research Diplomacy Team (FAIR) et al., 2022), agent-generated replay buffers and procedural environments (Gulcehre et al., 2020; Küttler et al., 2020), and algorithmic synthesis with verifiable labels (Tong et al., 2025; Wang et al., 2025b), substantially expand data volume and diversity. Yet they introduce new challenges: distribution shift from game updates, limited applicability of algorithmic synthesis to open-ended games, and the single-game focus that characterizes most competitive archives. These limitations motivate the cross-game scaling in the following section.

3.3. Act III — Scaling: Crossing the Single-Game Boundary

The datasets of Sections 3.1 and 3.2 share a common limitation: most are confined to a single game or a narrow family of related games. For Era 3’s goal of building generalist agents, this single-game isolation is a fundamental bottleneck. Two complementary strategies have emerged for constructing multi-game corpora: a breadth path that maximizes game coverage through automated collection, and a depth path that maximizes annotation quality within a carefully curated set of games.

The breadth path has progressed through several generations of increasingly ambitious collec-

tion. ALE (Bellemare et al., 2013a) first standardized 57 Atari 2600 titles under a uniform frame-action-reward interface, establishing the principle that a common data format spanning dozens of games could accelerate research. Genie (Bruce et al., 2024a) scaled this to the internet era, harvesting over 30,000 hours of unlabeled 2D platformer videos spanning hundreds of titles and using unsupervised learning to discover latent action spaces without action annotation. NitroGen (Magne et al., 2026b) and OpenP2P (Yue et al., 2026) represent the current frontier, assembling over 40,000 hours across more than 1,000 games through automated pipelines combining web-scale video harvesting with computer-vision-based action extraction. NitroGen’s significance lies in demonstrating that heterogeneous multi-game corpora can serve as the data foundation for generative pre-training architectures, including diffusion and autoregressive models). Yet breadth comes at a cost: as game coverage grows, action heterogeneity grows correspondingly. An input in CS carries entirely different semantics from the same in Dota II. Unifying these disparate action representations remains a pressing open challenge, directly motivating the unified action tokenization.

The depth path takes the opposite stance, focusing on fewer games where every task provides rich, verifiable training signals. Game-RL (Tong et al., 2025) instantiates this with a corpus spanning 30 games and 158 tasks, each paired with a Code2Logic verifiable reward function—eliminating the reward misspecification that destabilizes RL on noisy data. OpenHA (Wang et al., 2025e) and JARVIS-VLA (Li et al., 2025a) together contribute hierarchical VLA trajectories totaling approximately 4.2 billion tokens across 800+ tasks, with layered annotation from high-level natural-language goals to atomic keyboard commands. This hierarchy provides the training signal for hierarchical policy learning at multiple temporal and semantic granularities. The value of the depth path is a quality leverage effect: verifiable signals eliminate the ambiguity that leads to reward hacking, so even moderately sized datasets can produce training stability that much larger noisy corpora cannot match.

Surveying the cross-game coverage landscape reveals a highly uneven map. The Minecraft ecosystem (Baker et al., 2022; Cai et al., 2024b; Fan et al., 2022; Guss et al., 2019; Li et al., 2025b; Wang et al., 2025e) is the most mature, forming a multi-year chain of increasingly sophisticated resources. Text strategic games (Bailis et al., 2024; Grigsby et al., 2025; Karten et al., 2025; Meta Fundamental AI Research Diplomacy Team (FAIR) et al., 2022; Urbanek et al., 2019) have likewise received substantial coverage, whose text-native interfaces align naturally with LLM input formats. The Atari ecosystem (Bellemare et al., 2013a; Gulcehre et al., 2020; Küttler et al., 2020; Zhang et al., 2020) provides another mature data chain available for Era 3 reuse. By contrast, several important categories remain underrepresented. Real-time 3D games, including FPS, MOBAs, and action RPGs, demand millisecond-level response times and high-dimensional continuous observations yet lack large-scale annotated datasets comparable to Minecraft. WildWorld (Li et al., 2026d) and CombatVLA (Chen et al., 2025c) represent early steps toward filling this gap for the ARPG subdomain, but FPS and MOBA remain data-scarce. Continuous action-space games pose a related challenge: the majority of existing datasets encode discrete keyboard inputs, leaving the continuous control regime of racing games, flight simulators, and physics-based environments with almost no dedicated training data. Social deduction games, despite session-level datasets such as Werewolf Arena (Bailis et al., 2024), lack the temporally extended records needed for online adaptation and real-time belief modeling. These gaps reflect structural mismatches between data collection infrastructure and domain requirements, and closing them will likely require new recording frameworks for high-frequency, continuous-control, and socially interactive environments.

In summary, the breadth path (Bellemare et al., 2013a; Bruce et al., 2024a; Magne et al., 2026b; Yue et al., 2026) provides the data substrate for generative pre-training across a thousand games but exposes the action-space unification problem. The depth path (Fan et al., 2022; Li et al., 2025b;

Tong et al., 2025; Wang et al., 2025e) provides stable, verifiable reward signals for reliable RL but within a narrower scope. The human annotation or model-as-judge paradigm (Che et al., 2024; Fan et al., 2022; Park et al., 2025; Yu et al., 2025) offers a scalable middle ground but remains bounded by teacher-model capability. No single corpus achieves simultaneous leadership in scale, annotation quality, and game diversity. This three-way trade-off is the defining unresolved tension of the dataset pillar, driving the exploration of world-model-based data generation in the next section.

3.4. Coda — Creation: The World Model as Data Engine

Game datasets in Era 3 are still largely produced by human-authored, code-based game engines. This substrate naturally limits scale by engine-side collection efficiency, annotation quality by the gap between the AI learner and the game engine, and diversity by the closed set of already existing human-made games. Therefore, to push beyond the current boundary of game data acquisition, *the key is no longer to propose another dataset, but to propose another data engine*—one in which AI itself becomes the driver of data generation. This paradigm shift essentially unifies data generation and model learning, creating a self-amplifying data flywheel that can jointly expand scale, improve quality, and broaden diversity.

Breaking the Code-Engine Ceiling. Manual code engines are hard-coded systems to ensure a stable gameplay experience: they provide human-friendly control interfaces, maintain preset game world boundaries for robust interaction, and are typically optimized for a single game world. By contrast, game world models are learned neural simulators that model game dynamics and predict action-conditioned futures, thereby making data generation more flexible and opening a path to relaxing the ceilings imposed by code engines.

First, they relax the *interface ceiling*. In code-based engines, collectable trajectories are restricted to pre-defined keyboard, mouse, controller, or API interfaces. Recent world models begin to absorb action conditioning into the model itself. *Genie* (Bruce et al., 2024b) is an early landmark in this direction, as it learns interactive environments with latent actions from unlabeled videos rather than relying on explicitly recorded engine-side controls. *MineWorld* (Guo et al., 2025b) and *Matrix-Game 2.0* (He et al., 2025a) further strengthen this trend by emphasizing action following and action injection **under explicit control signals**, while *PAN* (PAN Team Institute of Foundation Models, 2025) extends the action channel beyond fixed **low-level** interfaces to language-conditioned actions. In this sense, what counts as an admissible action for data generation is **increasingly less confined to traditional low-level interfaces, even if some systems still rely on predefined controls**.

Second, world models loosen the *preset-boundary ceiling*. Code engines can only operate within mechanics, maps, and interaction patterns that have been explicitly authored in advance. By contrast, recent generative systems begin to instantiate and recombine interactive situations beyond the exact boundaries of logged gameplay. *GameFactory* (Yu et al., 2025) explicitly frames this as creating new games through generative interactive videos, while *GameGen-X* (Che et al., 2025) pushes toward open-world interactive generation with multimodal control. *PAN* (PAN Team Institute of Foundation Models, 2025) broadens this further to general, interactable, and long-horizon world simulation, and *IPR-1* (Zhang et al., 2025c) **complements this trend by moving toward** game-to-unseen generalization with prediction-reinforced physical reasoning. These works indicate that training worlds no longer need to remain fully confined to **the exact environments and interaction patterns specified in advance**.

Third, world models begin to push against the *single-game ceiling*. Traditional datasets are ultimately limited by the finite set of already existing human-made games, whereas recent neural game engines increasingly expand toward more diverse and open-ended worlds. *GameFactory* (Yu et al., 2025), *GameGen-X* (Che et al., 2025), and *PAN* (PAN Team Institute of Foundation Models, 2025) all broaden the scope of generated environments beyond narrow single-game settings, while *Solaris* (Savva et al., 2026) extends this expansion to *multiplayer shared-world simulation* with cross-player consistency.

Towards Self-Amplifying Data Flywheel. Beyond serving as a data engine that relaxes the constraints of code-based environments, the deeper promise of world models lies in entering the training loop itself as a self-amplifying data flywheel. In this setting, the world model is no longer a passive predictor of next states; instead, it becomes a training component that supports *adaptive generation* tailored to the player’s demands, while also enabling *closed-loop co-evolution* with the player model.

Early evidence in robotics has already shown that such a paradigm is not only feasible, but also powerful. On the one hand, recent works of *adaptive generation* suggest that: rather than producing a fixed pool of synthetic rollouts, the world model could generate data conditioned on the learner’s weaknesses and demands, thereby improving policy models in long-tail failure-prone scenarios. Ctrl-World (Guo et al., 2026) shows that imagined successful trajectories can be selectively synthesized for policy improvement, while WMPO (Zhu et al., 2026) demonstrates that on-policy post-training can be carried out inside a learned world model, allowing the generated training distribution to evolve together with the policy. On the other hand, world models can also support *closed-loop co-evolution* once they become part of the training loop itself. World-Gymnast (Sharma et al., 2026) explicitly couples world-model rollouts with policy refinement in a reinforcement learning loop, showing that simulator quality and policy capability can be iteratively improved together. A similar perspective is also emerging in embodied systems in Psi-R2 and Psi-W0 (Psibot Team, 2026), where the world model is not treated as a standalone predictor, but as an integrated component for policy evaluation, optimization, and autonomous self-improvement within a closed-loop pipeline. This evidence demonstrates the power of world models to function as a self-amplifying data flywheel that co-evolves with policy models.

Game LFM training naturally calls for a data flywheel that is tailored to and evolves with the model. Games are goal-directed, feedback-driven environments that naturally push players to improve through interaction. This makes adaptive training data especially valuable, as it can target the player’s current weaknesses, failure modes, and learning demands. Since game difficulty often evolves with player competence, games also naturally motivate player-simulator co-evolution. However, current game world models still face substantial challenges before they can serve as such a data flywheel. High-quality training data require neural game engines to maintain temporal persistence and spatial consistency across the game multiverse, rather than merely generating locally plausible clips. Moreover, shared-world modeling that supports coherent multi-player rollouts remains underexplored, yet it is essential for scaling the flywheel toward industrial-level rollout throughput and richer collaborative or competitive training scenarios. Further advances in game world models will enable neural data engines to internalize more faithful game dynamics, ultimately forming a self-amplifying data flywheel that is tailored to and co-evolves with the game LFMs.

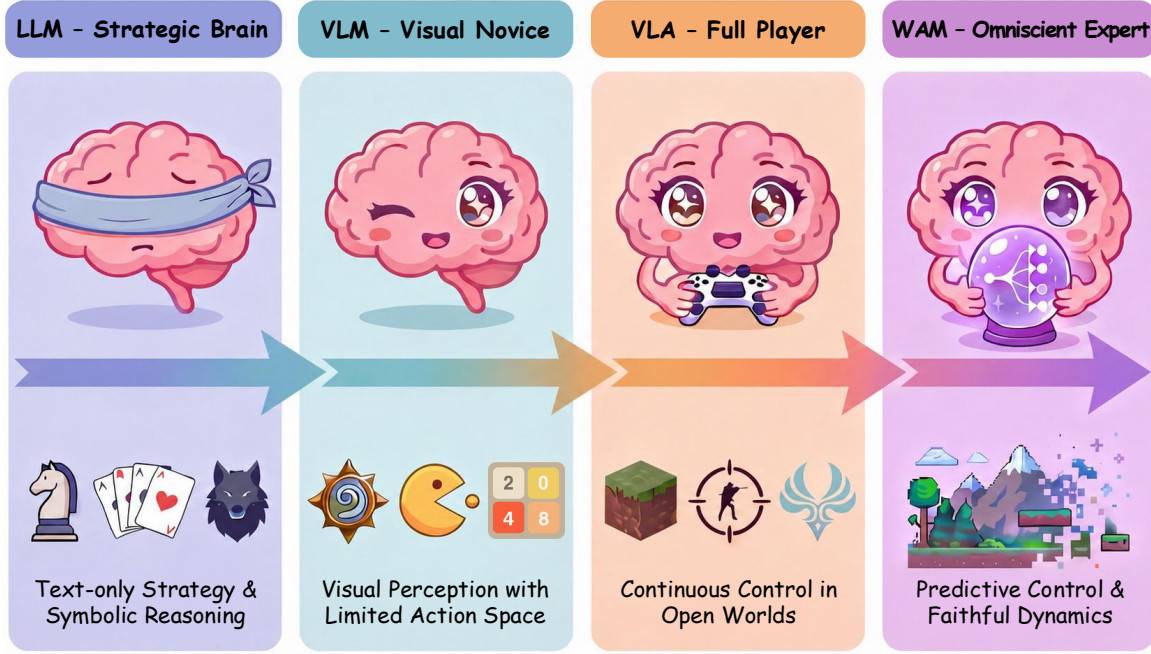


Fig. 3: Overview of Models in Game-playing AI. Game-playing models progress from LLMs as strategic brains for text-based reasoning, to VLMs as visual novices that perceive game screens, then to VLAs as full players that execute low-level controls, and finally to WAMs as predictive experts that internalize game dynamics for world-aware decision-making.

4. Models: The Evolving Brain of Generalist Game Player

Having established the paradigm shift towards generalist game players, we now turn to the cognitive core of these agents: the **Model**. In traditional paradigms, game AI was locked to rigid interfaces. A chess engine takes a fixed board vector and returns a move index; a StarCraft agent reads a state tensor and emits an action ID. These systems are powerful within their contracts, but the contract is the cage. Change the board size, swap the piece types, or ask the engine to explain its reasoning in natural language, and the entire architecture collapses. Its input parser, action head, and reward function are all fixed to a single game. In contrast, building foundation agents requires architectural unification, including designing a singular model capable of processing unconstrained multimodal observations (Ω), reasoning over open-ended goals (G), and executing through universal human-computer interfaces (A).

Within the current landscape, this pursuit of a unified architecture has evolved through four progressive stages. Each stage addresses a critical bottleneck in closing the perception-action loop:

- **Large Language Models (LLMs):** Acting as the foundational cognitive engine, LLMs introduce world knowledge and strategic planning to games. However, restricted to textual inputs and outputs, they remain visually disconnected—capable of complex reasoning but blind to raw visual environments.
- **Vision-Language Models (VLMs):** By integrating visual encoders, VLMs equip the agent with direct pixel-level perception. While achieving multimodal grounding, their outputs typically remain semantic, creating a disconnect between visual understanding and continuous motor

execution, such as keyboard, mouse and controller.

- **Vision-Language-Action Models (VLAs):** VLAs bridge the execution gap by mapping multi-modal observations directly to low-level continuous control (e.g., unified keyboard and mouse operations). This end-to-end architecture seamlessly unifies perception, reasoning, and action within a single stream.
- **World Action Models (WAMs):** Representing the emerging frontier, WAMs move beyond reactive control by internalizing the environment’s transition dynamics (T). By simulating future states and evaluating potential trajectories, these models enable predictive, model-based planning prior to execution.

In the following subsections, we systematically review these four architectural paradigms. We explore their structural designs, training methodologies, perception-action loop designs, and the inherent limitations that continue to drive the evolution of generalist models.

4.1. LLMs as Game Brain: Reasoning the Strategies in the Textual Multiverse

Large Language Models (LLMs) (Bai et al., 2023; Brown et al., 2020; Ouyang et al., 2022) remove the constraint in Deep Reinforcement Learning. They take free-form text in and produce free-form text out. The same model can read a chess position (Tong et al., 2025), a Werewolf transcript (Wu et al., 2024a), a Pokemon battle log (Zhang et al., 2026b), or a Minecraft inventory (Feng et al., 2024), all without architectural change. Both observation and action live in natural language, making them in principle unbounded. Moreover, LLMs also bring broad world knowledge. Pre-training on game manuals, strategy forums, tournament commentary, and community wikis means the model already understands bluffing before it plays its first poker hand, and knows where diamonds spawn before it enters a Minecraft cave (Wang et al., 2023a). These two properties, *flexible interface* and *pre-trained knowledge*, open the possibility of a single reasoning engine that generalizes across the textual multiverse: all games whose states and actions can be faithfully expressed in text.

This section traces how far training-based methods have pushed LLMs toward this goal, along two threads: natively textual games, where the challenge is strategic depth; and visual games accessible through text, where the challenge shifts to model design under perceptual constraints.

The Native Territory: Text Multiverse of Games. In text games, the observation is text, the action is text, and the core challenge is reasoning. This makes them the natural arena for sharpening LLM strategic capability through training.

Social deduction games demand unconstrained natural language as the action space: the agent persuades, deceives, and detects lies through dialogue, making LLMs irreplaceable. The landmark is CICERO (Meta Fundamental AI Research Diplomacy Team (FAIR) et al., 2022), which first achieved human-level performance in Diplomacy via game-theoretic planning. Subsequent shifts moved from modular designs (Xu et al., 2024) to unified alignment, where Werewolf LSPO (Xu et al., 2025b) and GameTalk (Vendrell et al., 2026) integrate strategies directly into LLMs via DPO and GRPO.

In imperfect information games (e.g., Poker), research focuses on long-horizon credit assignment. The GRPO (Guo et al., 2025a) family, including MS-GRPO (Dilkes et al., 2025) and ISO-GRPO (Xia et al., 2026), addresses sparse rewards, while SpinGPT (Maugin and Cazenave, 2025) standardizes the SFT-RL pipeline. Offline learning also thrives, with Metamon (Grigsby et al., 2025) reaching expert levels in Pokemon via ranked replays. For board games, models like Mastermind (Wang et al., 2025b) and Mixture of Masters (Frisoni et al., 2026) leverage algorithmic synthesis and MoE architectures to provide explainable strategic reasoning. VAM (Zhang et al., 2026c) masks illegal chess moves during

Towards Generalist Game Players : A Survey of Foundation Models in the Multiverse of Games

Table 3: Summary of representative large foundation models in gaming AI

Paper	Arch.	Base Model	Input	Action	Training Recipe	Game Set
CICERO (2022)	LLM	BART-like 2.7B	Text state + dialogue	NL messages + orders	RL + planning (piKL)	Diplomacy
Voyager (2023a)	LLM	GPT-4	Text API	Code (skills)	Training-free	Minecraft
Werewolf-RL (2024)	LLM	GPT-4 + RL Net	Text log	Dialogue	MAPPO	Werewolf
LLaMA-Rider (2024)	LLM	LLaMA-2-70B	Text API	Text API calls	SFT	Minecraft
Chain of Summ. (2024a)	LLM	GPT-3.5/4 + Qwen-7B	Text (TextSC2)	RTS commands	SFT (open-source)	StarCraft II
Werewolf LSPO (2025b)	LLM	LLaMA-3-8B	Text dialogue	Strategic utterances	CFR + DPO	Werewolf
Refl. of Episodes (2025a)	LLM	GPT-3.5-Turbo	Text (keyframes)	RTS commands	Training-free	StarCraft II
Mastermind (2025b)	LLM	LLaMA-2-7B	Text board state	Game moves	SFT (synth. data)	Doudizhu, Go
Metamon (2025)	LLM	Transformer 200M	Text battle state	Move selection	Offline RL	Pokémon
Orak (2025)	LLM	Qwen-2.5-72B	Text / image	Game actions	SFT	12 genres
Think in Games (2025)	LLM	Qwen-2.5-7B/32B	Text state	Game policies	SFT + RL	Multiple
SpinGPT (2025)	LLM	LLaMA-3	Text game state	Poker decisions	SFT + RL	Poker
MS-GRPO (2025)	LLM	Qwen2.5-3B	Text state	Sequential decisions	MS-GRPO	Snake, Frozen Lake
MARSHAL (2025)	LLM	Qwen3-4B	Text scenarios	Strategic decisions	Self-play RL	Multiple
Stronger-MAS (2025)	LLM	Qwen3-1.7B/8B	Text prompts	Multi-agent actions	AT-GRPO	Multiple
GameTalk (2026)	LLM	LLaMA-3-3B	Text dialogue	Strategic dialogue	GRPO + DPO	Dialogue games
VAM (2026c)	LLM	Qwen2.5-3B/7B	Text (chess pos.)	Chess moves	RL + action masking	Chess
ISO-GRPO (2026)	LLM	Qwen2.5-0.5B	Text game state	Poker decisions	SFT + ISO-GRPO	Poker, Pokémon
Mixture of Masters (2026)	LLM	Custom GPT (small)	Text (chess pos.)	Chess moves	RL + MoE routing	Chess
Creative Agents (2023)	VLM	GPT-4V + Diffusion	Image + Text + Goal image	Code action	Prompting	Minecraft
TeamCraft (2024)	VLM	Vicuna-7B/13B	Vision + text	Multi-agent actions	Imitation Learning	Minecraft
VARP (2024)	VLM	3 Models	Game Screenshot	Python kbd/mouse actions	Prompting+Retrieval	Black Myth: Wukong
ROCKET-1 (2024a)	VLM	GPT-4o + SAM-2	Pixel image+Mask	Keyboard&Mouse	SFT	Minecraft
GameSense (2025)	VLM	Qwen2.5-VL	Pixel image	Keyboard&Mouse	Prompting+Module RL	3 action games
Cradle (2025d)	VLM	GPT-4V	GUI Screenshot	Keyboard&Mouse code	Prompting	4 commercial games
ViGaL (2025)	VLM	Qwen2.5-VL-7B	Image + text	Game actions	RL	Arcade games
G1 (2025b)	VLM	Qwen2.5-VL-7B	Game Screenshot	Text action	SFT+GRPO	4 visual games
CoSo (2025)	VLM	LLaVA-1.6	Screenshot+Text	Text action	Online RL	Gym Cards
Game-RL (2025)	VLM	Qwen2.5-VL-7B	Game image+Text QA	Text answer	GRPO	30 games
VistaWise (2025)	VLM	GPT-4o	Text API + KG	API calls	Training-free	Minecraft
Orak (2025)	VLM	Qwen-2.5-72B	Text / image	Game actions	SFT	12 genres
VL-DAC (2025a)	VLM	Qwen2-VL-7B	Screenshot+Text	Text action	PPO	MiniWorld&Gym-Cards
Proact-VL (2026)	VLM	LiveCC-7B-Base	Video stream	Commentary / guidance	SFT	12 live-streamed games
BC-CSGO (2022)	VA	EfficientNet&ConvLSTM	Pixel image	Keyboard&Mouse	SFT	CSGO
VPT (2022)	VA	from scratch(Transformer-XL-style)	Pixel image	Keyboard&Mouse	PT+SFT+PPG	Minecraft
GROOT (2024b)	VA	VPT	Pixel image	Keyboard&Mouse	SFT	Minecraft
P2P(0.1) (2025a)	VA	from scratch(Decoder-only Transformer)	Pixel image	Keyboard&Mouse	PT	Roblox & MS-DOS games
NitroGen (2026b)	VA	Siglip&DiT	Pixel image	Gamepad	PT+SFT	1000+ video games
STEVE-1 (2023)	VLA	VPT&MineCLIP	Pixel image+Text	Keyboard&Mouse	SFT	Minecraft
Omnijarvis (2024)	VLA	LLaVA-7B	Pixel image+Text+Action	Action token	SFT	Minecraft
SIMA (2024)	VLA	SPARC&Phenaki&Transformer-XL	Pixel image+Text	Keyboard&Mouse	SFT	Multiple 3D games
P2P(0.3) (2025b)	VLA	from scratch(EfficientNet&Transformer)	Pixel image+Text	Keyboard&Mouse	PT+SFT	Roblox & MS-DOS & FPS games
SIMA2 (2025)	VLA	Gemini Flash-Lite	Pixel image+Text	Keyboard&Mouse	SFT+RL	Multiple 3D games
JarvisVLA (2025b)	VLA	Qwen2-VL-7B	Pixel image+Text	Action token	CPT+SFT	Minecraft
OpenHA (2025e)	VLA	Qwen2-VL-7B	Pixel image+Text	Multiple action space	PT+SFT	Minecraft
CombatVLA (2025c)	VLA	Qwen2.5-VL-3B	Pixel image+Text	Text Action	SFT	Wukong & Sekiro
Lumine (2025c)	VLA	Qwen2-VL-7B	Pixel image+Text	Text K&M Action	CPT+SFT	Genshin Impact
Game-TARS (2025g)	VLA	Qwen2.5-VL-7B&Seed-VL-1.5	Pixel image+Text	Code Action	CPT+SFT+RFT	500+ video games
UI-TARS2 (2025c)	VLA	Qwen2.5-VL-7B	Pixel image+Text	Text Action	SFT+PPO	15+ 2D games
OpenP2P (2026)	VLA	from scratch(Decoder-only Transformer)	Pixel image+Text	Keyboard&Mouse	PT+SFT	Multiple 3D games
Main-VLA (2026)	VLA	Qwen2-VL-7B	Pixel image+Text	Keyboard&Mouse	SFT	Minecraft & Peace & Valorant
Genie (2024b)	WM	Tokenizer + Transformer	Unlabeled video	Latent actions	Video pretraining	2D platformers
Oasis (2024)	WM	Autoregressive Transformer	Game frames	Keyboard&Mouse	Autoregressive video modeling	Minecraft-like world
GameNGen (2025)	WM	Diffusion model	DOOM frames + actions	Game actions	Diffusion next-frame modeling	DOOM
MineWorld (2025b)	WM	Autoregressive Transformer	Frames + actions	Keyboard&Mouse	Action-conditioned pretraining	Minecraft
GameFactory (2025)	WM	Video diffusion	Text/image/video prompts	Interactive actions	Diffusion fine-tuning	Generated games
Matrix-Game 3.0 (2026b)	WM	Diffusion Transformer	Frames + actions + memory	Keyboard&Mouse	AR diffusion + distillation	Minecraft
GameGen-X (2025)	WM	Video diffusion	Multimodal prompts	Interactive actions	Diffusion fine-tuning	Open-world games
PAN (2025)	WM	Long-horizon WM	History + text actions	Language actions	Long-horizon simulation	General worlds
WorldCam (2026)	WM	Autoregressive Transformer	Frames + camera pose	Camera controls	Pose-aware AR training	3D game worlds
DreamerV3 (2025)	WAM*	RSM	Pixels / states	Game actions	World-model RL	Atari, Minecraft, DMLab
JOWA (2025)	WAM*	World-action Transformer	Offline trajectories	Future actions	Joint WM-action pretraining	Atari
DIAMOND (2024)	WAM*	Diffusion WM	Atari pixels	Atari actions	In-model policy training	Atari

* Prototype WAMs: they couple world prediction with action/policy learning, but are not fully unified gaming WAMs with large foundation models.

RL to focus exploration on strategic quality. However, text games also expose reasoning limits. Credit assignment over hundreds of turns remains unsolved.

Moreover, MARSHAL (Yuan et al., 2025) and Stronger-MAS (Zhao et al., 2025) train on self-play from multiple games, testing cross-domain transfer. Game-RL (Tong et al., 2025) spans 30 games with Code2Logic verifiable rewards; ViGaL (Xie et al., 2025) and Think in Games (Liao et al., 2025) show that game RL improves general reasoning beyond games, suggesting games are training grounds for broader intelligence. Yet, multi-agent coordination degrades with agent count and belief-nesting depth. Most fundamentally, tracking what each player knows, believes, and intends across long interactions imposes a combinatorial cognitive load that current context windows struggle to support. These reasoning bottlenecks define the frontier in the textual multiverse.

Expanding the Frontier: LLMs in Visual Games. Most modern games are visual. LLMs cannot

see pixels but can play when game states reach them as text, through APIs (Mineflayer ([contributors, 2013](#)), TextStarCraft II ([Ma et al., 2024a](#))), text protocols (Pokemon Showdown), or perception modules. In this pipeline, the LLM acts as the reasoning core for planning and decomposition. Minecraft remains the central benchmark: Voyager ([Wang et al., 2023a](#)) introduced autonomous skill libraries, followed by LLaMA-Rider ([Feng et al., 2024](#)) for exploration and JARVIS-1 ([Wang et al., 2023c](#)) for memory-augmented planning. Advances in VistaWise ([Fu et al., 2025](#)) and TeamCraft ([Long et al., 2024](#)) further integrated cross-modal knowledge and multi-agent scaling. In complex RTS environments like StarCraft II, Chain of Summarization ([Ma et al., 2024a](#)) and Reflection of Episodes ([Xu et al., 2025a](#)) tackle context limits through hierarchical abstraction and episodic memory.

Despite successes in multi-game training ([Park et al., 2025](#)), *a perceptual ceiling persists*. Textual mediation often discards critical spatial relations and real-time visual cues, limiting the agent’s ability to master continuous dynamics. This bottleneck necessitates the transition to VLMs and VLAs, which integrate pixel-level perception and end-to-end motor control.

4.2. VLMs as Game Novice: Opening the Eyes to the Visual Multiverse

While LLM-based agents excel at planning and reasoning in text-mediated games, their reliance on text-converted state representations inevitably discards the spatial, temporal, and perceptual details that human players routinely extract from visual feedback. This perceptual bottleneck limits generalization across visually diverse game environments. Vision-Language Models (VLMs) ([Bai et al., 2025a,b](#); [Li et al., 2024](#)) address this gap by integrating visual encoders with language understanding, enabling agents to perceive and reason about game worlds directly from pixels.

VLMs as Perceptual Reasoners for Game Decision-Making. An early exploration in this direction is MineDojo ([Fan et al., 2022](#)), which trains a contrastive video-language model, MineCLIP, to score the alignment between an agent’s video trajectory and a language instruction, serving as a dense reward for reinforcement learning without manual reward design. This work demonstrates that vision-language alignment can effectively guide agents across diverse open-ended tasks, establishing the potential of joint vision-language learning for gameplay.

Subsequent works further leverage VLMs as the perceptual and reasoning backbone. Creative Agents ([Zhang et al., 2023](#)) pairs language and diffusion-based goal image generation with a VLM code controller for creative building in Minecraft. Cradle ([Tan et al., 2025d](#)) introduces the General Computer Control setting, where a VLM reasons over screenshots and outputs keyboard-mouse actions as code, generalizing across commercial games without game-specific APIs. VARP ([Chen et al., 2024](#)) applies VLM-driven action planning to ARPG combat in *Black Myth: Wukong*, combining a visual action planning module with a visual trajectory tracking system that jointly translates raw screen observations into timed combat decisions. ROCKET-1 ([Cai et al., 2024a](#)) addresses the spatial information bottleneck of language by introducing visual-temporal context prompting, where a VLM communicates interaction targets to a low-level policy via segmentation masks. GameSense ([Lu et al., 2025](#)) shifts the VLM from a per-step controller to a developer that synthesizes reusable action-feedback modules for real-time play. Proact-VL ([Yan et al., 2026](#)) further extends VLMs to real-time streaming, building a proactive video language model that processes continuous gameplay and autonomously decides when to respond, serving as an interactive AI companion.

Reinforcement Learning for Sharper Visual Game Reasoning. The recent success of reinforcement learning (RL) in improving VLM reasoning has also influenced game agent research. G1 ([Chen et al., 2025b](#)) trains VLMs via RL self-evolution in a multi-game environment, showing that perception and reasoning mutually bootstrap during training. CoSo ([Feng et al., 2025](#)) improves ex-

ploration efficiency by using counterfactual reasoning to focus RL updates on action-critical tokens. Game-RL (Tong et al., 2025) synthesizes verifiable reasoning data from game source code and applies GRPO-based training (Shao et al., 2024), finding that RL on game data alone transfers to broader vision-language benchmarks. VL-DAC (Bredis et al., 2025a) decouples token-level PPO (Schulman et al., 2017) updates from environment-step value estimation, demonstrating that RL training in lightweight simulators generalizes to real-image agentic control. Collectively, these efforts confirm that RL post-training consistently sharpens VLM decision-making in game settings and can yield benefits beyond the game domain.

Despite these advances, VLM-based game agents still rely on indirect execution mechanisms such as code generation, API calls, or predefined skill libraries and functions to translate high-level understanding into concrete actions. This decoupled architecture introduces latency, limits control granularity, and confines VLM agents to the role of slow-frequency planners. Realizing a truly general game agent thus still requires closing the perception-action loop within an end-to-end framework.

4.3. VLAs as Game Player: Closing the Perception-Action Loop

While VLMs provide visual perception, their outputs remain semantic and require external modules for motor execution. Vision-Language-Action models (VLAs) (Collaboration et al., 2025; Kim et al., 2024) close this gap by mapping multimodal observations directly to low-level control (Chi et al., 2023; Su et al., 2025) within end-to-end architectures. This paradigm builds on a sustained line of vision-to-action (VA) research that learns sensorimotor policies from gameplay.

Scaling Vision-to-Action Pre-training from Videos. Early VA research has explored video-based pre-training as a scalable route to sensorimotor policy learning. CSO (Pearce and Zhu, 2022) demonstrates that behavioral cloning (BC) (Jang et al., 2022) can scale to millions of human gameplay frames to produce competent FPS agents. VPT (Baker et al., 2022) then extends BC by leveraging an inverse dynamics model to pseudo-label massive internet gameplay videos and training a causal policy network on the resulting corpus, confirming the potential of internet-scale video pre-training for action generation. STEVE-1 (Lifshitz et al., 2023) builds on VPT with goal-conditioned fine-tuning, showing that the paradigm scales to multi-step task compositions. GROOT (Cai et al., 2024b) extends goal conditioning by using reference gameplay videos as instructions, demonstrating that video pre-training can simultaneously provide the policy prior. NitroGen (Magne et al., 2026b) further broadens the scope of VA paradigm, training a joint ViT-DiT vision-action architecture on 40K hours of gameplay spanning over 1,000 titles to achieve general-purpose continuous action generation for smoother and higher-resolution control. In light of the above VA advances, Pixels2Play (Yue et al., 2026) provides a systematic investigation of BC scaling laws, showing that increasing model capacity and data volume not only improves imitation fidelity but also promotes the emergence of causal reasoning over spurious correlation.

Leveraging Multimodal Understanding for Action Generation in VLAs. Building on the VA foundations, contemporary VLAs retain the end-to-end sensorimotor backbone while further integrating language comprehension into a unified multimodal policy, OmniJARVIS (Wang et al., 2024) pioneers the paradigm by processing multimodal inputs through a shared encoder-decoder architecture and training action outputs via BC, with SIMA (Team et al., 2024) validating the generality of this paradigm by training a language-conditioned agent across diverse commercial video games. This framework is quickly inherited by JARVIS-VLA (Li et al., 2025b) and OpenHA (Wang et al., 2025e), which further leverage pre-trained VLMs as the perceptual encoder and append lightweight policy decoders to generate actions.

The adoption of VLMs raises whether reinforcing their domain-specific understanding can benefit downstream action generation. CombatVLA (Chen et al., 2025c) distills tactical reasoning into learned representations through a progressive *Action-of-Thought* curriculum; MAIN-VLA (Zhou et al., 2026) prunes redundant visual and linguistic semantics into compact, action-critical features; and Lumine (Tan et al., 2025c) unifies language reasoning with action generation in a closed-loop framework, achieving instruction-adherent long-horizon control with cross-game transfer. These results indicate that deepening the perceptual and reasoning capacity of VLM encoders consistently improves both long-horizon interaction and cross-game generalizability.

Building on these supervised foundations, recent work introduces RL post-training to enable self-improvement beyond demonstration data—whether through self-generated gameplay across 3D virtual worlds (SIMA-team et al., 2025) or multi-turn RL within GUI environments at scale (Wang et al., 2025c). This supervised pre-training followed by RL post-training pipeline has become a prevalent recipe for generalist VLA agents, yielding policies that continuously adapt across heterogeneous interactive environments.

Toward Real-Time Control: Evolving Action Representations. To achieve better real-time control, the development of game VLAs has also focused on improving action representation. Early VLAs such as JARVIS-VLA (Li et al., 2025b) and Game-TARS (Wang et al., 2025g) discretize control into per-step tokens generated autoregressively alongside language output. To reduce the resulting inference overhead, subsequent work shifts toward *action chunks*: CombatVLA (Chen et al., 2025c) applies truncated decoding of action-of-thought sequences for a 50-fold speedup in 3D ARPG combat, while Lumine (Tan et al., 2025c) predicts six 33ms textual chunks per step to sustain 30Hz keyboard-mouse control. This trajectory culminates in NitroGen (Magne et al., 2026b), which bypasses language-space encoding entirely and employs a flow-matching diffusion transformer to generate 16-step chunks of *continuous* gamepad actions from a single RGB frame. Collectively, these advances chart a path from discrete, per-token action decoding toward unified continuous representations capable of accommodating diverse control modalities at real-time rates.

4.4. WAMs as Game Expert: Internalizing the Dynamics of the Game Multiverse

VLAs have achieved strong semantic reasoning and fine-grained control. However, their understanding of game-world evolution still remains largely semantic, and their control generation is predominantly reactive. WAMs offer a new paradigm to overcome these limitations in two aspects:

- **Predictive Control:** By internalizing action-conditioned dynamics, WAMs support *prediction before action*, so that decisions are guided by future anticipation rather than only current states.
- **Faithful Dynamics:** WAMs learn *world models* of games from visual and control signals, leading to more faithful dynamics modeling of how game worlds evolve under interaction.

More importantly, these two properties also make WAMs inherently more *general*: instead of merely fitting game-specific action patterns, they capture transferable regularities of how game worlds evolve under intervention across diverse games. In this sense, WAMs shift game agents from reactive perceive-and-act systems to predictive plan-and-execute systems, marking a crucial step toward more generalist game players.

Early Prototypes of Predictive Control. Early world-model-based game agents marked the first step toward predictive game playing by introducing dynamics internalization and future modeling into control. In this sense, they can be viewed as prototypes of gaming WAMs. *Action-Conditional*

Video Prediction (Oh et al., 2015) first made the dependency between future frames and control variables explicit, training convolutional and recurrent predictors to roll out Atari futures conditioned on candidate actions. *World Models* (Ha and Schmidhuber, 2018) compressed visual observations with a VAE, learned latent dynamics with an MDN-RNN, and evolved a compact controller inside the learned “dream” environment. *SimPLe* (Kaiser et al., 2020) turned this idea into an iterative Atari training loop, repeatedly fitting a stochastic video model from real interaction and training PPO on short imagined rollouts. The *Dreamer* series (Hafner et al., 2020, 2021, 2025) jointly trains an encoder, representation function, and dynamics function within recurrent latent state-space models, enabling implicit predictive control through imagined actor-critic learning across increasingly diverse control tasks. *IRIS* (Micheli et al., 2023) replaced recurrent latent prediction with a transformer world model over discrete image tokens, and *DIAMOND* (Alonso et al., 2024) further showed that diffusion-based dynamics can preserve action-relevant visual details and train Atari agents entirely inside a learned world model. Their shared contribution lies not in a modern unified world-action architecture that jointly maps historical observations and actions to future observations and actions, but in demonstrating that prediction-guided control can substantially improve control effectiveness. By grounding control in predicted futures, these works establish the early prototype of gaming WAMs. Nevertheless, limited generative capacity and insufficient dynamics-modeling ability prevented these early models from fully internalizing complex, visually grounded dynamics across a broader set of modern games with complex spatial layouts and rich interaction mechanisms.

Gaming World Models with Faithful Dynamics. While early prototypes had already demonstrated the effectiveness of predictive control, recent gaming WMs have strengthened the dynamics model itself. *Genie* (Bruce et al., 2024b) established this direction by learning latent actions and autoregressive dynamics from unlabeled videos, showing that interactive environments can emerge without explicit action labels. This autoregressive paradigm was later grounded in Minecraft-like sandbox worlds by *Oasis*, *MineWorld*, and the *Matrix-Game* series (Decart and Julian Quevedo, 2024; Guo et al., 2025b; He et al., 2025a; Wang et al., 2026b; Zhang et al., 2025d), which progressively improved action-conditioned generation, action following, streaming rollout, and long-horizon memory, moving gaming WMs from video continuation toward controllable neural environments. As domains become visually richer, diffusion- or flow-style neural engines further improve fidelity and controllability. *GameNGen* (Valevski et al., 2025) shows that a diffusion model trained on *DOOM* traces can operate as a real-time neural game engine for FPS dynamics, while *GameFactory* and *GameGen-X* (Che et al., 2025; Yu et al., 2025) use video-generation backbones and multi-stage training to extend world modeling toward open-domain, modern 3D, and open-world games. Beyond real-time controllability, recent works increasingly target persistent dynamics. *PAN* and *Model as a Game* (Chen et al., 2025a; PAN Team Institute of Foundation Models, 2025) emphasize long-horizon interaction, mechanic consistency, and numerical/spatial reliability; *WorldPlay* and *WorldCam* (Nam et al., 2026; Sun et al., 2025c) strengthen geometric consistency and spatial revisitation, with *WorldCam* using camera-pose control for long-horizon 3D grounding; and *Solaris* (Savva et al., 2026) extends world modeling to synchronized multiplayer Minecraft with coherent shared states. These advances suggest that dynamics modeling of modern games with complex spatial structures and interaction mechanisms is no longer fragile. Rather, world modeling for modern games is becoming increasingly reliable, providing the foundation on which gaming WAMs can internalize, predict, and ultimately control.

Towards Modern Gaming World Action Models. World Action Models (WAMs) jointly model future world states and future actions from interaction history, enabling prediction and control to be

Table 4: Summary of representative harness designing in gaming AI

Paper	Perception	Action	Reasoning	Real-time Reactivity	Memory	Adaptive Learning
GameVerse (2026a)	Raw	GUI/Semantic	Workflow	–	Both	Self-Reflection
PokeAgent Challenge (2026)	Textual Env.	Semantic	Multi-Agent	–	Both	Self-Reflection
Steve-Evolving (2026)	Visual Scaffolding	Code Control	Workflow	–	Both	Evolving-Skills
PERIL (2025)	Textual Env.	Semantic	Prompt Engineering	–	Working Only	–
AgileThinker (2025)	Textual Env.	Semantic	Workflow	Dual Thread	Working Only	–
CWM (2025)	Textual Env.	Code Control	Workflow	–	Both	Self-Reflection
TITAN (2025a)	Textual Env.	Semantic	Workflow	–	Both	Self-Reflection
FlashAdventure (2025)	Raw	GUI	Workflow	–	Both	Self-Reflection
DeepPHY (2026)	Visual Scaffolding	Semantic	Prompt Engineering	–	Working Only	Self-Reflection
VistaWise (2025)	Visual Scaffolding	GUI	Prompt Engineering	Pausing	Both	–
GMH (2025e)	Visual Scaffolding & Textual Env.	Semantic	Workflow	–	Working Only	Prompt Opt.
Orak (2025)	Textual Env.	Semantic	Workflow	Pausing	Both	Self-Reflection
Lmgame-Bench (2025a)	Textual Env.	Semantic	Prompt Engineering	Pausing	Working Only	Prompt Opt. & Self-Reflection
LPLH (2025)	Textual Env.	Semantic	Workflow	–	Both	Self-Reflection
VideoGameBench (2025a)	Raw	GUI	Workflow	Pausing	Working Only	–
GameSense (2025)	Visual Scaffolding	Code Control	Workflow	Decoupling Reasoning from Action	Both	Self-Reflection & Evolving Skills
DSGBench (2025)	Textual Env.	Semantic	Workflow	Pausing	Working Only	Self-Reflection
GAMEBoT (2025)	Textual Env.	Semantic	Prompt Engineering	–	Working Only	–
StarCraft Bench (2024b)	Textual Env.	Semantic	Prompt Engineering	–	Working Only	–
Cradle (2025d)	Raw	GUI	Workflow	Pausing	Both	Self-Reflection
Voyager (2023a)	Textual Env.	Code Control	Workflow	–	Both	Evolving Skills
AgentVerse (2023)	Textual Env.	Semantic	Multi-Agent	–	Working Only	–
DEPS (2023b)	Textual Env.	Semantic	Workflow	–	Working Only	Self-Reflection

learned within a unified framework rather than as separate modules. Promising evidence has already emerged in robotics, where works such as DreamZero (Ye et al., 2026) couple predictive world prediction with action generation, improving embodied control and cross-task generalization. In games, the Dreamer series (Hafner et al., 2020, 2021, 2025) provides an important prototype of this paradigm. Although it does not fully unify world-action anticipation, it demonstrates the effectiveness of acting on latent states shaped by internalized future dynamics. JOWA (Cheng et al., 2025) takes a more direct step toward modern gaming WAMs by jointly pretraining world and action predictions, showing that world-action coupling can improve decision-making in game environments. Nevertheless, gaming WAMs remain largely underexplored. Existing efforts have not yet fully leveraged the stronger world-model substrate emerging from recent gaming WMs, where faithful dynamics are learned from visual and control signals across richer environments. In this setting, WAMs could further enhance predictive control by grounding action generation in deeper anticipations of world evolution. More importantly, it may enable agents to learn transferable dynamics rather than game-specific action patterns, moving gaming WAMs toward a more fundamental world understanding and ultimately toward *omni-reality adaptability* across the game multiverse.

5. Harness: The Nervous System of Generalist Game Player

Classical cognitive science frameworks, such as the Unified Theories of Cognition (UTC), posit that intelligence is fundamentally a closed-loop system intertwined with essential functions like perception, memory, reasoning, and action. (Laird et al., 2017; Newell, 1994) However, although the foundation models demonstrate powerful representation capabilities, they are stateless by design and suffer from inherent limitations, such as memory volatility. (Kim, 2025; Tang et al., 2026) Consequently, they fail to function as complete intelligent systems capable of navigating dynamic and complex environments independently. To bridge this gap, researchers have engineered various Modular Harnesses designed to decouple and augment the foundational functions required by autonomous agents.

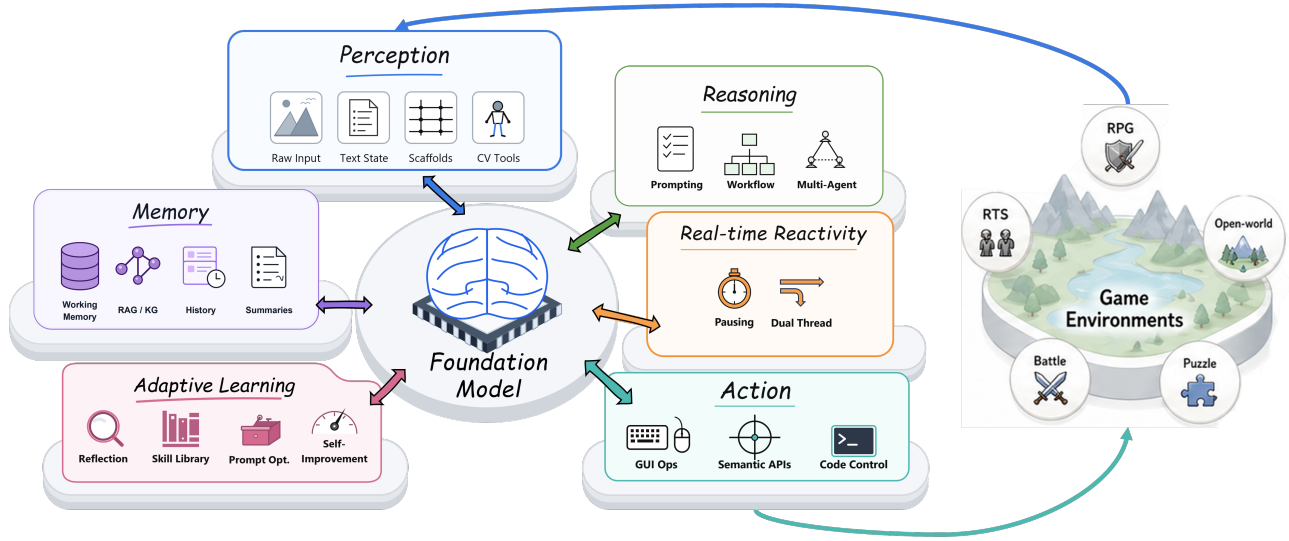


Fig. 4: Overview of Harness in Game-playing AI.

In the current landscape, the most prominent harnessing components can be categorized into six core dimensions:

- **Perception:** The perception module extracts multi-modal information from raw environmental observations (Ω) and transforms it into structured representations—such as textual prompts—that the underlying reasoning engine (e.g., LLMs) can process.
- **Action:** The action module translates the agent’s internal decisions into executable operations. It involves defining the allowable action space (A), designing the invocation mechanisms (e.g., APIs, code execution, or GUI operations), and applying these actions to alter the environment state.
- **Reasoning:** Serving as the cognitive core, the reasoning module synthesizes the predefined goals (G) with perceptual inputs to estimate the current state (S). Instead of executing actions directly, it performs task decomposition, multi-step planning, and selects the optimal action trajectory.
- **Real-time Reactivity:** This mechanism is specifically designed to mitigate the critical gap between the inherent inference latency of foundation models and the temporal constraints of dynamic, real-time environments, ensuring timely and robust responses.
- **Memory:** To overcome the stateless nature of foundation models, the memory module persistently stores and manages key interaction histories, environment knowledge, and procedural experiences. It provides the agent with a coherent context for consistent decision-making.
- **Adaptive Learning:** The adaptive learning module drives the continuous evolution of the agent. By reflecting on successes and failures, it abstracts interaction trajectories into reusable heuristic experiences, enabling the system to self-correct and autonomously adapt to novel or dynamic environments over time.

In the following sections, we systematically review the harness designs, practical efficacies, and current limitations corresponding to each of these six functional dimensions.

5.1. Perception:

In game interactions, vision serves as the most fundamental perceptual channel for humans, enabling them to observe dynamic environments, read textual information, confirm game states, and guide precise action execution. However, this natural process poses a significant challenge for foundation models. Large Language Models (LLMs) are inherently restricted to the text modality; meanwhile, although Vision-Language Models (VLMs) can process visual inputs and comprehend global semantics, they still exhibit a substantial gap compared to human visual adaptation, particularly in fine-grained spatial alignment. Recent studies reveal that when raw game screenshots are used directly as input, VLMs frequently encounter issues such as spatial localization bias, semantic ambiguity, and severe perceptual hallucinations(Zhang et al., 2025a, 2026a). To bridge this “spatial-semantic gap” and alleviate the models’ cognitive load, researchers have designed various perception harnesses. These module designs primarily include: **Textual Environment** and **Visual Scaffolding**.

Textual Environment. The textual environment paradigm bypasses the inherent spatial grounding limitations of vision models by converting complex, high-dimensional game states into structured natural language, tabular formats, or symbolic code representations. Recent studies implement this paradigm through distinct, framework-specific designs to optimize cognitive load. For instance, the LPLH framework utilizes Interactive Fiction games, which are natively text-based, thereby entirely circumventing visual perception challenges and allowing the agent to focus purely on semantic understanding and narrative reasoning(Zhang and Long, 2025). For visually complex Real-Time Strategy games, DSGBench and Reflection of Episodes leverage the TextStarCraft II environment to parse dense environmental states—such as unit counts, resource levels, and building statuses—into comprehensive textual summaries(Ma et al., 2024b; Tang et al., 2025; Xu et al., 2025a). Similarly, The PokeAgentChallenge and Pokéchamp Agent abstract the intricate battle mechanics of Pokémon into structured text via the Pokémon Showdown simulator(Karten et al., 2025, 2026). For spatially structured tasks, frameworks like LmgameBench extract backend data from grid-based games (e.g., Sokoban, 2048, Tetris) to convert visual layouts into textual tables, explicitly listing object coordinates and their attributes(Hu et al., 2025a). For open-world 3D sandbox games, frameworks like Voyager and the Orak benchmark utilize the Mineflayer JavaScript API to entirely bypass raw visual processing(contributors, 2013; Park et al., 2025; Wang et al., 2023a). Taking abstraction a step further, Code World Model translates natural language game rules and historical trajectories into an executable Python-based world model. This approach transforms the game environment into formal code, serving as a reliable simulator to support rigorous algorithmic planning, such as Monte Carlo Tree Search(Lehrach et al., 2025).

Visual Scaffolding. In scenarios where visual modalities need to be retained, frameworks deploy visual scaffolding or external computer vision (CV) tools to bridge the spatial-semantic gap. To directly enhance the raw visual input, General Modular Harness explicitly overlays grid lines and coordinate labels onto the game images to reduce perception errors(Zhang et al., 2025e). Alternatively, Steve-Evolving anchors continuous visual experiences into structured state snapshots (e.g., precise coordinates, inventory summaries, health, and GUI states), pairing them with fine-grained execution diagnoses to facilitate reliable recall and agent evolution(Xie et al., 2026). In privilege-free settings, modules employ external CV models as visual parsers. For example, GameSense operates entirely on real-time game screenshots without API access, deploying a suite of CV tools (e.g., Grounding DINO for object detection and OpenCV for state reading) to parse highly dynamic environments in FPS and

action games(Lu et al., 2025). Similarly, VistaWise and the Orak benchmark deploy lightweight object detection models (such as YOLOv10 and YOLOv11) to extract bounding boxes and entity positions from raw screenshots(Fu et al., 2025; Park et al., 2025). Crucially, they perform perception abstraction to translate continuous spatial data into discrete semantic concepts (e.g., categorizing the distance between fighters as “very close” or “far”).

5.2. Action:

When playing games, human players seamlessly translate high-level semantic intentions (e.g., “aim and shoot the enemy”) into fine-grained motor executions (e.g., moving the mouse to specific coordinates and left-clicking). While current foundation models exhibit robust reasoning and tool-invocation capabilities, they lack native motor skills to directly manipulate digital environments. Therefore, defining an appropriate action space and encapsulating environmental interactions into executable interfaces serves as the critical bridge connecting the model’s cognition to the game engine. As action spaces have rapidly evolved from predefined function calls toward more flexible interface-level navigation and code execution, we categorize current Action modules into three progressive paradigms: **GUI-level operations** via General Computer Control (GCC)(Tan et al., 2025d), **Semantic-designed action spaces** through API and skill encapsulation, and **Programmatic control** via automated code generation.

GUI-level Operations General Computer Control (GCC), a paradigm formalized by the Cradle framework, aims to build foundational agents that master computer tasks via a universal human-style interface—receiving inputs from screens and audio, and outputting native keyboard and mouse actions(Tan et al., 2025d). A representative design of such a GCC action space used by GameVerse is listed in Table.5 (Zhang et al., 2026a). Frameworks such as Cradle demonstrate that, by utilizing pure GUI actions, agents can navigate complex commercial games like Red Dead Redemption 2 without relying on any built-in APIs(Tan et al., 2025d). Similarly, benchmarks like FlashAdventure evaluate agents on completing full story arcs in diverse web-based games using mouse and keyboard interfaces(Ahn et al., 2025). While this paradigm offers ultimate generalizability across any software, it exposes a critical limitation of current Vision-Language Models (VLMs) known as the “knowing-doing gap” or “semantic-execution gap”(Zhang et al., 2025a, 2026a). As observed in evaluations, even when an agent correctly reasons about the optimal high-level strategy, it frequently fails to map this semantic intent to the precise action with the right parameters, such as coordinates. This profound disconnect between strategic reasoning and action execution leads to significant performance degradation, particularly in real-time or precision-heavy scenarios. Furthermore, GUI operations like drag and hold_key pose significant spatial and temporal challenges, as the drag trajectory and holding time require spatial and temporal intelligence, or even a combination of both. VideoGameBench noted that current foundation models perform poorly in the corresponding game environments (Zhang et al., 2025a).

Semantic-designed Actions Semantic actions are mostly pre-defined according to the common operational logic within the environment, utilizing conceptual or discrete parameters (e.g., target IDs or semantic locations) instead of precise pixel coordinates and physical distances(Lin et al., 2025; Wang et al., 2025a; Zhang et al., 2026a, 2025e). For instance, in open-world MMORPGs, the TITAN framework abstracts the overwhelmingly vast continuous action space into high-level templates such

Table 5: Example of GUI Action Space

Action Type	Description	Parameters
<i>Mouse Movement</i>		
MOVE_TO	Move to position	x, y (int, req): Target coordinates (window relative)
MOVE_BY	Move relatively	dx, dy (int, req): Offset; <code>duration</code> (float, opt): Seconds
<i>Mouse Click</i>		
CLICK	Click mouse	x, y (opt); <code>button</code> (left/right/middle); <code>num_clicks</code>
RIGHT_CLICK	Right click	x, y (int, opt): Click coordinates
DOUBLE_CLICK	Double click	x, y (int, opt): Click coordinates
<i>Mouse Drag</i>		
MOUSE_DOWN	Press button	<code>button</code> (str, opt): "left", "right", or "middle"; <code>duration</code> (float, opt)
MOUSE_UP	Release button	<code>button</code> (str, opt): "left", "right", or "middle";
DRAG_TO	Drag to target	x, y (int, req): Target coordinates
<i>Mouse Scroll</i>		
SCROLL	Scroll wheel	dx, dy (int, req): Scroll amount (+/-); <code>duration</code> (float, opt)
<i>Keyboard Input</i>		
TYPING	Type text	<code>text</code> (str, req); <code>interval</code> (float, opt)
PRESS	Press key	<code>key</code> (str, req); <code>duration</code> (float, opt)
KEY_DOWN	Key down	<code>key</code> (str, req): Key name; <code>duration</code> (float, opt)
KEY_UP	Key up	<code>key</code> (str, req): Key name;
HOTKEY	Key combo	<code>keys</code> (list, req): e.g., ['ctrl', 'c']; <code>duration</code> (float, opt)
<i>Control Flow</i>		
WAIT	Wait time	<code>duration</code> (float, req): Seconds to wait
DONE	Task success	No parameters
FAIL	Task failure	No parameters

as `Moveto(Location)` and `Attack(target)` (Wang et al., 2025a). Similarly, the Orak benchmark abstracts the exact frame-perfect joystick combinations required in fighting games (e.g., Street Fighter III) into interpretable, discrete semantic commands like "Move Closer" or character-specific skills like "Fireball" (Park et al., 2025). To systematically implement these semantic invocations, recent advancements leverage standardized tool-calling interfaces. In the PokéAgent Challenge and Orak, the semantic actions are wrapped into independent Model Context Protocol (MCP) server to achieve plug-and-play functionality (Karten et al., 2026; Park et al., 2025).

Programmatic Control Programmatic control decouples high-level reasoning from low-level execution by utilizing executable code as the action space. Unlike semantic APIs that rely on predefined functions, this paradigm allows foundation models to synthesize their own execution logic. For instance, to mitigate the inference latency associated with direct key-mouse control in fast-paced environments, the GameSense framework utilizes VLMs to generate specialized Python scripts known

as “Game Sense Modules” (GSMs). These scripts encapsulate real-time interactive logic and integrate external vision tools, running locally to manage tasks such as combat (Lu et al., 2025). In open-world settings, Voyager leverages the Mineflayer API to generate JavaScript control primitives, verifying and storing these code snippets in an extensible skill library for future reuse (Wang et al., 2023a). Expanding this approach to environment simulation, Code World Models (CWM) prompts LLMs to translate natural language game rules and trajectories into an executable Python world model. This synthesized code includes functions for state transitions, legal move enumeration, and termination checks, functioning as a simulation engine to facilitate planning algorithms like Monte Carlo Tree Search (MCTS) (Lehrach et al., 2025).

5.3. Reasoning:

While perception modules extract environmental states and action modules execute commands, reasoning and planning modules serve as the central cognitive engine that bridges the two. Foundation models typically excel at rapid, intuitive responses but often struggle with the deliberate, multi-step logic required for complex strategic tasks. To elicit higher-order reasoning, recent research leverages various inference-time cognitive harnesses. Based on the structural design of the reasoning process, we categorize current Reasoning modules into three progressive paradigms: **Prompt Engineering**, **Workflow Orchestration**, and **Multi-Agent Systems**.

Prompt Engineering At the most fundamental level, reasoning capabilities are elicited through carefully designed prompt formulations. Techniques such as Chain-of-Thought (CoT) prompting explicitly guide the model to generate intermediate reasoning traces before executing an action. For instance, the GAMEBoT framework leverages domain-specific CoT prompts to guide LLMs in addressing predefined modular subproblems prior to action selection (Lin et al., 2025). Similarly, DEPS provides few-shot demonstrations to prompt the model into generating self-explanations for complex open-world tasks (Wang et al., 2023b), while environments like 1mgame-Bench are specifically designed to support modern reasoning paradigms by allowing models to be evaluated with or without long chain-of-thought (long-CoT) reasoning (Hu et al., 2025a). Furthermore, to steer strategic reasoning in adversarial or cooperative settings, techniques like Persona Prompting inject specific strategic profiles into the model to reshape its decision-making process, as demonstrated in strategic board games like PERIL (Licato and Steinle, 2025).

Workflow Orchestration To tackle long-horizon tasks and dynamic environments, raw reasoning is structured into systematic workflows. A foundational paradigm is ReAct, which interleaves reasoning traces with executable actions, allowing the agent to continuously ground its thoughts in environmental observations rather than relying purely on internal logic (Yao et al., 2022). For complex goals, explicit task decomposition is employed to break down macro-objectives into manageable sub-goals. For instance, DEPS incorporates an interactive planning process with a trainable goal selector to rank parallel candidate sub-goals (Wang et al., 2023b), while Voyager utilizes an iterative prompting mechanism that incorporates environment feedback, execution errors, and self-verification to build and refine an executable skill library (Wang et al., 2023a). Similarly, TITAN employs a plan-execute-memorize-diagnose-replan workflow that tracks action histories and state coverage to diagnose execution stalls and dynamically adjust testing strategies in complex MMORPGs (Wang et al., 2025a). Taking structural reasoning a step further, frameworks incorporate traditional search algorithms directly into the LLM’s reasoning loop, such as PokéChamp, which utilizes LLMs for heuristic position

evaluation to guide depth-limited minimax search(Karten et al., 2025).

Multi-Agent Systems Shifting the focus from individual intelligence to collective rationality, reasoning is distributed across multi-agent ecosystems. In complex environments, architectures often adopt an orchestrator with specialized sub-agents. The PokéAgent Challenge implements a central orchestrator that maintains high-level route plans while dynamically dispatching specialized sub-agents for battle strategy, puzzle-solving, and self-reflection(Karten et al., 2026). Beyond internal cognitive division, AgentVerse deploys multiple independent agents in Minecraft(Chen et al., 2023). In this shared open-ended environment, this multi-player collaboration enables agents to efficiently tackle multi-step tasks while exhibiting emergent social behaviors, such as volunteering unallocated time and resources to accelerate team progress.

5.4. Real-time Reactivity Modules:

The inherent inference and execution latency of foundation models severely restricts their performance in real-time, dynamic environments. To mitigate this latency in interactive games, environment pausing is conventionally used to skip the reasoning time. For instance, Cradle, Orak, and VideoGameBench Lite pause the game during reasoning(Park et al., 2025; Tan et al., 2025d; Zhang et al., 2025a), while NitroGen intercepts the system clock of the game engine to freeze simulation time (Magne et al., 2026a). To bypass the reliance on environment pausing, recent frameworks focus on decoupling reasoning from action execution. In GameSense, by shifting the VLM’s role from a direct controller to a module developer, the VLM generates task-specific executable code, termed Game Sense Modules (GSMs), which autonomously take over high-frequency real-time operations(Lu et al., 2025). Besides, AgileThinker introduces a dual-thread architecture consisting of a planning thread for multi-step reasoning and a parallel reactive thread for outputting actions in real-time(Wen et al., 2025). Furthermore, to achieve real-time action generation, Lumine utilizes hybrid thinking to skip unnecessary reasoning and employs action chunking for high-frequency control, alongside latency optimizations including StreamingLLM, tensor parallelism, W8A8 quantization, and speculative decoding(Leviathan et al., 2023; Tan et al., 2025b; Xiao et al., 2023a,b).

5.5. Memory:

Since foundation models are inherently stateless, memory modules are essential to store and maintain information for multi-step interactions of agents with dynamic environments(Sumers et al., 2023). Following the Cognitive Architectures for Language Agents (CoALA) framework, we systematically organize memory modules into **Working Memory** and **Long-Term Memory**.

Working Memory Working memory maintains active and readily available information, persists across foundation model calls, and enables the models to be stateful agents(Sumers et al., 2023). A common approach for working memory is keeping a fixed steps of recent states and actions using a sliding window, such as the memory modules in lngame-Bench and GMH(Hu et al., 2025a; Zhang et al., 2025e). Furthermore, VistaWise designs the memory stack based on the concept of Last-In-First-Out (LIFO), allowing the agent to recall decisions from the most recent to earlier ones with controllable recall steps(Fu et al., 2025). Another form is history summarization, which maintains a fixed-length or compacted summary from history to avoid context overflow, as utilized in GameVerse and the PokéAgent Challenge(Karten et al., 2026; Zhang et al., 2026a).

Long-Term Memory Following the CoALA architecture, long-term memory can be systematically categorized into semantic, episodic, and procedural memory(Sumers et al., 2023). Semantic memory provides the agent with factual background knowledge about the environment. For instance, LPLH and VistaWise utilize knowledge graphs to store world knowledge, entities, and connections, dynamically evolving them with the agent’s exploration(Fu et al., 2025; Zhang and Long, 2025). Similarly, to mitigate the lack of domain-specific knowledge, the PokéAgent Challenge injects usage statistics and competitive strategies directly from Smogon, an online Pokémon community(Karten et al., 2026). Episodic memory records the agent’s past experiences and historical trajectories. To effectively maintain and retrieve these experiences, GameVerse employs a vector database (ChromaDB) to store the multimodal experiences that the agent decides to save(Zhang et al., 2026a). Finally, procedural memory stores the executable skills and action patterns. For example, Voyager maintains an ever-growing skill library of executable code to retain useful skills for future interactions(Wang et al., 2023a), while GameSense constructs a procedural memory database to store and retrieve historical action implementations and their corresponding execution codes(Lu et al., 2025).

5.6. Adaptive Learning:

In a training-free paradigm, agents must adapt to dynamic environments entirely through their external harness rather than parameter updates. We categorize these adaptive learning mechanisms into three hierarchical levels: **Self-Reflection** for text-based experiential learning, **Evolving Skills** for executable action-level evolution, and **Prompt Optimization** for system-level instruction refinement.

Self-Reflection Reflection utilizes the in-context learning capacity of foundational models, prompting the agent to analyze historical information and generate reflective text, which is then injected into the next-turn prompt to guide future actions. For state-level diagnosis, Orak compares the current state, previous state, and executed actions to generate reflections on state transitions(Park et al., 2025). Similarly, TITAN generates diagnostic reflections by evaluating the abstracted state, current screenshots, action history, and task objectives(Wang et al., 2025a). To further augment the reflection process, recent frameworks incorporate expert experiences as reference baselines. Reflection of Episodes (ROE) combines pre-defined expert experiences with self-experiences based on keyframe selection, generating updated self-experiences through post-episode reflection(Xu et al., 2025a). Furthermore, GameVerse employs Vision-Language Models (VLMs) to simultaneously analyze the agent’s failure videos and human expert tutorials, contrasting the two visual trajectories to generate concentrated experience prompts for policy refinement(Zhang et al., 2026a).

Evolving Skills To better adapt to dynamic game environments, agents can continuously learn and memorize new skills. For example, in Voyager, once a generated piece of code is self-verified to achieve its target goal, it is treated as a new skill and stored in an ever-growing skill library for future reuse(Wang et al., 2023a). Furthermore, Steve-Evolving advances this concept by structurally anchoring interactive experiences with fine-grained diagnostic signals, including state differences and enumerated failure causes(Xie et al., 2026). Based on these anchored experiences, successful actions are distilled into reusable skills with explicit preconditions, while failed trajectories are distilled into executable guardrails to strictly prevent the agent from repeating similar mistakes.

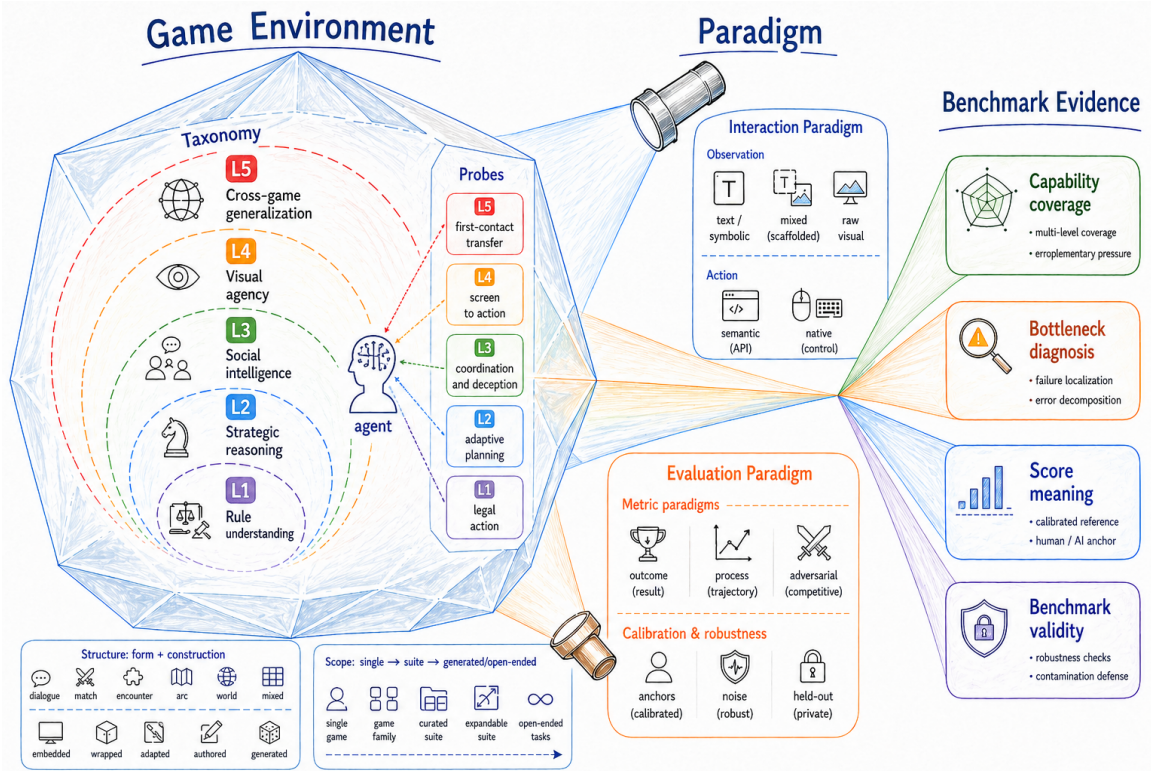


Fig. 5: Overview of conceptual framework for game benchmarks. We frame game benchmarks through three connected dimensions: five-level environment organization, capability probing over agents, and the interaction and evaluation paradigms that shape gameplay into benchmark evidence.

Prompt Optimization Agents’ performance heavily depends on the quality of prompts. To address this, the General Modular Harness (GMH) introduces a two-stage prompt optimization pipeline leveraging objective reward signals from the environment(Zhang et al., 2025e). In the first stage, GMH applies empirical prompt engineering to construct a baseline template that integrates trajectory histories (past states, actions, and rewards) and reflective summaries. In the second stage, it standardizes the optimization process using the DSPy framework and the SIMBA algorithm. By treating the game’s cumulative reward as the objective function, this process iteratively refines the baseline prompt templates.

6. Benchmarks: The Diverse Arena for Game-Playing Agents

The rapid advancement of large language and vision-language models has brought a fundamental evaluation challenge to the forefront: how should we assess systems that are increasingly studied not merely as answer generators, but as agents that must act over time? While many widely used benchmarks remain valuable for testing bounded knowledge, logical reasoning, or one-shot multi-modal understanding, they provide only partial evidence of the intelligence required for continuous interaction. True agentic competence emerges in closed-loop settings, where a system must track an evolving state, make decisions under uncertainty, adapt to feedback, and recover from mistakes while pursuing a goal across multiple steps. The central evaluation gap, therefore, is not simply that

existing tests are too easy or too narrow, but that many of them abstract away the dynamic structure in which intelligent behavior must actually unfold.

Games provide a compelling substrate for this kind of evaluation because they preserve that dynamic structure in forms natively designed for human play. Rather than mere collections of themed tasks, games are carefully engineered challenge systems characterized by explicit rules, goals, feedback loops, state progression, and consequences that accumulate over time. These properties make games natural probes of human-relevant capability. They effectively expose a broad range of reasoning demands, including rule understanding, planning, memory, uncertainty handling, coordination, and learning from interaction, while also supporting visually grounded agent tasks in which success depends on sustained perception, action, and progress toward an objective (Duan et al., 2024; Paglieri et al., 2025; Wu et al., 2024b). In this sense, games are valuable not because they are simply “harder” than static benchmarks, but because they retain the interactive, goal-directed, and human-oriented structure that static benchmarks frequently eliminate.

This perspective also changes how we should understand game diversity. In the context of this survey, the value of games does not lie in assembling a loose catalog of genres, but in viewing the multiverse of human games as a structured space of capability coverage. Different game structures impose different demands on reasoning, memory, coordination, adaptation, and long-horizon control. Thus, diversity matters not as surface-level variation, but as a principled mechanism for applying complementary cognitive and operational pressures on AI agents. This is precisely why games are unusually promising as benchmarks: they span a broad, continually expanding, and difficult-to-saturate design space. As highlighted by AI GameStore(Ying et al., 2026), the point is not to identify a small set of canonical titles to solve once and for all, but to treat the multiverse of games as an extensible evaluation substrate for increasingly general, human-relevant intelligence.

However, a game environment does not automatically constitute a benchmark. It becomes one only when a protocol specifies how gameplay is presented, what qualifies as progress, and how behavior is converted into evidence. This section develops that argument through three dimensions:

- **Taxonomy** examines the types of game benchmarks the field has developed, organizing them into an evolutionary design space that spans rule-bounded settings, broader reasoning abilities, and increasingly open-ended environments.
- **Purpose** investigates what these environments are designed to measure, mapping different game structures to their motivating capability targets—ranging from rule grounding and strategic reasoning to social intelligence, visual agency, and cross-game generalization.
- **Paradigm** explores how gameplay is operationalized into benchmark evidence, analyzing the interaction interfaces and evaluation metrics that determine what a benchmark score truly represents.

Together, these three sections move from benchmark structure, to capability target, to measurement logic. This framing provides a comprehensive understanding of how existing game benchmarks are organized and how the field has evolved. It also helps clarify how benchmark design can more effectively reveal model capabilities and limitations, thereby informing future model and benchmark development.

6.1. Taxonomy: The Evolutionary Levels of Game Environments

While conventional game genres categorize titles by human player experience, benchmark taxonomies must prioritize the functional demands imposed on an AI agent. For benchmarking, the critical ques-

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Table 6: Taxonomy of representative game benchmarks for LLM and VLM agents

Paper	Date	Level	Form	Build	Scope	Input	Action	Game Set
Jericho (2020)	2019/09	L2 Strategy	Arc	Wrapped	Curated	Text	Semantic	56 IF games
NLE (2020)	2020/06	L2 Strategy	World	Wrapped	Single	Text	Semantic/native	NetHack
Crafter (2022)	2021/09	L4 Visual	World	Authored	Single	Visual	Native	22 achievements
CICERO (2022)	2022/11	L3 Social	Dialogue	Embedded	Single	Text	Semantic	Diplomacy
climbench (2023)	2023/05	L3 Social	Dialogue	Authored	Curated	Text	Semantic	7 dialogue datasets
SmartPlay (2024b)	2023/10	L1 Rule	Mixed	Wrapped	Curated	Text	Semantic	6 games
AvalonBench (2023)	2023/10	L3 Social	Dialogue	Adapted	Single	Text	Semantic	Avalon
LLM StarCraft II (2024b)	2023/12	L2 Strategy	Match	Wrapped	Single	Text	Semantic	StarCraft II
CivRealm (2024)	2024/01	L2 Strategy	World	Wrapped	Open-ended	Text	Semantic	Civilization
GTBench (2024)	2024/02	L1 Rule	Match	Adapted	Curated	Text	Semantic	10 game-theory tasks
GAMABench (2025)	2024/05	L2 Strategy	Match	Adapted	Curated	Text	Semantic	8 scenarios
GameBench (2024)	2024/06	L2 Strategy	Match	Wrapped	Curated	Text	Semantic	9 games
Werewolf Arena (2024)	2024/07	L3 Social	Dialogue	Adapted	Single	Text	Semantic	Werewolf
BALROG (2025)	2024/11	L4 Visual	Mixed	Wrapped	Curated	Mixed	Semantic	6 env. families
TeamCraft (2024)	2024/12	L3 Social	World	Adapted	Single	Mixed	Semantic	Minecraft
GameArena (2025b)	2024/12	L3 Social	Dialogue	Adapted	Curated	Text	Semantic	3 live games
GAMEBot (2025)	2024/12	L2 Strategy	Match	Wrapped	Curated	Text	Semantic	8 games
PokerBench (2025)	2025/01	L2 Strategy	Puzzle	Adapted	Single	Text	Semantic	11K poker spots
Collab-Overcooked (2025a)	2025/02	L3 Social	Encounter	Adapted	Single	Text	Semantic	30 tasks
DSGBench (2025)	2025/03	L2 Strategy	Mixed	Wrapped	Curated	Text	Semantic	6 games
LVLm Game Players (2025d)	2025/03	L4 Visual	Mixed	Adapted	Curated	Mixed	Semantic	6 games
LLM-Coordination (2025)	2025/04	L3 Social	Match	Adapted	Curated	Text	Semantic	4 games
TextArena (2025)	2025/04	L5 General	Mixed	Authored	Expandable	Text	Semantic	57+ games
KORGym (2025)	2025/05	L2 Strategy	Mixed	Authored	Curated	Mixed	Semantic	51 games
LMGAME-BENCH (2025a)	2025/05	L5 General	Mixed	Wrapped	Curated	Mixed	Semantic	6 games
VideoGameBench (2025a)	2025/05	L4 Visual	Mixed	Embedded	Curated	Visual	Native	23 games
MCU (2025)	2025/05	L5 General	World	Generated	Open-ended	Visual	Native	3452 tasks
Orak (2025)	2025/06	L5 General	Mixed	Wrapped	Curated	Mixed	Semantic	12 games
TextAtari (2025c)	2025/06	L5 General	Encounter	Adapted	Curated	Text	Semantic	23 Atari games
TextQuests (2025)	2025/07	L5 General	Arc	Wrapped	Curated	Text	Semantic	25 IF games
StarDojo (2025a)	2025/07	L5 General	World	Adapted	Open-ended	Mixed	Semantic	1000 tasks
GVGAI-LLM (2025d)	2025/08	L5 General	Mixed	Generated	Expandable	Text	Semantic	118 games
FlashAdventure (2025)	2025/09	L4 Visual	Arc	Embedded	Family	Visual	Native	34 games
StarBench (2025b)	2025/10	L4 Visual	Encounter	Embedded	Single	Visual	Mixed	RPG battles
LLM-Hanabi (2025)	2025/10	L3 Social	Match	Adapted	Single	Text	Semantic	Hanabi
PuzzlePlex (2025)	2025/10	L2 Strategy	Puzzle	Authored	Curated	Mixed	Semantic	15 puzzle types
WOLF (2025)	2025/12	L3 Social	Dialogue	Adapted	Single	Text	Semantic	Werewolf
BotzoneBench (2026a)	2026/01	L1 Rule	Match	Wrapped	Curated	Text	Semantic	8 games
Strategic Hanabi (2026)	2026/01	L3 Social	Match	Adapted	Single	Text	Semantic	Hanabi
EMemBench (2026b)	2026/01	L4 Visual	Puzzle	Generated	Curated	Mixed	Mixed	16 games
MineNPC-Task (2026)	2026/01	L3 Social	World	Adapted	Single	Text	Semantic	Minecraft
AI GAMESTORE (2026)	2026/02	L5 General	Mixed	Generated	Expandable	Mixed	Native	100 games
Beyond Scaling (2026c)	2026/03	L2 Strategy	Match	Authored	Single	Text	Semantic	Zero-sum game
GameVerse (2026a)	2026/03	L5 General	Mixed	Wrapped	Curated	Mixed	Mixed	15 games
ARC-AGI-3 (2026)	2026/03	L5 General	Puzzle	Authored	Curated	Visual	Mixed	135 envs.
GameplayQA (2026a)	2026/03	L4 Visual	Puzzle	Adapted	Curated	Visual	Semantic	9 games
GameWorld (2026)	2026/04	L4 Visual	Mixed	Wrapped	Curated	Visual	Mixed	34 games
PokeGym (2026b)	2026/04	L4 Visual	World	Embedded	Single	Visual	Mixed	30 tasks

tion is not simply what kind of game a title is, but what cognitive and operational burdens it preserves or abstracts away. Two benchmarks drawn from distant genres may serve nearly identical evaluative roles, whereas two superficially similar games may demand fundamentally different capacities for perception, memory, inference, or execution. Therefore, we conceptualize the taxonomy of game benchmarks as a multidimensional design space rather than a traditional genre catalog. To systematically deconstruct these environments, our taxonomy is anchored by an evolutionary spine tracing the historical development of the field, supported by secondary coding axes—structure, scope, and modality—that describe how specific environments are instantiated.

The evolutionary spine of this literature traces how game environments have historically advanced in tandem with foundation model capabilities, categorizing benchmark ambition across five distinct levels. Level 1 corresponds to early benchmarks built around formal, rule-grounded interaction, evaluating whether an agent can simply maintain state and make legal moves within highly

controlled interfaces (e.g., SmartPlay, GTBench). Level 2 reflects a shift toward broader strategic decision-making in evolving environments, pushing the ambition from basic rule compliance to dynamic planning and interactive reasoning (e.g., PokerBench, DSGBench, BeyondScaling). Level 3 marks the integration of multi-agent social reasoning, where the evaluation target becomes the ability to cooperate, negotiate, or deceive within language-mediated arenas (e.g., Werewolf Arena, WOLF). Level 4 represents a leap toward true ecological validity, shifting the environmental form to preserve the visual and interface burdens of human play, testing whether models can operate as grounded visual agents (e.g., BALROG, StarBench). Finally, Level 5 extends the frontier from competence in a single environment to adaptability across a multiverse of games, evaluating cross-game generalization and first-contact adaptation (e.g., GameVerse, AI GAMESTORE, ARC-AGI-3). This progression highlights a continuous unified trajectory: as model architectures evolve, the game environments used to benchmark them must abstract away fewer of the native complexities of human play.

While the five-level evolutionary spine provides the historical backbone, it does not by itself specify how a benchmark instantiates the game as an evaluation environment. Benchmarks occupying the same evolutionary level often impose different operational pressures by varying along secondary coding axes.

The first secondary axis is Structure. It records how a benchmark turns gameplay into an evaluable object through two fields: Form and Construction. Form identifies the playable unit on which progress, completion, and failure are defined. Match benchmarks organize evaluation around bounded contests with clear outcomes, as in formal and strategic game suites (GTBench; GameBench). Dialogue benchmarks treat language interaction itself as the game loop, where debate, negotiation, or reference games drive the episode forward (WerewolfArena; Clembench). Encounter benchmarks focus on bounded tactical segments, while Arc benchmarks evaluate longer story or quest progressions (StarBench; FlashAdventure). World benchmarks use persistent environments where exploration, resources, tasks, or achievement graphs define progress (MCU; CivRealm). Mixed is reserved for suites that deliberately combine several of these playable units rather than fitting one dominant form (SmartPlay; BALROG; GameWorld; Orak).

Construction records the benchmark’s relationship to the original game substrate. Embedded benchmarks preserve the native client or live play surface as much as possible, increasing ecological fidelity but also increasing execution noise and evaluation cost (StarBench; FlashAdventure). Wrapped benchmarks reuse existing games or environments through a shared harness, API, or sandbox, improving comparability and instrumentation while abstracting part of the native interface (SmartPlay; BALROG; GameWorld; Orak). Adapted benchmarks convert existing games, rules, or scenes into targeted evaluation tasks, such as game-theoretic settings, social-deduction arenas, or grounded rule questions (GTBench; WerewolfArena; RuleOracles). Authored benchmarks design game-like environments specifically for measurement, while Generated benchmarks make task, rule, level, or game-instance generation part of the benchmark’s scalability strategy (BeyondScaling; PuzzlePlex; AIGameStore; MCU; GVGAILLM).

A separate axis is Benchmark Scope, which describes how a benchmark packages diversity to evaluate distinct operational scales. ‘Single game’ benchmarks concentrate on one environment to facilitate deep task analysis and expert capability evaluation, as seen in Werewolf Arena, StarBench, and Beyond Scaling. (Bailis et al., 2024; Li et al., 2026c; Zhang et al., 2025b). ‘Game family’ benchmarks expand within a relatively coherent family of related environments, avoiding overfitting and maintaining recognizable interfaces and consistent evaluation logic (eg:FlashAdventure (?)). ‘Curated suites’ intentionally assemble a fixed set of diverse games to evaluate a model’s general competence and cover multiple capability slices under a unified protocol (e.g., SmartPlay, GameBench,

DSGBench, BALROG, Orak) ‘Expandable suite’ treat continuous growth as a core design principle, shifting the generalization focus from mastering historical game lists to handling new instances and rules generated by the platform over time (e.g. (?)). Finally, ‘open-ended tasks’ derive their breadth from large, combinatorial, or effectively unbounded task spaces within a persistent environment (e.g., MCU, TeamCraft). It is critical to clarify that “open-ended tasks” is a scope property, not merely an environment label; it refers to the benchmark’s use of long-tail task diversity rather than the mere presence of a sandbox world.

The final axis is ‘Modality’, which records how the benchmark exposes observation and action channels. On the observation side, benchmarks may be text-symbolic, converting game states into natural language or structured data to maximize diagnostic clarity; visual, utilizing raw screenshots or pixel streams; or mixed, providing multi-modal inputs or offering both raw and scaffolded tracks for comparison. On the action side, benchmarks typically rely on semantic controls, where the agent outputs high-level semantic commands or API calls, or native control, requiring human-like execution such as GUI interactions or direct keyboard-mouse inputs. At the taxonomic level, modality serves to objectively document which segments of the human perception-action loop are preserved, scaffolded, or entirely abstracted away.

Taken together, these dimensions elevate the taxonomy table from a superficial genre catalog into a comprehensive design-space map. The five-level evolutionary spine traces the field’s historical progression toward increasingly unconstrained generalization, while the structural, scope, and modal axes deconstruct exactly how these environments are engineered. With this architectural framework established, the next section moves a step forward to address the core objective of these environments: what specific capabilities game benchmarks actually measure, and why they serve as an unparalleled substrate for rigorous capability probes.

6.2. Purpose: Core Capabilities Evaluated by Games

6.3. Paradigm: From Interaction to Assessment

A game environment does not automatically constitute a benchmark merely by being interactive or difficult. It becomes an evaluation instrument only when gameplay is mediated by an interaction contract and an evaluation contract. The former determines what part of the human play loop is exposed to the model, while the latter determines what kind of evidence can be extracted from the resulting behavior. In this sense, the methodological challenge of game benchmarking is not simply to select harder games, but to transform gameplay into a measurement pipeline that is interpretable, comparable, and sustainable as model capabilities evolve.

6.3.1. Interaction

Interface design operates as the front end of this evaluation pipeline. It specifies what the model is allowed to see, how it is allowed to act, and which cognitive or operational burdens of human play are preserved, simplified, or removed. Consequently, two benchmarks may use similar game substrates but probe fundamentally different capabilities if one presents a symbolic state and semantic actions while the other requires visual grounding and native control. In this context, interaction design should be treated as a core part of benchmark validity rather than as an implementation detail.

Observation channels trace one of the clearest paradigm shifts in game benchmarks: the transition from abstract textual or symbolic state representations to raw visual streams. Textified interfaces

align naturally with language-centric LLMs and excel in reasoning-oriented evaluations. By explicitly exposing rules, state variables, histories, and action constraints, they support highly controlled diagnostic probing (SmartPlay; GTBench; DSGBench). However, scaling such interfaces to complex visual or commercial games necessitates bespoke APIs, wrappers, or state-extraction procedures. This fundamentally bypasses the challenge of inferring game states directly from the play surface.

Therefore, the shift toward raw visual input is not simply a change in modality; it is a commitment to more human-like, end-to-end agent evaluation. Models are forced to interpret screen pixels, track dynamic visual changes, and make decisions under severe perceptual uncertainty. Unsurprisingly, current systems struggle in these settings, as raw visual observations reintroduce the state noise, grounding ambiguity, and temporal instability that text abstractions had neatly excised (VideoGameBench; PokeGym; GameplayQA). To bridge this gap, hybrid designs—such as OCR-assisted, detector-assisted, or dual text-and-image tracks—occupy a critical middle ground. They recognize that while pure vision best approximates human play, providing intermediate perceptual scaffolding yields a more diagnostic benchmark, isolating higher-order reasoning failures from low-level perceptual bottlenecks (StarBench; Orak).

Action channels determine how a model’s decisions manifest within the game environment. Semantic actions regularize the agent’s output space by restricting choices to high-level moves, skills, targets, or predefined commands that are seamlessly parsed and executed by the benchmark harness. By eliminating the friction of motor execution and UI manipulation, this design directs evaluation toward pure reasoning, planning, and decision quality (SmartPlay; GTBench; DSGBench; Orak).

Native control, by contrast, requires the model to interact through interfaces originally designed for human players, such as mouse, keyboard, controller, or GUI actions. This elevates the agentic requirements of the task: the model must not only formulate a strategy but also ground that strategy in executable, low-level interactions with the game’s surface. Furthermore, native-control interfaces provide a scalable pathway for evaluating commercial and browser games. By leveraging standardized, human-facing control channels, researchers can bypass the prohibitive engineering cost of developing custom symbolic APIs for every new game environment (FlashAdventure; VideoGameBench; GameWorld; StarBench).

The intersection of these observation and action choices establishes an interface privilege hierarchy, underscoring why benchmark results must be interpreted with caution. Highly privileged textified and API-mediated environments strip away the burdens of perception, UI grounding, and precise action execution. While this yields stable, automatically verifiable instruments ideal for isolating high-level strategic reasoning, it fundamentally weakens claims about whether a model can understand and act within an unconstrained environment as presented to humans (SmartPlay; GTBench; DSGBench).

Low-privilege, human-like interfaces preserve the visual grounding, interface comprehension, and temporal coordination essential for end-to-end agent evaluation. However, this ecological validity comes at the cost of entangled failure modes. In a native visual environment, a low score could stem from flawed planning, but it is equally likely to result from perceptual hallucinations, latency mismatches, invalid control formatting, or the inability to recover from a minor UI misclick (VideoGameBench; PokeGym; StarBench; GameWorld). Thus, interface privilege is not an inherent flaw to be eradicated, but a conscious design variable that delineates the specific capabilities a benchmark can validly claim to measure. A rigorous game benchmark must explicitly document its privilege level and, ideally, provide comparative tracks between assisted and human-like settings to successfully disentangle reasoning competence from grounding and execution failures.

6.3.2. Evaluation

The evaluation contract defines what can be validly inferred from gameplay. If the interaction contract determines what kind of game loop the model faces, the evaluation contract determines how agent behavior is synthesized into evidence. Games naturally contain goals, rewards, progress, and failure states, but these native signals do not automatically constitute a rigorous benchmark. A game becomes a benchmark only when its outcomes are translated into metrics that are interpretable, comparable, and robust against the noise of stochastic interactions.

Rule-Based Metrics: The most direct paradigm is result-based evaluation. Games provide native outcome signals such as win rate, score, reward, progress, survival, or task completion, making this paradigm highly natural to assess end-to-end agentic performance. It preserves the original objective structure of the game and is usually easy to automate, which makes it attractive for large-scale benchmark construction. Many game benchmarks rely on this logic because final outcomes remain the clearest evidence of whether an agent can actually achieve the goal of play (SmartPlay; GTBench; Balrog; GameWorld).

However, result-based metrics are often “black-box” indicators that lack diagnostic details. A low score in a game typically conflates multiple failure modes ranging from poor planning, weak rule understanding, visual misrecognition, invalid action, latency or memory collapse. This problem becomes especially severe in visually grounded or long-horizon games, where current models often achieve extremely low native progress and the metric loses diagnostic resolution (VideoGameBench; StarBench). Furthermore, native scores suffer from weak semantic portability: a score in one game rarely rarely maps to the same score in another, complicating the aggregation of results in multi-game suites

This motivates a second paradigm: process-level and diagnostic evaluation. These metrics decompose gameplay into intermediate milestones, trajectory analysis, sub-skill scores, and reasoning checks. By tracking “how” an agent behaves along the path to a goal, these instruments remain informative even when final success is sparse. Benchmarks that add intermediate reasoning or capability-specific metrics show that models with similar outcomes can differ substantially in the quality of their decisions, state tracking, or recovery behavior (GAMEBoT; DSGBench; GameWorld; MCU; GameVerse).

The strength of process-level evaluation is interpretability, but this paradigm introduces design dependency. Once a benchmark decomposes gameplay into subskills or milestones, the decomposition itself reflects the benchmark creator’s prior assumptions about positive behavior. Therefore, while these metrics improve failure analysis, they trade off some of the objectivity. They are most useful when treated as an explanatory layer around outcome metrics, rather than as a universal replacement for them.

Adversarial and Relative Evaluation: A third paradigm evaluates performance dynamically through adversarial interaction or multi-agent social networks. Unlike static test sets, adversarial environments resist rapid saturation because the benchmark’s difficulty scales in tandem with the capabilities of the opposing agents. This dynamic pressure is uniquely suited for probing opponent modeling, deceptive negotiation and cooperative strategy capabilities that strictly emerge through live interaction (GTBench; WerewolfArena; GameArena). Consequently, adversarial frameworks maintain benchmark freshness and capture dimensions of social intelligence that isolated, single-agent tasks inherently miss. However, arena-style evaluations fundamentally alter the semantics of a score. Results derived from tournaments are inherently relative: they quantify how a model performs against a specific opponent pool under a fixed protocol, but they do not isolate the absolute

source of that capability. Rankings can fluctuate wildly if the opponent pool, prompting strategy, or match structure is modified.

Calibration is therefore necessary for giving game scores external meaning. A calibrated benchmark does not simply report a 60% win rate or an arbitrary numerical score; it articulates exactly what that proficiency represents against a known, stable baseline. By introducing fixed AI anchors, explicitly tiered opponent ladders, or calibrated solver references, benchmark designers can transform fluctuating relative performances into interpretable claims. Human baselines, expert anchors, solver references, random baselines, novice/expert comparisons, or human-normalized efficiency are critical for grounding model performance.(SmartPlay; BotzoneBench; ARCAGI3; GameWorld). This ensures that the same numerical score is not overinterpreted across different games or varying interface privileges within a suite.

Robustness is the complementary requirement. A benchmark score must not only be interpretable but also remain stable and meaningful under reasonable variations in the testing environment. Beyond statistical stability, robustness demands that the benchmark actively defends against the broader systemic issues of dataset contamination and benchmark-specific optimization. To address this, recent rigorous benchmarks have begun implementing explicit defensive mechanisms. These include utilizing private explicit by using private or dynamically held-out evaluation content instituting strict contamination checks, enforcing scaffold-ablated comparisons, and maintaining a clear separation between public development environments and official, closed-door leaderboard evaluations (ARCAGI3; LMGAMEBench; Clembench2024; GameWorld). These practices show that evaluation design must defend not only against statistical noise, but also against the broader problem that rapidly improving agents may optimize for the benchmark protocol rather than the underlying capability.

Ultimately, evaluation within game environments must be approached as a pluralistic measurement system rather than a monolithic scoreboard. Result-based metrics honor the native objective of the game; process-level diagnostics explain the mechanics of behavior; adversarial protocols introduce dynamic stress tests; calibration applies external semantics; and robustness preserves those semantics against optimization drift. The most rigorous game benchmarks do not merely select difficult games; they transparently articulate what their scores represent, what blind spots remain, and how their protocols will sustain validity as the frontier of agentic intelligence advances.

7. Challenge and Future

The preceding sections have traced the full pipeline of large foundation models in the multiverse of games, from datasets through models and harness to benchmarks. At every stage, a consistent pattern has emerged: progress on one axis exposes a bottleneck on another. This section distills these recurring bottlenecks into five fundamental trade-offs that define the current frontier, and then charts a five-level roadmap toward the generalist game player envisioned by this survey.

7.1. Five Fundamental Trade-Offs

The challenges facing game-playing foundation models are not isolated problems awaiting point solutions. They are structural tensions in which improving one side inevitably stresses the other. We identify five such trade-offs, each grounded in evidence from the preceding sections.



Fig. 6: Overview of five-level roadmap towards generalist game player. The progressive roadmap illustrating the evolution from narrow, single-game mastery (Level 1) to fully generalist agents capable of cross-task transfer within genres (Level 2), cross-genre generalization (Level 3), and lifelong adaptation to unseen environments in human-craft game universe (Level 4), culminating in the “Demiurge” stage (Level 5), where agents transcend gameplay to generate, simulate, and evolve entire game worlds. This hierarchy reflects an increasing degree of agency, from solving tasks to constructing and expanding the game multiverse.

7.1.1. Scale vs. Fidelity vs. Diversity: The Data Trilemma

Section 3 established that no existing dataset simultaneously leads in scale, annotation quality, and game diversity. NitroGen (Magne et al., 2026b) assembles 40,000 hours across 1,000+ games through automated controller-overlay extraction, but its button accuracy of 0.96 and joystick R^2 of 0.84 mean that roughly one in twenty-five discrete actions and one in six continuous inputs are wrong. These errors compound over long trajectories, precisely the regime where fidelity matters most. At the other extreme, OpenP2P (Yue et al., 2026) invests 8,300 hours of meticulous human annotation across 45 games and demonstrates that behavior cloning follows a predictable scaling law: increasing both model capacity and data volume leads to the emergence of causal reasoning. Yet the linear cost of human annotation makes NitroGen-scale labeling economically prohibitive. Meanwhile, VPT’s (Baker et al., 2022) 70,000 hours of IDM-pseudo-labeled data cover only Minecraft, and PLAICraft’s (He et al., 2025b) five-modality temporally aligned corpus, though high in quality, is equally confined to a single game.

The tension is not merely practical. OpenP2P (Yue et al., 2026) shows that annotation quality enables qualitative capability jumps; NitroGen (Magne et al., 2026b) shows that data scale enables cross-game transfer. Neither substitutes for the other. Bridging this gap will likely require either

learned annotation models that approach human fidelity at machine cost, or world-model-based data engines that generate trajectories with built-in ground-truth labels.

7.1.2. Breadth vs. Depth: The Heterogeneity Wall

A human player adapts to a new game within minutes. Current models cannot. The core obstacle is game heterogeneity: the same keyboard key carries entirely different semantics across games. "W" means move forward in an FPS, build a worker in an RTS, and nothing at all in a card game. This action-space fragmentation means that scaling to more games does not automatically yield a better agent in any single game.

The evidence is consistent. Per-game specialists reach strong performance ceilings: Metamon (Grigsby et al., 2025) achieves human-level Pokemon through offline RL on a decade of ranked replays; CombatVLA (Chen et al., 2025c) attains human-level ARPG combat with sub-15ms action alignment. Cross-game systems cover far more titles but at substantially lower competence: NitroGen (Magne et al., 2026b) spans 1,000+ games yet produces only reactive motor mimicry without goal-directed behavior; Game-TARS (Wang et al., 2025g) trains on 500+ games and approaches fresh-human performance on unseen web games, but requires game pausing during reasoning in fast-paced settings. Lumine (Tan et al., 2025c) represents the most promising middle ground, achieving zero-shot transfer of multi-hour storyline completion across structurally similar anime RPGs (Genshin Impact, Honkai: Star Rail, Wuthering Waves), but this transfer is confined to games sharing similar UI conventions and interaction patterns, and has not been demonstrated on mechanically distinct titles, such as Black Myth: Wukong.

Attempts to unify action representations, whether through NitroGen's (Magne et al., 2026b) 20-dimensional gamepad vector or OmniJARVIS's (Wang et al., 2024) FSQ tokenization, inevitably lose game-specific precision. The field has not yet found an action abstraction that preserves enough information for expert play while remaining general enough for cross-game transfer.

7.1.3. Reasoning vs. Reactivity: The Latency-Intelligence Dilemma

Strategic games and action games impose opposite demands on the same model. CICERO's (Meta Fundamental AI Research Diplomacy Team (FAIR) et al., 2022) multi-turn Diplomacy campaigns and LSPO's (Xu et al., 2025b) equilibrium-finding in Werewolf require deep, multi-step reasoning that consumes hundreds of tokens. FPS and MOBA games require responses within 100 milliseconds.

Current systems cannot satisfy both demands. On VideoGameBench (Zhang et al., 2025a), the strongest frontier VLM, Gemini 2.5 Pro (Comanici et al., 2025), completes only 0.48% of game tasks when operating in real time; the benchmark had to introduce a game-pausing mode to reach even 1.6%. Existing solutions trade one end for the other: CombatVLA (Chen et al., 2025c) truncates its Action-of-Thought chain to achieve a 50 $\times$ speedup but sacrifices interpretability and strategic depth; Lumine (Tan et al., 2025c) reasons at 5Hz and acts at 30Hz by invoking deliberation only when needed; NitroGen's (Magne et al., 2026b) flow-matching DiT generates 16-step action chunks in parallel, eliminating the autoregressive bottleneck but also eliminating reasoning entirely. Game-TARS (Wang et al., 2025g) introduces a "Greedy Thinking" strategy, but finds that excessive reasoning in fast-paced games triggers hallucinated reasoning loops that degrade performance below the no-reasoning baseline.

The fundamental issue is architectural. Autoregressive language models couple reasoning depth to inference latency: more thinking means more tokens means more time. No current architecture

provides a principled mechanism for an agent to deliberate deeply on strategic decisions and react reflexively to split-second events within a single forward pass.

7.1.4. Modular Workflow vs. Model-as-Whole: The Harness Paradox

Section 5 documented the harness as the nervous system connecting foundation models to game environments. The paradox is that current models need this external scaffolding to function, yet the scaffolding itself introduces fragility and limits generality.

End-to-end models struggle without harness support. On VideoGameBench (Zhang et al., 2025a) and GameVerse (Zhang et al., 2026a), frontier VLMs applied directly to game screenshots fail to progress beyond the opening minutes of most games. By contrast, modular harnesses achieve qualitatively stronger results: Cradle (Tan et al., 2025d) completes 40-minute missions in Red Dead Redemption II by composing perception, memory, planning, and action modules around a VLM core. Voyager (Wang et al., 2023a) builds an open-ended skill library in Minecraft through retrieval-augmented generation and self-verification. The gap demonstrates that capabilities such as persistent memory, self-correction, and low-latency reflexes remain beyond the native capacity of current foundation models. Yet modularity has costs. Each module introduces extra engineering, and the interfaces create information bottlenecks. Section 5.2 documented the “knowing-doing gap”: even when an agent correctly identifies the optimal strategy, it frequently fails to translate that strategy into the precise action parameters required. This gap between semantic understanding and motor execution is a direct consequence of the modular boundary between the reasoning and action components.

The ideal resolution is a model that natively possesses memory, reflection, and fine-grained control, making the external harness unnecessary. Current VLAs represent a step toward this goal by unifying perception and action, but they still lack persistent memory and self-correction. Closing this gap is a prerequisite for moving from game-specific agent pipelines to a universal game player.

7.1.5. Code Engine vs. World Model: The Simulation Gap

Most training and evaluation depend on code-based game engines. These engines provide solid physics, deterministic signals, and arbitrarily long rollouts, but they impose three ceilings: the action space is limited to predefined interfaces, the game diversity is limited to what humans have built, and each new game requires dedicated integration effort. World models offer a path beyond these ceilings. Genie (Bruce et al., 2024a) discovers latent action spaces from unlabeled video, removing the interface constraint. GameFactory (Yu et al., 2025) and GameGen-X (Che et al., 2024) generate interactive environments beyond the boundaries of existing games, pushing toward the theoretical multiverse of Era 4. PAN (PAN Team Institute of Foundation Models, 2025) extends action conditioning to natural language, and Solaris (Savva et al., 2026) introduces multi-agent shared worlds with cross-player consistency. Most significantly, SIMA 2 (SIMA-team et al., 2025) has been successfully positioned inside Genie 3 generated worlds (Ball et al., 2025) and showed positive transfer to held-out tasks, providing the first evidence that world models can serve as viable training environments.

However, current world models remain far from replacing code engines. Oasis (Decart and Julian Quevedo, 2024) maintains consistency for only a few seconds before errors compound into visible drift. Genie 3 (Ball et al., 2025) generates playable environments but is limited to 60-second rollouts with imperfect physics adherence. GameNGen (Valevski et al., 2025) sustains human-indistinguishable quality for several minutes in DOOM, but only for a single, visually simple game. More fundamentally, world models lack the deterministic reward signals that code engines provide. When the envi-

ronment itself is generated by a neural network, the definition of success becomes ambiguous: how does one verify that an agent has “won” a game whose rules are themselves approximate?

Bridging this gap requires advances on three fronts: temporal consistency over rollouts of thousands of steps, verifiable reward signals within model-generated environments, and multi-game world models that can instantiate diverse game universes rather than imitating a single title.

7.2. Five-Level Roadmap to the Future

The trade-offs above define why the challenges is hard. This section defines what progress looks like. We organize the path toward the generalist game player into five levels of increasing generalization, from mastering tasks within a single game to becoming the game environment itself. At each level, we identify the current frontier, its limits, and the trade-offs that must be resolved to advance further.

7.2.1. Level 1: Single-Game Task Mastery

“Complete all tasks within a single game, from atomic actions to long-horizon objectives.”

In developmental psychology ([Anderson, 1982](#); [Fitts and Posner, 1967](#)), skill acquisition within a single rule system is the foundation of all higher-order transfer. A child must first master the rules, controls, and feedback loops of one game before any cross-game generalization becomes meaningful. For AI agents, this corresponds to achieving robust competence across the full task distribution of a single game environment, including both short-horizon atomic skills and long-horizon composite objectives that chain these skills over hundreds of sequential decisions.

The most extensively studied game environment is Minecraft, where over seven years of sustained research have produced an unparalleled data-model ecosystem (MineRL ([Guss et al., 2019](#)), MineDojo ([Fan et al., 2022](#)), VPT ([Baker et al., 2022](#)), GROOT ([Cai et al., 2024b](#)), OpenHA ([Wang et al., 2025e](#)), JARVIS-VLA ([Li et al., 2025a](#)), PLAICraft). Yet even in this richest ecosystem, single-game mastery remains incomplete. OpenHA’s inference speed of 0.98 FPS is far below real-time play. Out-of-distribution tasks cause performance drops of up to 30 percentage points. Combat scenarios exhibit high variance ($\pm 43.5\%$), with the agent unable to track fast-moving targets. Most critically, no existing system has attempted a full long-horizon objective such as progressing from an empty world to defeating the Ender Dragon, a task requiring thousands of sequential decisions spanning resource gathering, crafting, exploration, and combat. Even in the most mature game ecosystem, the gap between completing atomic tasks and mastering the full game experience remains wide.

Advancing to single-game mastery requires resolving Trade-off 3, reasoning vs. reactivity for long-horizon planning and Trade-off 4, harness-supported memory and self-correction vs. end-to-end efficiency.

7.2.2. Level 2: Cross-Task Transfer Within Genre

“Transfer learned competence across games that share visual style, interface conventions, or interaction patterns.”

Analogical reasoning research ([Gentner, 1983](#); [Gick and Holyoak, 1983](#)) shows that humans transfer skills most readily between domains that share surface features and relational structure. A player fluent in one open-world RPG navigates a new RPG almost immediately, because they share the similar interaction grammar. Level 2 tests whether AI agents can exploit this same structural overlap: given mastery of one game, can they generalize to a second game within the same genre without retraining?

Lumine ([Tan et al., 2025c](#)) demonstrates the current frontier at this level. Trained on the Mondstadt region of Genshin Impact, it achieves zero-shot completion of the five-hour main storyline in Honkai: Star Rail and a 100-minute mission in Wuthering Waves. Its 5Hz perception and 30Hz action pipeline operates in real time without game-specific adaptation, and its selective reasoning mechanism activates deliberation only when needed.

The limits are equally clear. Simple instruction-following tasks succeed at over 80%, but performance drops sharply on puzzles, flying enemies, and non-flat terrain. Approximately half of all errors stem from failures in multimodal understanding, particularly the detection of small or environmentally blended objects. More fundamentally, these three games share a common genre of anime-style open-world RPGs with similar minimap layouts, dialogue systems, and quest markers. Transfer to mechanically distinct titles such as action RPGs with different combat systems, inventory designs, and camera conventions has failed. And even within the supported genre, multi-hour storyline completion covers only a fraction of the hundreds of hours of content that each game offers.

Breaking through Level 2 requires addressing Trade-off 2 (breadth vs. depth): the agent must generalize beyond UI-similar games without losing the depth needed for complex in-game tasks.

7.2.3. Level 3: Cross-Genre Generalization

“Operate across games with fundamentally different action spaces, visual styles, time scales, and mechanics.”

This level corresponds to what game studies call “ludic literacy” ([Juul, 2005](#); [Salen and Zimmerman, 2003](#)): the ability to parse an unfamiliar rule system by recognizing abstract patterns (resource management, spatial navigation, turn economy) beneath surface-level differences. A chess expert picking up a new strategy board game does not start from zero, because the underlying combinatorial reasoning transfers even when pieces, boards, and victory conditions change entirely. For AI, Level 3 demands that a single model handles games in which $\mathcal{A}$, $\mathcal{O}$, and $\mathcal{T}$ are all different, requiring an invariant representation of agency that abstracts over game-specific details.

Several systems in this level have reached varying degrees of competence. Game-TARS ([Wang et al., 2025g](#)), pre-trained on 500B+ tokens from 500+ games, doubles prior state-of-the-art performance on Minecraft, exceeds GPT-5 and Gemini-2.5-Pro ([Comanici et al., 2025](#)) on ViZDoom FPS maps, and approaches fresh-human performance on unseen web games. NitroGen ([Magne et al., 2026b](#)) trains a DiT-based architecture on 40,000 hours of gameplay across 1,000+ titles, demonstrat-

ing that heterogeneous multi-game corpora can support generative pre-training. OpenP2P (Yue et al., 2026) validates behavior-cloning scaling laws across 45 3D games and observes the emergence of causal reasoning with increasing model and data scale.

However, competence at this level is qualitatively shallower than at Levels 1 and 2. NitroGen (Magne et al., 2026b) produces plausible motor patterns, such as combat reactions and navigation, but has no capacity for goal-directed planning, language understanding, or strategic reasoning. GameTARS requires game pausing during reasoning steps, and its “Greedy Thinking” strategy triggers hallucinated reasoning loops in fast-paced settings. OpenP2P (Yue et al., 2026) follows only simple instructions and exhibits behavioral artifacts including wall collisions and off-target shooting. The consistent pattern is that as game coverage increases, the level of mastery in each game decreases. No current system achieves both broad coverage and deep competence.

Advancing beyond Level 3 requires resolving Trade-off 1, large-scale, high-fidelity data for diverse games and Trade-off 2, a unified action representation that preserves game-specific precision.

7.2.4. Level 4: Lifelong Adaptation

“Rapidly adapt to entirely new environments through self-directed exploration and continuous self-improvement.”

Levels 1 through 3 evaluate competence on a fixed distribution of games seen during training or closely related to them. Level 4 introduces a qualitatively different requirement: *online learning in an unknown environment*. This mirrors what cognitive scientists call “learning to learn” (Flavell, 1979; Harlow, 1949), the meta-cognitive ability to form hypotheses about new rules, test them through exploration, and revise them from sparse feedback. A human player dropped into a game with no manual does exactly this: within minutes, she infers the control mapping, the objective structure, and the penalty conditions through trial and error. Current foundation models, by contrast, are static after training and cannot update their policies from a handful of in-game experiences.

SIMA 2 (SIMA-team et al., 2025) provides the clearest prototype for this level. Its bootstrapped improvement cycle begins with human demonstrations augmented by Gemini-generated labels, transitions to self-directed play in unseen games without human data, and uses its own experience to train successive agent generations. On held-out environments never seen during training, SIMA 2 roughly doubles the success rate of its predecessor, completing 26 out of 50 MineDojo task categories (versus 2 for SIMA 1 (Team et al., 2024)) and autonomously progressing through 15 to 20 minutes of a previously unseen story-driven game, demonstrating skills that were never present in its training data.

The gap to true lifelong adaptation remains substantial. SIMA 2 (SIMA-team et al., 2025) still requires a full SFT and RL training phase before self-improvement becomes possible; it is not a zero-shot or few-shot learner. Its context window limits working memory, causing performance degradation on tasks requiring long-horizon reasoning. Precise motor control, particularly in combat requiring split-second timing, remains a weakness. And the self-improvement loop depends on Gemini-based feedback, inheriting the reasoning costs and failure modes of the evaluator model. The broader challenge is that current foundation models lack the mechanisms for rapid online adaptation: they cannot update their behavior from a handful of in-game experiences in the way a human player adjusts strategy after a single failed attempt.

Reaching Level 4 requires resolving Trade-off 1 to 4. An agent who reaches Level 4 is a really generalist player.

7.2.5. Level 5: The Demiurge, From Player to Creator

“Become the environment itself: generate, reshape, and evolve game worlds.”

The preceding levels concern agents that play within fixed game environments. Level 5 envisions a qualitative shift in the agent’s ontological role. In game design theory (Huizinga, 1938; Salen and Zimmerman, 2003), the distinction between player and designer is fundamental: a player optimizes within the rules; a designer constructs the rules themselves. This is also the distinction between intelligence that adapts to a given world and intelligence that creates worlds. Within the POMDP formalism, the agent no longer optimizes a policy within a predefined game $\mathcal{M}$, but actively generates the state space $\mathcal{S}$, action space $\mathcal{A}$, transition dynamics $\mathcal{T}$, and reward structures $\mathcal{R}$. This is the Era 4 Demiurge outlined in Figure 1.

Early steps toward this vision have begun to converge. Genie 3 (Ball et al., 2025) generates diverse, interactive 3D environments in real time across multiple locomotion modes, producing worlds that are navigable by both humans and autonomous agents. SIMA 2 (SIMA-team et al., 2025) has explored inside Genie 3-generated worlds and demonstrated positive transfe, providing the first closed-loop evidence that a world model can serve as both training environment and evaluation substrate for an embodied agent.

The distance to the full vision remains large. Genie 3 (Ball et al., 2025) is limited to 60-second rollouts, lacks persistent world state, and does not guarantee physical consistency. No current world model supports the generation of verifiable game rules, win conditions, or reward functions. The multi-agent setting, in which multiple agents co-inhabit and co-shape a generated world, has been explored only by Solaris (Savva et al., 2026) and remains at the proof-of-concept stage. Realizing the Demiurge requires bridging all five trade-offs simultaneously: large-scale and high-fidelity data to train the world model (Trade-off 1), broad and deep game understanding to generate diverse yet coherent worlds (Trade-off 2), real-time generation with long-horizon consistency (Trade-off 3), native model capabilities that replace external scaffolding (Trade-off 4), and world models that match code-engine reliability while surpassing their diversity (Trade-off 5).

The five levels above are not merely a ladder of increasing difficulty. They represent a progression in the degree of agency: from executing predefined tasks, through adapting to new contexts, to constructing the contexts themselves. Each level subsumes the capabilities of those below it. An agent that can generate and evolve game worlds (Level 5) must also be able to adapt to new environments (Level 4), generalize across genres (Level 3), transfer within a genre (Level 2), and master individual games (Level 1). In this sense, the roadmap defines both the incremental milestones and the ultimate destination of the future.

References

M. Agarwal, S. Rana, T. Sundoro, H. Berhe, S. Kim, V. Sharma, S. O’Brien, and K. Zhu. Wolf: Werewolf-based observations for llm deception and falsehoods, 2025. URL <https://arxiv.org/abs/2512.09187>.

- S. Agashe, Y. Fan, A. Reyna, and X. E. Wang. LLM-coordination: Evaluating and analyzing multi-agent coordination abilities in large language models. In *Findings of the Association for Computational Linguistics: NAACL 2025*, pages 8053–8072, 2025. doi: 10.18653/v1/2025.findings-naacl.448. URL <https://aclanthology.org/2025.findings-naacl.448/>.
- J. Ahn, J. Kim, H. Yun, J. Son, D. Park, J. Cho, and G. Kim. Flashadventure: A benchmark for gui agents solving full story arcs in diverse adventure games. In *EMNLP*, 2025.
- E. Alonso, A. Jelley, V. Micheli, A. Kanervisto, A. J. Storkey, T. Pearce, and F. Fleuret. Diffusion for world modeling: Visual details matter in atari. In *Advances in Neural Information Processing Systems*, 2024.
- J. R. Anderson. Acquisition of cognitive skill. *Psychological Review*, 89:369–406, 1982.
- J. Bai, S. Bai, Y. Chu, Z. Cui, K. Dang, X. Deng, Y. Fan, W. Ge, Y. Han, F. Huang, B. Hui, L. Ji, M. Li, J. Lin, R. Lin, D. Liu, G. Liu, C. Lu, K. Lu, J. Ma, R. Men, X. Ren, X. Ren, C. Tan, S. Tan, J. Tu, P. Wang, S. Wang, W. Wang, S. Wu, B. Xu, J. Xu, A. Yang, H. Yang, J. Yang, S. Yang, Y. Yao, B. Yu, H. Yuan, Z. Yuan, J. Zhang, X. Zhang, Y. Zhang, Z. Zhang, C. Zhou, J. Zhou, X. Zhou, and T. Zhu. Qwen technical report. *arXiv:2309.16609*, 2023.
- S. Bai, Y. Cai, R. Chen, and K. C. et al. Qwen3-vl technical report. *arXiv:2511.21631*, 2025a.
- S. Bai, K. Chen, X. Liu, J. Wang, W. Ge, S. Song, K. Dang, P. Wang, S. Wang, J. Tang, H. Zhong, Y. Zhu, M. Yang, Z. Li, J. Wan, P. Wang, W. Ding, Z. Fu, Y. Xu, J. Ye, X. Zhang, T. Xie, Z. Cheng, H. Zhang, Z. Yang, H. Xu, and J. Lin. Qwen2.5-vl technical report, 2025b. URL <https://arxiv.org/abs/2502.13923>.
- S. Bailis, J. Friedhoff, and F. Chen. Werewolf arena: A case study in llm evaluation via social deduction, 2024. URL <https://arxiv.org/abs/2407.13943>.
- B. Baker, I. Akkaya, P. Zhokov, J. Huizinga, J. Tang, A. Ecoffet, B. Houghton, R. Sampedro, and J. Clune. Video pretraining (VPT): Learning to act by watching unlabeled online videos. In A. H. Oh, A. Agarwal, D. Belgrave, and K. Cho, editors, *Advances in Neural Information Processing Systems*, 2022.
- P. J. Ball, J. Bauer, F. Belletti, B. Brownfield, A. Ephrat, S. Fruchter, A. Gupta, K. Holsheimer, A. Holynski, J. Hron, C. Kaplanis, M. Limont, M. McGill, Y. Oliveira, J. Parker-Holder, F. Perbet, G. Scully, J. Shar, S. Spencer, O. Tov, R. Villegas, E. Wang, J. Yung, C. Baetu, J. Berbel, D. Bridson, J. Bruce, G. Buttimore, S. Chakera, B. Chandra, P. Collins, A. Cullum, B. Damoc, V. Dasagi, M. Gazeau, C. Gbadamosi, W. Han, E. Hirst, A. Kachra, L. Kerley, K. Kjems, E. Knoepfel, V. Koriakin, J. Lo, C. Lu, Z. Mehring, A. Moufarek, H. Nandwani, V. Oliveira, F. Pardo, J. Park, A. Pierson, B. Poole, H. Ran, T. Salimans, M. Sanchez, I. Saprykin, A. Shen, S. Sidhwani, D. Smith, J. Stanton, H. Tomlinson, D. Vijaykumar, L. Wang, P. Wingfield, N. Wong, K. Xu, C. Yew, N. Young, V. Zubov, D. Eck, D. Erhan, K. Kavukcuoglu, D. Hassabis, Z. Gharamani, R. Hadsell, A. van den Oord, I. Mosseri, A. Bolton, S. Singh, and T. Rocktäschel. Genie 3: A new frontier for world models, 2025. URL <https://deepmind.google/blog/genie-3-a-new-frontier-for-world-models/>.
- M. G. Bellemare, Y. Naddaf, J. Veness, and M. Bowling. The arcade learning environment: An evaluation platform for general agents. *Journal of Artificial Intelligence Research*, 47:253–279, 2013a.

- M. G. Bellemare, Y. Naddaf, J. Veness, and M. Bowling. The arcade learning environment: An evaluation platform for general agents. *Journal of Artificial Intelligence Research*, 47:253–279, June 2013b. ISSN 1076-9757.
- C. Berner, G. Brockman, B. Chan, and V. C. et al. Dota 2 with large scale deep reinforcement learning. *arXiv*, abs/1912.06680, 2019.
- G. Bredis, S. Dereka, V. Sinii, R. Rakhimov, and D. Gavrilov. Enhancing vision-language model training with reinforcement learning in synthetic worlds for real-world success, 2025a. URL <https://arxiv.org/abs/2508.04280>.
- G. Bredis, S. Dereka, V. Sinii, R. Rakhimov, and D. Gavrilov. Enhancing vision-language model training with reinforcement learning in synthetic worlds for real-world success. *arXiv preprint arXiv:2508.04280*, 2025b. URL <https://arxiv.org/abs/2508.04280>.
- T. B. Brown, B. Mann, N. Ryder, M. Subbiah, J. Kaplan, P. Dhariwal, A. Neelakantan, P. Shyam, G. Sastry, A. Askell, S. Agarwal, A. Herbert-Voss, G. Krueger, T. Henighan, R. Child, A. Ramesh, D. M. Ziegler, J. Wu, C. Winter, C. Hesse, M. Chen, E. Sigler, M. Litwin, S. Gray, B. Chess, J. Clark, C. Berner, S. McCandlish, A. Radford, I. Sutskever, and D. Amodei. Language models are few-shot learners. In *Proceedings of the 34th International Conference on Neural Information Processing Systems*, NIPS ’20, Red Hook, NY, USA, 2020. Curran Associates Inc. ISBN 9781713829546.
- J. Bruce, M. Dennis, A. Edwards, J. Parker-Holder, Y. Shi, E. Hughes, M. Lai, A. Mavalankar, R. Steigerwald, C. Apps, Y. Aytar, S. Bechtle, F. Behbahani, S. Chan, N. Heess, L. Gonzalez, S. Osindero, S. Ozair, S. Reed, J. Zhang, K. Zolna, J. Clune, N. de Freitas, S. Singh, and T. Rocktäschel. Genie: Generative interactive environments. In *International Conference on Machine Learning (ICML 2024)*, 2024a.
- J. Bruce, M. D. Dennis, A. Edwards, J. Parker-Holder, Y. Shi, E. Hughes, M. Lai, A. Mavalankar, R. Steigerwald, C. Apps, Y. Aytar, S. M. E. Bechtle, F. Behbahani, S. C. Y. Chan, N. Heess, L. Gonzalez, S. Osindero, S. Ozair, S. Reed, J. Zhang, K. Zolna, J. Clune, N. de Freitas, S. Singh, and T. Rocktäschel. Genie: Generative interactive environments. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 4603–4623, 2024b. URL <https://proceedings.mlr.press/v235/bruce24a.html>.
- S. Cai, Z. Wang, K. Lian, Z. Mu, X. Ma, A. Liu, and Y. Liang. Rocket-1: Mastering open-world interaction with visual-temporal context prompting. *arXiv preprint arXiv: 2410.17856*, 2024a.
- S. Cai, B. Zhang, Z. Wang, X. Ma, A. Liu, and Y. Liang. GROOT: Learning to follow instructions by watching gameplay videos. In *The Twelfth International Conference on Learning Representations*, 2024b.
- M. Campbell, A. J. Hoane, and F.-h. Hsu. Deep blue. *Artif. Intell.*, 134(1–2):57–83, Jan. 2002. ISSN 0004-3702.
- K. Chalamalasetti, J. Götze, S. Hakimov, B. Madureira, P. Sadler, and D. Schlangen. clembench: Using game play to evaluate chat-optimized language models as conversational agents. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 11174–11219, 2023. doi: 10.18653/v1/2023.emnlp-main.689. URL <https://aclanthology.org/2023.emnlp-main.689/>.

- H. Che, X. He, Q. Liu, C. Jin, and H. Chen. Gamegen-x: Interactive open-world game video generation, 2024.
- H. Che, X. He, Q. Liu, C. Jin, and H. Chen. Gamegen-x: Interactive open-world game video generation. In *International Conference on Learning Representations*, 2025.
- J. Chen, Y. Zhao, Y. Huang, L. Cui, L. Dong, T. Lv, Q. Chen, and F. Wei. Model as a game: On numerical and spatial consistency for generative games. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV) Workshops*, pages 1937–1946, 2025a.
- L. Chen, H. Gao, T. Liu, Z. Huang, F. Sung, X. Zhou, Y. Wu, and B. Chang. G1: Bootstrapping perception and reasoning abilities of vision-language model via reinforcement learning, 2025b. URL <https://arxiv.org/abs/2505.13426>.
- P. Chen, P. Bu, J. Song, Y. Gao, and B. Zheng. Can vlms play action role-playing games? take black myth wukong as a study case. *arXiv:2409.12889*, 2024.
- P. Chen, P. Bu, Y. Wang, X. Wang, Z. Wang, J. Guo, Y. Zhao, Q. Zhu, J. Song, S. Yang, J. Wang, and B. Zheng. Combatvla: An efficient vision-language-action model for combat tasks in 3d action role-playing games. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 10919–10928, October 2025c.
- P. Chen, P. Bu, Y. Wang, and X. e. a. Wang. Combatvla: An efficient vision-language-action model for combat tasks in 3d action role-playing games. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 10919–10928, October 2025d.
- T. Chen, X. Hu, Z. Ding, and C. Jin. Learning world models for interactive video generation. *arXiv preprint arXiv:2505.21996*, 2025e. URL <https://arxiv.org/abs/2505.21996>.
- W. Chen, Y. Su, J. Zuo, C. Yang, C. Yuan, C.-M. Chan, H. Yu, Y. Lu, Y.-H. Hung, C. Qian, et al. Agent-verse: Facilitating multi-agent collaboration and exploring emergent behaviors. In *The Twelfth International Conference on Learning Representations*, 2023.
- J. Cheng, R. Qiao, Y. Ma, B. Li, G. Xiong, Q. Miao, Y. Li, and Y. Lv. Scaling offline model-based rl via jointly-optimized world-action model pretraining. In *International Conference on Learning Representations*, 2025.
- C. Chi, S. Feng, Y. Du, Z. Xu, E. Cousineau, B. Burchfiel, and S. Song. Diffusion policy: Visuomotor policy learning via action diffusion. *The International Journal of Robotics Research*, 44:1684 – 1704, 2023.
- E. Collaboration, A. O’Neill, A. Rehman, A. Gupta, A. Maddukuri, A. Gupta, A. Padalkar, A. Lee, A. Pooley, A. Gupta, A. Mandlekar, A. Jain, A. Tung, A. Bewley, A. Herzog, A. Irpan, A. Khazatsky, A. Rai, A. Gupta, A. Wang, A. Kolobov, A. Singh, A. Garg, A. Kembhavi, A. Xie, A. Brohan, A. Raffin, A. Sharma, A. Yavary, A. Jain, A. Balakrishna, A. Wahid, B. Burgess-Limerick, B. Kim, B. Schölkopf, B. Wulfe, B. Ichter, C. Lu, C. Xu, C. Le, C. Finn, C. Wang, C. Xu, C. Chi, C. Huang, C. Chan, C. Agia, C. Pan, C. Fu, C. Devin, D. Xu, D. Morton, D. Driess, D. Chen, D. Pathak, D. Shah, D. Büchler, D. Jayaraman, D. Kalashnikov, D. Sadigh, E. Johns, E. Foster, F. Liu, F. Ceola, F. Xia, F. Zhao, F. V. Fruejri, F. Stulp, G. Zhou, G. S. Sukhatme, G. Salhotra, G. Yan, G. Feng, G. Schiavi, G. Berseth, G. Kahn, G. Yang, G. Wang, H. Su, H.-S. Fang, H. Shi, H. Bao, H. B. Amor, H. I.

- Christensen, H. Furuta, H. Bharadhwaj, H. Walke, H. Fang, H. Ha, I. Mordatch, I. Radosavovic, I. Leal, J. Liang, J. Abou-Chakra, J. Kim, J. Drake, J. Peters, J. Schneider, J. Hsu, J. Vakil, J. Bohg, J. Bingham, J. Wu, J. Gao, J. Hu, J. Wu, J. Wu, J. Sun, J. Luo, J. Gu, J. Tan, J. Oh, J. Wu, J. Lu, J. Yang, J. Malik, J. Silvério, J. Hejna, J. Booher, J. Tompson, J. Yang, J. Salvador, J. J. Lim, J. Han, K. Wang, K. Rao, K. Pertsch, K. Hausman, K. Go, K. Gopalakrishnan, K. Goldberg, K. Byrne, K. Oslund, K. Kawaharazuka, K. Black, K. Lin, K. Zhang, K. Ehsani, K. Lekkala, K. Ellis, K. Rana, K. Srinivasan, K. Fang, K. P. Singh, K.-H. Zeng, K. Hatch, K. Hsu, L. Itti, L. Y. Chen, L. Pinto, L. Fei-Fei, L. Tan, L. J. Fan, L. Ott, L. Lee, L. Weihs, M. Chen, M. Lepert, M. Memmel, M. Tomizuka, M. Itkina, M. G. Castro, M. Spero, M. Du, M. Ahn, M. C. Yip, M. Zhang, M. Ding, M. Heo, M. K. Srirama, M. Sharma, M. J. Kim, M. Z. Irshad, N. Kanazawa, N. Hansen, N. Heess, N. J. Joshi, N. Sunderhauf, N. Liu, N. D. Palo, N. M. M. Shafiullah, O. Mees, O. Kroemer, O. Bastani, P. R. Sanketi, P. T. Miller, P. Yin, P. Wohlhart, P. Xu, P. D. Fagan, P. Mitrano, P. Sermanet, P. Abbeel, P. Sundaresan, Q. Chen, Q. Vuong, R. Rafailov, R. Tian, R. Doshi, R. Martín-Martín, R. Baijal, R. Scalise, R. Hendrix, R. Lin, R. Qian, R. Zhang, R. Mendonca, R. Shah, R. Hoque, R. Julian, S. Bustamante, S. Kirmani, S. Levine, S. Lin, S. Moore, S. Bahl, S. Dass, S. Sonawani, S. Tulsiani, S. Song, S. Xu, S. Haldar, S. Karamcheti, S. Adebola, S. Guist, S. Nasiriany, S. Schaal, S. Welker, S. Tian, S. Ramamoorthy, S. Dasari, S. Belkhale, S. Park, S. Nair, S. Mirchandani, T. Osa, T. Gupta, T. Harada, T. Matsushima, T. Xiao, T. Kollar, T. Yu, T. Ding, T. Davchev, T. Z. Zhao, T. Armstrong, T. Darrell, T. Chung, V. Jain, V. Kumar, V. Vanhoucke, V. Guizilini, W. Zhan, W. Zhou, W. Burgard, X. Chen, X. Chen, X. Wang, X. Zhu, X. Geng, X. Liu, X. Liangwei, X. Li, Y. Pang, Y. Lu, Y. J. Ma, Y. Kim, Y. Chebotar, Y. Zhou, Y. Zhu, Y. Wu, Y. Xu, Y. Wang, Y. Bisk, Y. Dou, Y. Cho, Y. Lee, Y. Cui, Y. Cao, Y.-H. Wu, Y. Tang, Y. Zhu, Y. Zhang, Y. Jiang, Y. Li, Y. Li, Y. Iwasawa, Y. Matsuo, Z. Ma, Z. Xu, Z. J. Cui, Z. Zhang, Z. Fu, and Z. Lin. Open x-embodiment: Robotic learning datasets and rt-x models, 2025. URL <https://arxiv.org/abs/2310.08864>.
- G. Comanici, E. Bieber, M. Schaekermann, I. Pasupat, and N. S. et al. Gemini 2.5: Pushing the frontier with advanced reasoning, multimodality, long context, and next generation agentic capabilities. *arXiv:2507.06261*, 2025.
- P. contributors. PrismarineJS/mineflayer: Create Minecraft bots with a powerful, stable, and high-level JavaScript API. <https://github.com/PrismarineJS/mineflayer>, 2013. Accessed: 2025-09-22.
- L. Cosmides and J. Tooby. Beyond intuition and instinct blindness: Toward an evolutionarily rigorous cognitive science. *Cognition*, 50:41–77, 04 1994. doi: 10.1016/0010-0277(94)90020-5.
- A. Costarelli, M. Allen, R. Hauksson, G. Sodunke, S. Hariharan, C. Cheng, W. Li, J. Clymer, and A. Yadav. Gamebench: Evaluating strategic reasoning abilities of llm agents, 2024. URL <https://arxiv.org/abs/2406.06613>.
- Decart and S. C. X. C. R. W. Julian Quevedo, Quinn McIntyre. Oasis: A universe in a transformer. 2024. URL <https://oasis-model.github.io/>.
- J. Dilkes, V. Yazdanpanah, and S. Stein. Reinforced language models for sequential decision making, 2025.
- T. S. M. Doss, M. Xu, S. Rao, A. D. Wilson, and B. T. Kumaravel. Minenpc-task: Task suite for memory-aware minecraft agents, 2026. URL <https://arxiv.org/abs/2601.05215>.

- J. Duan, R. Zhang, J. Diffenderfer, B. Kailkhura, L. Sun, E. Stengel-Eskin, M. Bansal, T. Chen, and K. Xu. Gtbench: Uncovering the strategic reasoning limitations of llms via game-theoretic evaluations, 2024. URL <https://arxiv.org/abs/2402.12348>.
- R. Dubey, P. Agrawal, D. Pathak, T. Griffiths, and A. Efros. Investigating human priors for playing video games. In J. Dy and A. Krause, editors, *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, pages 1349–1357. PMLR, 10–15 Jul 2018.
- L. Fan, G. Wang, Y. Jiang, A. Mandlekar, Y. Yang, and etc. Minedojo: Building open-ended embodied agents with internet-scale knowledge. In *Thirty-sixth Conference on Neural Information Processing Systems Datasets and Benchmarks Track*, 2022.
- L. Feng, W. Tan, Z. Lyu, L. Zheng, H. Xu, M. Yan, F. Huang, and B. An. Towards efficient online tuning of vlm agents via counterfactual soft reinforcement learning. *arXiv preprint arXiv:2505.03792*, 2025.
- Y. Feng, Y. Wang, J. Liu, S. Zheng, and Z. Lu. LLaMA-rider: Spurring large language models to explore the open world. In *Findings of the Association for Computational Linguistics: NAACL 2024*, pages 4705–4724, Mexico City, Mexico, June 2024. Association for Computational Linguistics.
- P. M. Fitts and M. I. Posner. *Human Performance*. Brooks/Cole, Belmont, CA, 1967.
- J. H. Flavell. Metacognition and cognitive monitoring: A new area of cognitive-developmental inquiry. *American Psychologist*, 34:906–911, 1979.
- L. Fluri, L. Lang, A. Abate, P. Forré, D. Krueger, and J. Skalse. The perils of optimizing learned reward functions: Low training error does not guarantee low regret. *arXiv preprint arXiv:2406.15753*, 2024. URL <https://arxiv.org/abs/2406.15753>.
- A. P. Foundation. Arc-agi-3: A new challenge for frontier agentic intelligence, 2026. URL <https://arxiv.org/abs/2603.24621>.
- G. Frisoni, L. Molfetta, D. Freddi, and G. Moro. Mixture of masters: Sparse chess language models with player routing, 2026.
- H. Fu, J. Ren, Q. Chai, D. Ye, Y. Cai, and H. Wang. Vistawise: Building cost-effective agent with cross-modal knowledge graph for minecraft. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 21895–21909, 2025.
- D. Gentner. Structure-mapping: A theoretical framework for analogy. *Cogn. Sci.*, 7:155–170, 1983.
- M. Gick and K. J. Holyoak. Schema induction and analogical transfer. *Cognitive Psychology*, 15:1–38, 1983.
- G. Gigerenzer and D. Goldstein. Reasoning the fast and frugal way: Models of bounded rationality. *Psychological Review*, 62:650–669, 10 1996. doi: 10.1093/acprof:oso/9780199744282.003.0002.
- S. Greydanus, A. Koul, J. Dodge, and A. Fern. Visualizing and understanding atari agents. *Proceedings of the 35th International Conference on Machine Learning*, 2018.

- J. Grigsby, Y. Xie, J. Sasek, S. Zheng, and Y. Zhu. Human-level competitive pokémon via scalable offline reinforcement learning with transformers. In *Reinforcement Learning Conference (RLC 2025)*, 2025.
- L. Guertler, B. Cheng, S. Yu, B. Liu, L. Choshen, and C. Tan. Textarena, 2025. URL <https://arxiv.org/abs/2504.11442>.
- C. Gulcehre, Z. Wang, A. Novikov, T. L. Paine, S. G. Colmenarejo, K. Zolna, R. Agarwal, J. Merel, D. Mankowitz, C. Paduraru, G. Dulac-Arnold, J. Li, M. Norouzi, M. Hoffman, O. Nachum, G. Tucker, N. Heess, and N. de Freitas. RL unplugged: A suite of benchmarks for offline reinforcement learning. In *Advances in Neural Information Processing Systems 33 (NeurIPS 2020)*, 2020.
- D. Guo, D. Yang, H. Zhang, J. Song, P. Wang, Q. Zhu, R. Xu, R. Zhang, S. Ma, X. Bi, X. Zhang, X. Yu, Y. Wu, Z. F. Wu, Z. Gou, Z. Shao, Z. Li, Z. Gao, A. Liu, B. Xue, B. Wang, B. Wu, B. Feng, C. Lu, C. Zhao, C. Deng, C. Ruan, D. Dai, D. Chen, D. Ji, E. Li, F. Lin, F. Dai, F. Luo, G. Hao, G. Chen, G. Li, H. Zhang, H. Xu, H. Ding, H. Gao, H. Qu, H. Li, J. Guo, J. Li, J. Chen, J. Yuan, J. Tu, J. Qiu, J. Li, J. L. Cai, J. Ni, J. Liang, J. Chen, K. Dong, K. Hu, K. You, K. Gao, K. Guan, K. Huang, K. Yu, L. Wang, L. Zhang, L. Zhao, L. Wang, L. Zhang, L. Xu, L. Xia, M. Zhang, M. Zhang, M. Tang, M. Zhou, M. Li, M. Wang, M. Li, N. Tian, P. Huang, P. Zhang, Q. Wang, Q. Chen, Q. Du, R. Ge, R. Zhang, R. Pan, R. Wang, R. J. Chen, R. L. Jin, R. Chen, S. Lu, S. Zhou, S. Chen, S. Ye, S. Wang, S. Yu, S. Zhou, S. Pan, S. S. Li, S. Zhou, S. Wu, T. Yun, T. Pei, T. Sun, T. Wang, W. Zeng, W. Liu, W. Liang, W. Gao, W. Yu, W. Zhang, W. L. Xiao, W. An, X. Liu, X. Wang, X. Chen, X. Nie, X. Cheng, X. Liu, X. Xie, X. Liu, X. Yang, X. Li, X. Su, X. Lin, X. Q. Li, X. Jin, X. Shen, X. Chen, X. Sun, X. Wang, X. Song, X. Zhou, X. Wang, X. Shan, Y. K. Li, Y. Q. Wang, Y. X. Wei, Y. Zhang, Y. Xu, Y. Li, Y. Zhao, Y. Sun, Y. Wang, Y. Yu, Y. Zhang, Y. Shi, Y. Xiong, Y. He, Y. Piao, Y. Wang, Y. Tan, Y. Ma, Y. Liu, Y. Guo, Y. Ou, Y. Wang, Y. Gong, Y. Zou, Y. He, Y. Xiong, Y. Luo, Y. You, Y. Liu, Y. Zhou, Y. X. Zhu, Y. Huang, Y. Li, Y. Zheng, Y. Zhu, Y. Ma, Y. Tang, Y. Zha, Y. Yan, Z. Z. Ren, Z. Ren, Z. Sha, Z. Fu, Z. Xu, Z. Xie, Z. Zhang, Z. Hao, Z. Ma, Z. Yan, Z. Wu, Z. Gu, Z. Zhu, Z. Liu, Z. Li, Z. Xie, Z. Song, Z. Pan, Z. Huang, Z. Xu, Z. Zhang, and Z. Zhang. Deepseek-r1 incentivizes reasoning in llms through reinforcement learning. *Nature*, 645(8081):633–638, 2025a. ISSN 1476-4687. doi: 10.1038/s41586-025-09422-z. URL <http://dx.doi.org/10.1038/s41586-025-09422-z>.
- J. Guo, Y. Ye, T. He, H. Wu, Y. Jiang, T. Pearce, and J. Bian. Mineworld: a real-time and open-source interactive world model on minecraft. *arXiv preprint arXiv:2504.08388*, 2025b.
- Y. Guo, L. X. Shi, J. Chen, and C. Finn. Ctrl-world: A controllable generative world model for robot manipulation. In *The Fourteenth International Conference on Learning Representations*, 2026.
- W. H. Guss, B. Houghton, N. Topin, P. Wang, C. Codel, M. Veloso, and R. Salakhutdinov. Minerl: A large-scale dataset of minecraft demonstrations. In *Proceedings of the Twenty-Eighth International Joint Conference on Artificial Intelligence, IJCAI-19*, pages 2442–2448. International Joint Conferences on Artificial Intelligence Organization, 7 2019. doi: 10.24963/ijcai.2019/339. URL <https://doi.org/10.24963/ijcai.2019/339>.
- D. Ha and J. Schmidhuber. Recurrent world models facilitate policy evolution. In S. Bengio, H. Wallach, H. Larochelle, K. Grauman, N. Cesa-Bianchi, and R. Garnett, editors, *Advances in Neural Information Processing Systems*, volume 31. Curran Associates, Inc., 2018.

- D. Hafner. Benchmarking the spectrum of agent capabilities, 2022. URL <https://arxiv.org/abs/2109.06780>.
- D. Hafner, T. Lillicrap, J. Ba, and M. Norouzi. Dream to control: Learning behaviors by latent imagination. In *International Conference on Learning Representations*, 2020.
- D. Hafner, T. Lillicrap, M. Norouzi, and J. Ba. Mastering atari with discrete world models. In *International Conference on Learning Representations*, 2021.
- D. Hafner, J. Pasukonis, J. Ba, and T. Lillicrap. Mastering diverse control tasks through world models. *Nature*, 640:647–653, 2025. doi: 10.1038/s41586-025-08744-2.
- H. F. Harlow. The formation of learning sets. *Psychological Review*, 56(1):51–65, 1949.
- M. Hausknecht, P. Ammanabrolu, M.-A. Côté, and X. Yuan. Interactive fiction games: A colossal adventure. *Proceedings of the AAAI Conference on Artificial Intelligence*, 34(05):7903–7910, Apr. 2020. ISSN 2159-5399. doi: 10.1609/aaai.v34i05.6297. URL <https://doi.org/10.1609/aaai.v34i05.6297>.
- X. He, C. Peng, Z. Liu, B. Wang, Y. Zhang, Q. Cui, F. Kang, B. Jiang, M. An, Y. Ren, B. Xu, H.-X. Guo, K. Gong, C. Wu, W. Li, X. Song, Y. Liu, E. Li, and Y. Zhou. Matrix-game 2.0: An open-source, real-time, and streaming interactive world model. *arXiv preprint arXiv:2508.13009*, 2025a.
- Y. He, C. D. Weillbach, M. E. Wojciechowska, Y. Zhang, and F. Wood. Plaicraft: Large-scale time-aligned vision-speech-action dataset for embodied ai, 2025b. URL <https://arxiv.org/abs/2505.12707>.
- J. Ho, A. Jain, and P. Abbeel. Denoising diffusion probabilistic models. In H. Larochelle, M. Ranzato, R. Hadsell, M. Balcan, and H. Lin, editors, *Advances in Neural Information Processing Systems*, volume 33, pages 6840–6851. Curran Associates, Inc., 2020.
- L. Hu, M. Huo, Y. Zhang, H. Yu, E. P. Xing, I. Stoica, T. Rosing, H. Jin, and H. Zhang. Imgame-bench: How good are llms at playing games? *arXiv preprint arXiv:2505.15146*, 2025a.
- L. Hu, Q. Li, A. Xie, N. Jiang, I. Stoica, H. Jin, and H. Zhang. Gamearena: Evaluating LLM reasoning through live computer games. In *The Thirteenth International Conference on Learning Representations*, 2025b.
- J. Huizinga. *Homo Ludens: A Study of the Play-Element in Culture*. Wolters-Noordhoff, Groningen, 1938. English translation: Routledge & Kegan Paul, London, 1949.
- E. Jang, A. Irpan, M. Khansari, D. Kappler, F. Ebert, C. Lynch, S. Levine, and C. Finn. Bc-z: Zero-shot task generalization with robotic imitation learning, 2022. URL <https://arxiv.org/abs/2202.02005>.
- Z. Jiang, S. Zhou, Y. Jiang, Z. Huang, M. Wei, Y. Chen, T. Zhou, Z. Guo, H. Lin, Q. Zhang, Y. Wang, H. Li, C. Yu, and D. Zhao. WoVR: World models as reliable simulators for post-training VLA policies with RL. *arXiv preprint arXiv:2602.13977*, 2026. URL <https://arxiv.org/abs/2602.13977>.
- M. Johnson, K. Hofmann, T. Hutton, and D. Bignell. The malmo platform for artificial intelligence experimentation. In *Proceedings of the Twenty-Fifth International Joint Conference on Artificial Intelligence, IJCAI’16*, page 4246–4247. AAAI Press, 2016. ISBN 9781577357704.

- J. Juul. *Half-Real: Video Games between Real Rules and Fictional Worlds*. MIT Press, Cambridge, MA, 2005. ISBN 978-0-262-10110-3.
- L. Kaiser, M. Babaeizadeh, P. Milos, B. Osinski, R. H. Campbell, K. Czechowski, D. Erhan, C. Finn, P. Kozakowski, S. Levine, A. Mohiuddin, R. Sepassi, G. Tucker, and H. Michalewski. Model-based reinforcement learning for atari. In *International Conference on Learning Representations*, 2020.
- S. Karten, A. L. Nguyen, and C. Jin. Pokéchamp: an expert-level minimax language agent, 2025.
- S. Karten, J. Grigsby, T. Upaa Jr, J. Bae, S. Hong, H. Jeong, J. Jung, K. Kerdthaisong, G. Kim, H. Kim, et al. The pokeagent challenge: Competitive and long-context learning at scale. *arXiv preprint arXiv:2603.15563*, 2026.
- M. H. Kim. Bridging symbolic control and neural reasoning in llm agents: The structured cognitive loop. *arXiv preprint arXiv:2511.17673*, 2025.
- M. J. Kim, K. Pertsch, S. Karamcheti, T. Xiao, A. Balakrishna, S. Nair, R. Rafailov, E. Foster, G. Lam, P. Sanketi, Q. Vuong, T. Kollar, B. Burchfiel, R. Tedrake, D. Sadigh, S. Levine, P. Liang, and C. Finn. Openvla: An open-source vision-language-action model, 2024. URL <https://arxiv.org/abs/2406.09246>.
- D. E. Knuth and R. W. Moore. An analysis of alpha-beta pruning. *Artificial Intelligence*, 6(4):293–326, 1975. ISSN 0004-3702. doi: [https://doi.org/10.1016/0004-3702\(75\)90019-3](https://doi.org/10.1016/0004-3702(75)90019-3).
- S. Kulinski, N. R. Waytowich, J. Z. Hare, and D. I. Inouye. Starcraftimage: A dataset for prototyping spatial reasoning methods for multi-agent environments. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2023)*, pages 22004–22013, 2023.
- H. Küttler, N. Nardelli, A. H. Miller, R. Raileanu, M. Selvatichi, E. Grefenstette, and T. Rocktäschel. The nethack learning environment. In *Advances in Neural Information Processing Systems 33 (NeurIPS 2020)*, 2020.
- H. Küttler, N. Nardelli, A. H. Miller, R. Raileanu, M. Selvatichi, E. Grefenstette, and T. Rocktäschel. The nethack learning environment, 2020. URL <https://arxiv.org/abs/2006.13760>.
- J. Laird and M. Lent. Human-level ai’s killer application: Interactive computer games. *AI Magazine*, 22:15–26, 06 2001.
- J. E. Laird, C. Lebiere, and P. S. Rosenbloom. A standard model of the mind: Toward a common computational framework across artificial intelligence, cognitive science, neuroscience, and robotics. *Ai Magazine*, 38(4):13–26, 2017.
- B. M. Lake, T. D. Ullman, J. B. Tenenbaum, and S. J. Gershman. Building machines that learn and think like people. *Behavioral and Brain Sciences*, 40:e253, 2017. doi: 10.1017/S0140525X16001837.
- S. Legg and M. Hutter. Universal intelligence: A definition of machine intelligence. *Minds and Machines*, 17:391–444, 2007.
- W. Lehrach, D. Hennes, M. Lazaro-Gredilla, X. Lou, C. Wendelken, Z. Li, A. Dedieu, J. Grau-Moya, M. Lanctot, A. Iscen, et al. Code world models for general game playing. *arXiv preprint arXiv:2510.04542*, 2025.

- Y. Leviathan, M. Kalman, and Y. Matias. Fast inference from transformers via speculative decoding. In *International Conference on Machine Learning*, pages 19274–19286. PMLR, 2023.
- B. Li, Y. Zhang, D. Guo, R. Zhang, F. Li, H. Zhang, K. Zhang, Y. Li, Z. Liu, and C. Li. Llava-onevision: Easy visual task transfer. *arXiv preprint arXiv:2408.03326*, 2024.
- L. Li, Y. Lu, Y. Zhang, J. Yao, Y. Zhu, K. Cheng, Y. Wang, Q. Zheng, X. Yang, and W. Li. Botzonebench: Scalable llm evaluation via graded ai anchors, 2026a. URL <https://arxiv.org/abs/2602.13214>.
- M. Li, Z. Wang, K. He, X. Ma, and Y. Liang. JARVIS-VLA: Post-training large-scale vision language models to play visual games with keyboards and mouse. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 17878–17899, 2025a.
- M. Li, Z. Wang, K. He, X. Ma, and Y. Liang. JARVIS-VLA: Post-training large-scale vision language models to play visual games with keyboards and mouse. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 17878–17899, Vienna, Austria, 2025b. Association for Computational Linguistics. doi: 10.18653/v1/2025.findings-acl.920. URL <https://aclanthology.org/2025.findings-acl.920/>.
- W. Li, W. Li, C. Shen, J. Sheng, Z. Huang, D. Wu, Y. Hua, W. Yin, X. Wang, H. Zha, and B. Jin. Textatari: 100k frames game playing with language agents, 2025c. URL <https://arxiv.org/abs/2506.04098>.
- X. Li, Z. Zhu, S. Liu, Y. Ma, Y. Zang, Y. Cao, and A. Sun. Emembench: Interactive benchmarking of episodic memory for vlm agents, 2026b. URL <https://arxiv.org/abs/2601.16690>.
- Y. Li, C. Lin, M. U. Nasir, P. Bontrager, J. Liu, and J. Togelius. Gvgai-llm: Evaluating large language model agents with infinite games, 2025d. URL <https://arxiv.org/abs/2508.08501>.
- Y. Li, X. Chen, Y. Liu, G. Qi, Y. Bi, Z. Wang, Y. Zhang, and Y. Zhu. Beyond scaling: Assessing strategic reasoning and rapid decision-making capability of llms in zero-sum environments, 2026c. URL <https://arxiv.org/abs/2603.09337>.
- Z. Li, Z. Meng, S. Shi, W. Peng, Y. Wu, B. Zheng, C. Li, and K. Zhang. Wildworld: A large-scale dataset for dynamic world modeling with actions and explicit state toward generative arpg, 2026d.
- F. Liang, T. Zheng, C. Chan, Y. Yim, and Y. Song. Llm-hanabi: Evaluating multi-agent gameplays with theory-of-mind and rationale inference in imperfect information collaboration game, 2025. URL <https://arxiv.org/abs/2510.04980>.
- Y. Liao, Y. Gu, Y. Sui, Z. Zhu, Y. Lu, G. Tang, Z. Sun, and W. Yang. Think in games: Learning to reason in games via reinforcement learning with large language models, 2025.
- J. Licato and S. Steinle. Do persona-infused llms affect performance in a strategic reasoning game? In *Proceedings of the 14th International Joint Conference on Natural Language Processing and the 4th Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics*, pages 3497–3528, 2025.
- S. Lifshitz, K. Paster, H. Chan, J. Ba, and S. McIlraith. STEVE-1: A generative model for text-to-behavior in minecraft. In *NeurIPS 2023 Foundation Models for Decision Making Workshop*, 2023.

- J. Light, M. Cai, S. Shen, and Z. Hu. Avalonbench: Evaluating llms playing the game of avalon, 2023. URL <https://arxiv.org/abs/2310.05036>.
- W. Lin, J. Roberts, Y. Yang, S. Albanie, Z. Lu, and K. Han. Gamebot: Transparent assessment of llm reasoning in games. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 7656–7682, 2025.
- Q. Long, Z. Li, R. Gong, Y. N. Wu, D. Terzopoulos, and X. Gao. TeamCraft: A benchmark for multi-modal multi-agent systems in Minecraft, 2024.
- Y. Long, Y. Jiang, H. Liu, Y. Zhao, J. Sun, Y. Shen, C. Zhao, A. Cohan, and D. Shasha. Puzzleplex: Benchmarking foundation models on reasoning and planning with puzzles, 2025. URL <https://arxiv.org/abs/2510.06475>.
- W. Lu, J. He, Z. Zhang, Y. Guo, and T. Zang. Cultivating game sense for yourself: Making vlms gaming experts. *arXiv preprint arXiv:2503.21263*, 2025.
- W. Ma, Q. Mi, Y. Zeng, X. Yan, Y. Wu, R. Lin, H. Zhang, and J. Wang. Large language models play StarCraft II: Benchmarks and a chain of summarization approach. In *Advances in Neural Information Processing Systems 37 (NeurIPS)*, 2024a.
- W. Ma, Q. Mi, Y. Zeng, X. Yan, Y. Wu, R. Lin, H. Zhang, and J. Wang. Large language models play starcraft ii:benchmarks and a chain of summarization approach. In A. Globerson, L. Mackey, D. Belgrave, A. Fan, U. Paquet, J. Tomczak, and C. Zhang, editors, *Advances in Neural Information Processing Systems*, volume 37, pages 133386–133442. Curran Associates, Inc., 2024b. doi: 10.52202/079017-4240.
- L. Magne, A. Awadalla, G. Wang, Y. Xu, and etc. Nitrogen: An open foundation model for generalist gaming agents. *arXiv:2601.02427*, 2026a.
- L. Magne, A. Awadalla, G. Wang, Y. Xu, et al. Nitrogen: An open foundation model for generalist gaming agents. *arXiv:2601.02427*, 2026b.
- N. Maugin and T. Cazenave. SpinGPT: A large-language-model approach to playing poker correctly. In *Advances in Computer Games (ACG 2025)*, Lecture Notes in Computer Science. Springer, 2025.
- Meta Fundamental AI Research Diplomacy Team (FAIR), A. Bakhtin, N. Brown, E. Dinan, G. Farina, C. Flaherty, D. Fried, A. Goff, J. Gray, H. Hu, A. P. Jacob, M. Komeili, K. Konath, M. Kwon, A. Lerer, M. Lewis, A. H. Miller, S. Mitts, A. Renduchintala, S. Roller, D. Rowe, W. Shi, J. Spisak, A. Wei, D. Wu, H. Zhang, and M. Zijlstra. Human-level play in the game of diplomacy by combining language models with strategic reasoning. *Science*, 378(6624):1067–1074, 2022. doi: 10.1126/science.ade9097.
- V. Micheli, E. Alonso, and F. Fleuret. Transformers are sample-efficient world models. In *The Eleventh International Conference on Learning Representations*, 2023.
- A. Mott, D. Zoran, M. Chrzanowski, D. Wierstra, and D. Jimenez Rezende. Towards interpretable reinforcement learning using attention augmented agents. In H. Wallach, H. Larochelle, A. Beygelzimer, F. d'Alché-Buc, E. Fox, and R. Garnett, editors, *Advances in Neural Information Processing Systems*, volume 32, 2019.

- J. Nam, Y. Hong, C.-H. P. Huang, F. Liu, J. Lee, J. Kim, S. Jin, Y. Lee, J. Jung, S. Choi, S. Kim, and Y. Zhou. Worldcam: Interactive autoregressive 3d gaming worlds with camera pose as a unifying geometric representation. *arXiv preprint arXiv:2603.16871*, 2026.
- A. Newell. *Unified theories of cognition*. Harvard University Press, 1994.
- J. Oh, X. Guo, H. Lee, R. L. Lewis, and S. Singh. Action-conditional video prediction using deep networks in atari games. In *Advances in Neural Information Processing Systems*, 2015.
- OpenAI, :, A. Hurst, A. Lerer, A. P. Goucher, and A. P. et al. Gpt-4o system card. *arXiv:2410.21276*, 2024.
- Y. Oshima, Y. Iwasawa, M. Suzuki, Y. Matsuo, and H. Furuta. Worldpack: Compressed memory improves spatial consistency in video world modeling. *arXiv preprint arXiv:2512.02473*, 2025. URL <https://arxiv.org/abs/2512.02473>.
- L. Ouyang, J. Wu, X. Jiang, D. Almeida, C. L. Wainwright, P. Mishkin, C. Zhang, S. Agarwal, K. Slama, A. Ray, J. Schulman, J. Hilton, F. Kelton, L. Miller, M. Simens, A. Askell, P. Welinder, P. Christiano, J. Leike, and R. Lowe. Training language models to follow instructions with human feedback. In *Proceedings of the 36th International Conference on Neural Information Processing Systems, NIPS '22*, Red Hook, NY, USA, 2022. Curran Associates Inc. ISBN 9781713871088.
- M. Ouyang, S. Hu, K. Q. Lin, H. T. Ng, and M. Z. Shou. Gameworld: Towards standardized and verifiable evaluation of multimodal game agents, 2026. URL <https://arxiv.org/abs/2604.07429>.
- D. Paglieri, B. Cupiał, S. Coward, U. Piterbarg, M. Wolczyk, A. Khan, E. Pignatelli, Łukasz Kuciński, L. Pinto, R. Fergus, J. N. Foerster, J. Parker-Holder, and T. Rocktäschel. Balrog: Benchmarking agentic llm and vlm reasoning on games, 2025. URL <https://arxiv.org/abs/2411.13543>.
- PAN Team Institute of Foundation Models. Pan: A world model for general, interactable, and long-horizon world simulation. *arXiv preprint arXiv:2511.09057*, 2025.
- D. Park, M. Kim, B. Choi, J. Kim, and etc. Orak: A foundational benchmark for training and evaluating llm agents on diverse video games. *arXiv:2506.03610*, 2025.
- T. Pearce and J. Zhu. Counter-strike deathmatch with large-scale behavioural cloning. In *2022 IEEE Conference on Games (CoG)*, pages 104–111, 2022. doi: 10.1109/CoG51982.2022.9893617.
- L. Phan, M. Mazeika, A. Zou, and D. Hendrycks. Textquests: How good are llms at text-based video games?, 2025. URL <https://arxiv.org/abs/2507.23701>.
- Psibot Team. From human skill to robotic mastery, Mar. 2026. URL <https://cypypccpy.github.io/tech-blog.github.io/>.
- S. Qi, S. Chen, Y. Li, X. Kong, J. Wang, B. Yang, P. Wong, Y. Zhong, X. Zhang, Z. Zhang, N. Liu, Y. Yang, and S.-C. Zhu. Civrealm: A learning and reasoning odyssey in civilization for decision-making agents. In *The Twelfth International Conference on Learning Representations*, 2024.
- M. Ramesh, K. Jayakumar, A. Ramkumar, P. Thodima, A. Rege, and E.-V. Vlatakis-Gkaragkounis. Sparks of cooperative reasoning: Llms as strategic hanabi agents, 2026. URL <https://arxiv.org/abs/2601.18077>.

- K. Salen and E. Zimmerman. *Rules of Play: Game Design Fundamentals*. MIT Press, Cambridge, MA, 2003. ISBN 978-0-262-24045-1.
- G. Savva, O. Michel, D. Lu, S. Waiwitlikhit, T. Meehan, D. Mishra, S. Poddar, J. Lu, and S. Xie. Solaris: Building a multiplayer video world model in minecraft. *arXiv preprint arXiv:2602.22208*, 2026.
- J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017.
- Z. Shao, P. Wang, Q. Zhu, R. Xu, J. Song, X. Bi, H. Zhang, M. Zhang, Y. K. Li, Y. Wu, and D. Guo. Deepseekmath: Pushing the limits of mathematical reasoning in open language models, 2024.
- A. K. Sharma, Y. Sun, N. Lu, Y. Zhang, J. Liu, and S. Yang. World-gymnast: Training robots with reinforcement learning in a world model, 2026.
- J. Shi, J. Yang, J. Liu, X. Bu, J. Chen, J. Zhou, K. Ma, Z. Wen, B. Wang, Y. He, L. Song, H. Zhu, S. Li, X. Wang, W. Zhang, R. Yuan, Y. Yao, W. Yang, Y. Wang, S. Fang, S. Yuan, Q. He, X. Tang, Y. Tan, W. Zhou, Z. Zhang, Z. Li, W. Huang, and G. Zhang. Korgym: A dynamic game platform for llm reasoning evaluation, 2025. URL <https://arxiv.org/abs/2505.14552>.
- H.-A. M. C. e. a. Silver, D. Mastering the game of go with deep neural networks and tree search. *Nature*, 529:484–489, 2016.
- SIMA-team, A. Bolton, A. Lerchner, A. Cordell, A. Moufarek, A. Bolt, A. Lampinen, A. Mitenkova, A. O. Hallingstad, B. Vujatovic, B. Li, C. Lu, D. Wierstra, D. P. Sawyer, D. Slater, D. Reichert, D. Vercelli, D. Hassabis, D. A. Hudson, D. Williams, E. Hirst, F. Pardo, F. Hill, F. Besse, H. Openshaw, H. Chan, H. Soyer, J. X. Wang, J. Clune, J. Agapiou, J. Reid, J. Marino, J. Kim, K. Gregor, K. Sridhar, K. McKinney, L. Kampis, L. M. Zhang, L. Matthey, L. Wang, M. A. Raad, M. Loks-Thompson, M. Engelcke, M. Kecman, M. Jackson, M. Gazeau, O. Purkiss, O. Knagg, P. Stys, P. Mendolicchio, R. Hadsell, R. Ke, R. Faulkner, S. Chakera, S. S. Baveja, S. Legg, S. Kashem, T. Terzi, T. Keck, T. Harley, T. Scholtes, T. Roberts, V. Mnih, Y. Liu, Z. Wang, and Z. Ghahramani. Sima 2: A generalist embodied agent for virtual worlds. *arXiv:2512.04797*, 2025.
- R. D. Smallwood and E. J. Sondik. The optimal control of partially observable markov processes. *Stanford University*, 1971.
- Y. Su, X. Zhan, H. Fang, H. Xue, H.-S. Fang, Y.-L. Li, C. Lu, and L. Yang. Dense policy: Bidirectional autoregressive learning of actions. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 14486–14495, October 2025.
- T. Sumers, S. Yao, K. R. Narasimhan, and T. L. Griffiths. Cognitive architectures for language agents. *Transactions on Machine Learning Research*, 2023.
- H. Sun, S. Zhang, L. Niu, L. Ren, H. Xu, H. Fu, F. Zhao, C. Yuan, and X. Wang. Collab-overcooked: Benchmarking and evaluating large language models as collaborative agents. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 4922–4951, 2025a. doi: 10.18653/v1/2025.emnlp-main.249. URL <https://aclanthology.org/2025.emnlp-main.249/>.
- W. Sun, H. Zhang, H. Wang, J. Wu, Z. Wang, Z. Wang, Y. Wang, J. Zhang, T. Wang, and C. Guo. Worldplay: Towards long-term geometric consistency for real-time interactive world modeling. *arXiv preprint arXiv:2512.14614*, 2025b. URL <https://arxiv.org/abs/2512.14614>.

- W. Sun, H. Zhang, H. Wang, J. Wu, Z. Wang, Z. Wang, Y. Wang, J. Zhang, T. Wang, and C. Guo. Worldplay: Towards long-term geometric consistency for real-time interactive world modeling. *arXiv preprint arXiv:2512.14614*, 2025c.
- W. Tan, C. Jiang, Y. Duan, M. Lei, J. Li, Y. Hong, X. Wang, and B. An. Stardojo: Benchmarking open-ended behaviors of agentic multimodal llms in production-living simulations with stardew valley. *arXiv:2507.07445*, 2025a.
- W. Tan, X. Li, Y. Fang, H. Yao, and S. Y. et al. Lumine: An open recipe for building generalist agents in 3d open worlds. *arXiv:2511.08892*, 2025b.
- W. Tan, X. Li, Y. Fang, H. Yao, S. Yan, H. Luo, T. Ao, H. Li, H. Ren, B. Yi, Y. Qin, B. An, L. Liu, and G. Shi. Lumine: An open recipe for building generalist agents in 3d open worlds. *arXiv preprint arXiv:2511.08892*, 2025c. URL <https://arxiv.org/abs/2511.08892>.
- W. Tan, W. Zhang, X. Xu, H. Xia, Z. Ding, and etc. Cradle: Empowering foundation agents towards general computer control. In *Forty-second International Conference on Machine Learning*, 2025d.
- G. Tang, Q. Jia, G. Chen, T. Li, Z. Huang, Z. Lv, and N. Ji. Vlm-dewm: Dynamic external world model for verifiable and resilient vision-language planning in manufacturing. *arXiv preprint arXiv:2602.15549*, 2026.
- W. Tang, Y. Zhou, E. Xu, K. Cheng, M. Li, and L. Xiao. Dsgbench: A diverse strategic game benchmark for evaluating llm-based agents in complex decision-making environments. *arXiv preprint arXiv:2503.06047*, 2025.
- S. Team, M. A. Raad, A. Ahuja, C. Barros, F. Besse, A. Bolt, A. Bolton, B. Brownfield, G. Buttimore, M. Cant, S. Chakera, S. C. Y. Chan, J. Clune, A. Collister, V. Copeman, A. Cullum, I. Dasgupta, D. de Cesare, J. D. Trapani, Y. Donchev, E. Dunleavy, M. Engelcke, R. Faulkner, F. Garcia, C. Gbadamosi, Z. Gong, L. Gonzales, K. Gupta, K. Gregor, A. O. Hallingstad, T. Harley, S. Haves, F. Hill, E. Hirst, D. A. Hudson, J. Hudson, S. Hughes-Fitt, D. J. Rezende, M. Jasarevic, L. Kampis, R. Ke, T. Keck, J. Kim, O. Knagg, K. Kopparapu, R. Lawton, A. Lampinen, S. Legg, A. Lerchner, M. Limont, Y. Liu, M. Loks-Thompson, J. Marino, K. M. Cussons, L. Matthey, S. Mcloughlin, P. Mendolicchio, H. Merzic, A. Mitenkova, A. Moufarek, V. Oliveira, Y. Oliveira, H. Openshaw, R. Pan, A. Pappu, A. Platonov, O. Purkiss, D. Reichert, J. Reid, P. H. Richemond, T. Roberts, G. Ruscoe, J. S. Elias, T. Sandars, D. P. Sawyer, T. Scholtes, G. Simmons, D. Slater, H. Soyer, H. Strathmann, P. Stys, A. C. Tam, D. Teplyashin, T. Terzi, D. Vercelli, B. Vujatovic, M. Wainwright, J. X. Wang, Z. Wang, D. Wierstra, D. Williams, N. Wong, S. York, and N. Young. Scaling instructable agents across many simulated worlds, 2024. URL <https://arxiv.org/abs/2404.10179>.
- J. Tong, J. Tang, H. Li, Y. Mou, M. Zhang, J. Zhao, Y. Wen, F. Song, J. Zhan, Y. Lu, C. Tao, Z. Guo, J. Yu, T. Cheng, Z. Xi, C. Jiang, Z. Yin, Y. Zheng, W. Ge, G. Chen, T. Gui, X. Qiu, Q. Zhang, and X. Huang. Game-rl: Synthesizing multimodal verifiable game data to boost vlms’ general reasoning, 2025.
- M. Towers, A. Kwiatkowski, J. Terry, and J. U. B. et al. Gymnasium: A standard interface for reinforcement learning environments. *arXiv:2407.17032*, 2024.
- J. tse Huang, E. J. Li, M. H. Lam, T. Liang, W. Wang, Y. Yuan, W. Jiao, X. Wang, Z. Tu, and M. R. Lyu. How far are we on the decision-making of llms? evaluating llms’ gaming ability in multi-agent environments, 2025. URL <https://arxiv.org/abs/2403.11807>.

- A. M. Turing. Digital computers applied to games. In *Faster Than Thought*. Pitman, London, 1953.
- J. Urbanek, A. Fan, S. Karamcheti, S. Jain, S. Humeau, E. Dinan, T. Rocktäschel, D. Kiela, A. Szlam, and J. Weston. Learning to speak and act in a fantasy text adventure game. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP 2019)*, pages 673–683, 2019.
- D. Valevski, Y. Leviathan, M. Arar, and S. Fruchter. Diffusion models are real-time game engines. *arXiv preprint arXiv:2408.14837*, 2024. URL <https://arxiv.org/abs/2408.14837>.
- D. Valevski, Y. Leviathan, M. Arar, and S. Fruchter. Diffusion models are real-time game engines. In *International Conference on Learning Representations*, 2025.
- V. C. Vendrell, M. R. Luyten, and M. van der Schaar. GameTalk: Training LLMs for strategic conversation, 2026.
- B. I. C. W. e. a. Vinyals, O. Grandmaster level in starcraft ii using multi-agent reinforcement learning. *Nature*, 575:350–354, 2019.
- O. Vinyals, T. Ewalds, S. Bartunov, P. Georgiev, A. S. Vezhnevets, M. Yeo, A. Makhzani, H. Küttler, J. Agapiou, J. Schrittwieser, J. Quan, S. Gaffney, S. Petersen, K. Simonyan, T. Schaul, H. van Hasselt, D. Silver, T. Lillicrap, K. Calderone, P. Keet, A. Brunasso, D. Lawrence, A. Ekermo, J. Repp, and R. Tsing. Starcraft II: A new challenge for reinforcement learning, 2017.
- C. Wang, L. Tang, M. Yuan, J. Yu, X. Xie, and J. Bu. Leveraging llm agents for automated video game testing. *arXiv preprint arXiv:2509.22170*, 2025a.
- G. Wang, Y. Xie, Y. Jiang, A. Mandlekar, C. Xiao, Y. Zhu, L. Fan, and A. Anandkumar. Voyager: An open-ended embodied agent with large language models. *arXiv preprint arXiv: Arxiv-2305.16291*, 2023a.
- H. Wang, X. Li, Y. Niu, S. Hu, and H. Li. Empowering llms in decision games through algorithmic data synthesis, 2025b.
- H. Wang, H. Zou, H. Song, and J. F. et al. Ui-tars-2 technical report: Advancing gui agent with multi-turn reinforcement learning. *arXiv:2509.02544*, 2025c.
- X. Wang, B. Zhuang, and Q. Wu. Are large vision language models good game players? In *The Thirteenth International Conference on Learning Representations*, 2025d.
- Y. Wang, R. Xu, K. Zheng, T. Zhang, J. N. Kogundi, S. Hans, and V. Ustun. Gameplayqa: A benchmarking framework for decision-dense pov-synced multi-video understanding of 3d virtual agents, 2026a. URL <https://arxiv.org/abs/2603.24329>.
- Z. Wang, S. Cai, G. Chen, A. Liu, X. Ma, and Y. Liang. Describe, explain, plan and select: Interactive planning with large language models enables open-world multi-task agents. *arXiv preprint arXiv:2302.01560*, 2023b.
- Z. Wang, S. Cai, A. Liu, Y. Jin, J. Hou, B. Zhang, H. Lin, Z. He, Z. Zheng, Y. Yang, X. Ma, and Y. Liang. JARVIS-1: Open-world multi-task agents with memory-augmented multimodal language models, 2023c.

- Z. Wang, S. Cai, Z. Mu, H. Lin, C. Zhang, X. Liu, Q. Li, A. Liu, X. Ma, and Y. Liang. Omnijarvis: Unified vision-language-action tokenization enables open-world instruction following agents. In *Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*, 2024. doi: 10.52202/079017-2331.
- Z. Wang, M. Li, K. He, X. Wang, Z. Mu, A. Liu, and Y. Liang. Openha: A series of open-source hierarchical agentic models in minecraft. *arXiv preprint arXiv:2509.13347*, 2025e. URL <https://arxiv.org/abs/2509.13347>.
- Z. Wang, X. Li, Y. Ye, J. Fang, and H. W. et al. Game-tars: Pretrained foundation models for scalable generalist multimodal game agents. *arXiv:2510.23691*, 2025f.
- Z. Wang, X. Li, Y. Ye, J. Fang, H. Wang, et al. Game-tars: Pretrained foundation models for scalable generalist multimodal game agents. *arXiv:2510.23691*, 2025g.
- Z. Wang, Z. Liu, J. Li, K. Huang, B. Xu, F. Kang, M. An, P. Wang, B. Jiang, Y. Wei, Y. Xietian, J. Pei, L. Hu, B. Jiang, H. Xue, Z. Wang, H. Sun, W. Li, W. Ouyang, X. He, Y. Liu, Y. Li, and Y. Zhou. Matrix-game 3.0: Real-time and streaming interactive world model with long-horizon memory. *arXiv preprint arXiv:2604.08995*, 2026b.
- Y. Wen, Y. Ye, Y. Zhang, D. Yang, and H. Zhu. Real-time reasoning agents in evolving environments. *arXiv preprint arXiv:2511.04898*, 2025.
- R. Wu, X. He, M. Cheng, T. Yang, Y. Zhang, Z. Kang, X. Cai, X. Wei, C. Guo, C. Li, and M.-M. Cheng. Infinite-world: Scaling interactive world models to 1000-frame horizons via pose-free hierarchical memory. *arXiv preprint arXiv:2602.02393*, 2026. URL <https://arxiv.org/abs/2602.02393>.
- S. Wu, L. Zhu, T. Yang, S. Xu, Q. Fu, Y. Wei, and H. Fu. Enhance reasoning for large language models in the game werewolf, 2024a.
- Y. Wu, X. Tang, T. M. Mitchell, and Y. Li. Smartplay: A benchmark for llms as intelligent agents, 2024b. URL <https://arxiv.org/abs/2310.01557>.
- B. Xia, W. Tian, Q. Ren, J. Huang, J. Xiao, S. Lu, K. Wang, L. Ai, E. Yang, and B. Shi. Implicit strategic optimization: Rethinking long-horizon decision-making in adversarial poker environments, 2026.
- G. Xiao, J. Lin, M. Seznec, H. Wu, J. Demouth, and S. Han. Smoothquant: Accurate and efficient post-training quantization for large language models. In *International conference on machine learning*, pages 38087–38099. PMLR, 2023a.
- G. Xiao, Y. Tian, B. Chen, S. Han, and M. Lewis. Efficient streaming language models with attention sinks. *arXiv preprint arXiv:2309.17453*, 2023b.
- Y. Xie, Y. Ma, S. Lan, A. Yuille, J. Xiao, and C. Wei. Play to generalize: Learning to reason through game play, 2025.
- Z. Xie, Z. Chen, Z. Weng, T. Wu, C. Li, V. Zhang, and K. Wang. Steve-evolving: Open-world embodied self-evolution via fine-grained diagnosis and dual-track knowledge distillation. *arXiv preprint arXiv:2603.13131*, 2026.
- X. Xu, Z. Li, C. Lu, R. Qi, and etc. Reflection of episodes: Learning to play game from expert and self experiences. *arXiv:2502.13388*, 2025a.

- X. Xu, P. Bu, Y. Wang, B. F. Karlsson, Z. Wang, T. Song, Q. Zhu, J. Song, Z. Ding, and B. Zheng. DeepPHY: Benchmarking Agentic VLMs on Physical Reasoning. In *Proceedings of the 40th Annual AAAI Conference on Artificial Intelligence (AAAI-26)*, 2026.
- Z. Xu, C. Yu, F. Fang, Y. Wang, and Y. Wu. Language agents with reinforcement learning for strategic play in the werewolf game. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 55434–55464. PMLR, 21–27 Jul 2024.
- Z. Xu, W. Gu, C. Yu, Y. Wu, and Y. Wang. Learning strategic language agents in the Werewolf game with iterative latent space policy optimization. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, 2025b.
- W. Yan, Y. Dai, Q. Ran, H. Li, W. Lin, H. Liao, X. Xie, T. Jin, and J. Lian. Proact-vl: A proactive videollm for real-time ai companions, 2026.
- A. Yang, A. Li, B. Yang, B. Zhang, B. Hui, B. Zheng, B. Yu, C. Gao, C. Huang, C. Lv, C. Zheng, D. Liu, F. Zhou, F. Huang, F. Hu, H. Ge, H. Wei, H. Lin, J. Tang, J. Yang, J. Tu, J. Zhang, J. Yang, J. Yang, J. Zhou, J. Zhou, J. Lin, K. Dang, K. Bao, K. Yang, L. Yu, L. Deng, M. Li, M. Xue, M. Li, P. Zhang, P. Wang, Q. Zhu, R. Men, R. Gao, S. Liu, S. Luo, T. Li, T. Tang, W. Yin, X. Ren, X. Wang, X. Zhang, X. Ren, Y. Fan, Y. Su, Y. Zhang, Y. Zhang, Y. Wan, Y. Liu, Z. Wang, Z. Cui, Z. Zhang, Z. Zhou, and Z. Qiu. Qwen3 technical report. *arXiv:2505.09388*, 2025.
- M. Yang, J. Li, Z. Fang, S. Chen, Y. Yu, Q. Fu, W. Yang, and D. Ye. Playable game generation. *arXiv preprint arXiv:2412.00887*, 2024.
- S. Yao, J. Zhao, D. Yu, N. Du, I. Shafran, K. R. Narasimhan, and Y. Cao. React: Synergizing reasoning and acting in language models. In *The eleventh international conference on learning representations*, 2022.
- S. Ye, Y. Ge, K. Zheng, S. Gao, S. Yu, G. Kurian, S. Indupuru, Y. L. Tan, C. Zhu, J. Xiang, A. Malik, K. Lee, W. Liang, N. Ranawaka, J. Gu, Y. Xu, G. Wang, F. Hu, A. Narayan, J. Bjorck, J. Wang, G. Kim, D. Niu, R. Zheng, Y. Xie, J. Wu, Q. Wang, R. Julian, D. Xu, Y. Du, Y. Chebotar, S. Reed, J. Kautz, Y. Zhu, L. J. Fan, and J. Jang. World action models are zero-shot policies, 2026.
- L. Ying, R. Truong, P. Sharma, K. I. Zhao, N. Cloos, K. R. Allen, T. L. Griffiths, K. M. Collins, J. Hernández-Orallo, P. Isola, S. J. Gershman, and J. B. Tenenbaum. Ai gamestore: Scalable, open-ended evaluation of machine general intelligence with human games. *arXiv:2602.17594*, 2026.
- J. Yu, Y. Qin, X. Wang, P. Wan, D. Zhang, and X. Liu. Gamefactory: Creating new games with generative interactive videos. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 11590–11599, 2025.
- H. Yuan, Z. Xu, Z. Tan, X. Yi, M. Guang, K. Long, H. Hui, B. Li, X. Chen, B. Zhao, X.-P. Zhang, C. Yu, and Y. Wang. MARSHAL: Incentivizing multi-agent reasoning via self-play with strategic LLMs, 2025.
- Y. Yue, C. Green, S. Hunt, I. Salia, W. Shi, and J. J. Hunt. Pixels to play: A foundation model for 3d gameplay. In *2025 IEEE Conference on Games (CoG)*, pages 1–4. IEEE, 2025a.
- Y. Yue, I. Salia, S. Hunt, C. Green, W. Shi, and J. J. Hunt. Learning to play: A multimodal agent for 3d game-play. *arXiv preprint arXiv:2510.16774*, 2025b.

- Y. Yue, I. Salia, S. Hunt, C. Green, W. Shi, and J. J. Hunt. Scaling behavior cloning improves causal reasoning: An open model for real-time video game playing. *arXiv:2601.04575*, 2026.
- A. L. Zhang, T. L. Griffiths, K. R. Narasimhan, and O. Press. Videogamebench: Can vision-language models complete popular video games? *arXiv:2505.18134*, 2025a.
- C. Zhang, P. Cai, Y. Fu, H. Yuan, and Z. Lu. Creative agents: Empowering agents with imagination for creative tasks. *arXiv preprint arXiv:2312.02519*, 2023.
- H. Zhang, C. Zhu, S. Guo, H. Guo, H. Li, and D. Yu. Starbench: A turn-based rpg benchmark for agentic multimodal decision-making and information seeking, 2025b. URL <https://arxiv.org/abs/2510.18483>.
- J. Zhang and Y. Long. Learning to play like humans: A framework for llm adaptation in interactive fiction games. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 10188–10205, 2025.
- K. Zhang, D. Liu, Q. Zhao, J. Hou, X. Zhang, Q. Xie, M. Liu, and Y. Li. Gameverse: Can vision-language models learn from video-based reflection? *arXiv preprint arXiv:2603.06656*, 2026a.
- M. Zhang, L. Zhuo, T. Tan, G. Xie, X. Nie, Y. Li, R. Zhao, Z. He, Z. Wang, J. Cai, and Y.-L. Li. Ipr-1: Interactive physical reasoner. *arXiv preprint arXiv:2511.15407*, 2025c.
- R. Zhang, C. Walshe, Z. Liu, L. Guan, K. S. Muller, J. A. Whritner, L. Zhang, M. M. Hayhoe, and D. H. Ballard. Atari-head: Atari human eye-tracking and demonstration dataset. In *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI 2020)*, pages 6811–6820, 2020. doi: 10.1609/aaai.v34i04.6161.
- R. Zhang, Y. Huang, Y. Pan, C. Shen, Z. Liu, T. Xie, W. Li, and L. Duan. Pokegym: A visually-driven long-horizon benchmark for vision-language models, 2026b. URL <https://arxiv.org/abs/2604.08340>.
- Y. Zhang, C. Peng, B. Wang, P. Wang, Q. Zhu, F. Kang, B. Jiang, Z. Gao, E. Li, Y. Liu, and Y. Zhou. Matrix-game: Interactive world foundation model. *arXiv preprint arXiv:2506.18701*, 2025d.
- Y. Zhang, H. Yu, L. Hu, H. Jin, and H. Zhang. General modular harness for llm agents in multi-turn gaming environments. *arXiv preprint arXiv:2507.11633*, 2025e.
- Z. Zhang, Z. Wang, Y. Du, and F. Fang. VAM: Verbalized action masking for controllable exploration in RL post-training – a chess case study, 2026c.
- Y. Zhao, L. Hu, Y. Wang, M. Hou, H. Zhang, K. Ding, and J. Zhao. Stronger-MAS: Multi-agent reinforcement learning for collaborative LLMs, 2025.
- X. Zheng, H. Lin, K. He, Z. Wang, Z. Zheng, and Y. Liang. Mcu: An evaluation framework for open-ended game agents, 2025. URL <https://arxiv.org/abs/2310.08367>.
- S. Zhou, Y. Du, Y. Yang, L. Han, P. Chen, D.-Y. Yeung, and C. Gan. Learning 3D persistent embodied world models. *arXiv preprint arXiv:2505.05495*, 2025. URL <https://arxiv.org/abs/2505.05495>.

- Z. Zhou, L. Du, Z. Sun, X. Zhou, R. Ye, Q. Chen, Y. Chen, and L. Qiu. MAIN-VLA: Modeling abstraction of intention and environment for vision-language-action models. *arXiv preprint arXiv:2602.02212*, 2026. URL <https://arxiv.org/abs/2602.02212>.
- F. Zhu, Y. Zhengyang, Z. Hong, Q. Shou, X. Ma, and S. Guo. WMPO: World model-based policy optimization for vision-language-action models. In *The Fourteenth International Conference on Learning Representations*, 2026.
- R. Zhuang, A. Gupta, R. Yang, A. Rahane, Z. Li, and G. Anumanchipalli. Pokerbench: Training large language models to become professional poker players, 2025. URL <https://arxiv.org/abs/2501.08328>.
- B. Zitkovich, T. Yu, S. Xu, P. Xu, T. Xiao, F. Xia, J. Wu, P. Wohlhart, S. Welker, A. Wahid, Q. Vuong, V. Vanhoucke, H. Tran, R. Soricut, A. Singh, J. Singh, P. Sermanet, P. R. Sanketi, G. Salazar, M. S. Ryoo, K. Reymann, K. Rao, K. Pertsch, I. Mordatch, H. Michalewski, Y. Lu, S. Levine, L. Lee, T.-W. E. Lee, I. Leal, Y. Kuang, D. Kalashnikov, R. Julian, N. J. Joshi, A. Irpan, B. Ichter, J. Hsu, A. Herzog, K. Hausman, K. Gopalakrishnan, C. Fu, P. Florence, C. Finn, K. A. Dubey, D. Driess, T. Ding, K. M. Choromanski, X. Chen, Y. Chebotar, J. Carbajal, N. Brown, A. Brohan, M. G. Arenas, and K. Han. Rt-2: Vision-language-action models transfer web knowledge to robotic control. In J. Tan, M. Toussaint, and K. Darvish, editors, *Proceedings of The 7th Conference on Robot Learning*, volume 229 of *Proceedings of Machine Learning Research*, pages 2165–2183. PMLR, 06–09 Nov 2023.

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Acknowledgments goes here.

Table S1: Summary of representative world models in gaming AI (to be filled)

Model	Arch.	Duration	Controllability	Action Space	Game Set
Name	DiT/etc.	n-seconds/real-time	character/environment	actions	game and OOD?

S1. Supplementary Materials

Here, we provide all supplementary materials used in our analysis.

S1.1. Foundation Models as Game Engine

Beyond serving as generalist game players that perceive and act, foundation models are increasingly being developed as **intelligent game engines** for the multiverse of games. At present, this role is primarily instantiated through generative, interactive **world models (WMs)**, which model the environment’s transition dynamics T and observation dynamics O under action-conditioned interaction, while maintaining a persistent world state S from multimodal observations Ω and control inputs A . In this sense, Gaming WMs are not merely about video continuation; they are about approximating a stateful, playable, and consistent intelligent game engine, pushing foundation models toward stronger **omni-reality adaptability** across diverse game worlds. Their development can be characterized along two axes:

- **World modeling scope:** the **range of game worlds and interaction settings** a gaming WM can support, including simulation duration, environment diversity, control/action spaces, and the number of interacting agents. This axis reflects broader **multiverse adaptability** on the input side.
- **World modeling fidelity:** the **faithfulness of simulated world dynamics** a gaming WM can maintain, including temporal persistence, spatial consistency, entity permanence, and interaction causality. This axis reflects how faithfully the model approximates game mechanics and state evolution on the output side.

These two axes co-evolve: broader game settings demand higher fidelity, while improved fidelity enables further expansion across the multiverse of games, jointly driving Gaming WMs toward more advanced **intelligent game engines**. Below, we detail this progression step by step.

Note. The current draft only focuses on the WM development narrative. Details like citations, method, and game-set information have **not yet been added**. And we will add a table to summarize the duration, resolution, action space, architecture and game set of each discussed model.

L1. From Auxiliary Dynamics Models to Playable Game Engines. Early Gaming WMs learned action-conditioned game dynamics, but mainly as auxiliary models for model-based policy learning rather than as playable game engines. Representative examples include action-conditional video prediction in Atari, *World Models*, *SimPLe*, *Dreamer Series*, and *IRIS*, where learned dynamics primarily improved planning, policy learning, or sample efficiency. A clearer turn toward intelligent game engines emerged with *GameGAN*, which offered an early prototype of a runnable game learned

from gameplay videos, and later with *Genie*, which made generative interactive environments explicit through latent action-based control learned from unlabeled videos. This shift redefined the key requirement at this stage: not merely predicting the next frame, but enabling **playability** and **interaction causality**, so that user actions can meaningfully drive world evolution.

L2. From Reactive Simulation to Persistent Spatiotemporal Worlds. Once established as playable game engines, Gaming WMs expanded toward more persistent, real-time, long-horizon, and spatially grounded worlds. This expansion raised new fidelity requirements: models must preserve **temporal persistence**, **spatial consistency**, and **entity permanence** as interaction becomes longer and spatially richer. Early signs of this trend appeared in MarioVGG and *Oasis*, which moved beyond short reactive rollouts and highlighted the need for longer-horizon continuity. *Matrix-Game 2.0* made this direction more explicit by targeting real-time, streaming interactive world modeling, while *PAN* framed long-horizon simulation as a core property of a general and interactable world model. As spatiotemporal complexity increased, *Model as a Game* identified numerical and spatial consistency as central bottlenecks, showing that local continuation is insufficient once world state and structure must remain stable over time. Subsequent works further strengthened this line through memory and geometry: *WorldPlay* emphasized long-term geometric consistency, *WorldCam* unified action control and 3D consistency through camera pose, and *Matrix-Game 3.0* pushed long-horizon memory in high-resolution real-time generation. Together, these works show that, after interactivity is established, the next step is to sustain a **persistent and spatially grounded world**, rather than merely producing short sequences of locally plausible reactions.

L3. From Single-World Fitting to Multiverse Generalization. After achieving more persistent and spatially grounded simulation, Gaming WMs began to generalize beyond single-game fitting toward the broader **multiverse of games**. The key question here is whether a model can retain faithful game mechanics and causal world dynamics across more diverse, open, and even out-of-distribution game settings. *GameGen-X* is an early representative in this direction, scaling interactive open-world game video generation to a large and diverse set of games with multimodal control. *GameFactory* further emphasizes scene generalization, aiming to transfer action-conditioned generation beyond fixed training worlds and even support new game creation. *PAN* pushes this trend toward a more explicit notion of a general, interactable, and long-horizon world model, while *IPR-1* suggests that world-model rollouts can support more transferable physical and causal reasoning in unseen settings. Together, these works indicate that Gaming WMs are moving beyond single-world neural simulators toward more generalizable models of interactive world dynamics across the multiverse of games.

L4. From Ego-Centric Worlds to Shared Multi-Agent Worlds. After improving generalization across game worlds, Gaming WMs further evolved from ego-centric worlds to shared multi-agent worlds. This shift introduces a new requirement: instead of maintaining only a consistent observation stream for one player, the model must preserve a **shared and coherent world state** across multiple agents and viewpoints. *Solaris* represents this transition most clearly by explicitly targeting multiplayer world modeling in Minecraft. Unlike earlier Gaming WMs that mainly simulate the world from a single-player perspective, *Solaris* models environments where multiple agents move, observe, build, and interact within the same underlying world. To support this setting, it introduces synchronized multi-agent data collection, staged training from single-player to multiplayer, and evaluations centered on movement, memory, grounding, building, and cross-view consistency. Together,

these efforts show that Gaming WMs are beginning to move beyond simulating “the world as seen by one player” toward simulating **a world jointly accessed, modified, and experienced by multiple agents**.

Note. There will be a summary of discussed models and messages, and conclude with the trend to **era 4** as mentioned in the introduction.

Current Approximations and Open Challenges. Recent progress provides the first credible approximation to the *world-model-as-data-engine* vision, but still falls short of a true replacement for code-based game engines. More generally, synthetic-world reinforcement learning has begun to show that training in cheap generated environments can transfer back to real-image agentic benchmarks (Bredis et al., 2025b). Within gaming and interactive simulation, *Genie* and *GameNGen* illuminate two complementary ends of the current design space: *Genie* prioritizes breadth, learning action-controllable environments from large-scale unlabeled video, while *GameNGen* demonstrates that a neural simulator can already achieve striking short-horizon fidelity for a specific game (Bruce et al., 2024b; Valevski et al., 2024). Yet this very contrast exposes the first major bottleneck, namely *fidelity versus diversity*. As world models are pushed toward broader coverage, weaker grounding and hallucinated transitions become increasingly difficult to avoid. The consequence is not merely perceptual degradation. In a training loop, physically inconsistent or strategically invalid rollouts can distort the effective data distribution, causing agents to optimize against artifacts of the simulator rather than the structure of the underlying game (Chen et al., 2025e; Jiang et al., 2026). In other words, scaling synthetic diversity without preserving transition faithfulness risks reproducing the central weakness of noisy pseudo-labeled datasets, except now inside the simulator itself.

A second challenge is *long-horizon consistency*. Current world models can often produce rollouts that appear convincing over short clips, but local plausibility is not equivalent to stable temporal reasoning over hundreds of steps. Once rollouts become extended, small predictive errors compound, task-relevant details are forgotten, and delayed causal dependencies begin to drift away from the true game dynamics. Recent work increasingly treats this not as an implementation bug, but as a structural limitation of current autoregressive video generation: explicit memory retrieval, compressed history representations, and specialized long-context training objectives are introduced precisely because naive rollout is too brittle for persistent control (Chen et al., 2025e; Oshima et al., 2025; Sun et al., 2025b; Wu et al., 2026). This limitation is especially severe for games, where the value of an action is often revealed only after a long chain of downstream consequences. A simulator that is impressive at the level of moment-to-moment realism but unstable at strategic horizons may generate enormous quantities of data while contributing little to the learning of planning, credit assignment, or delayed recovery behaviors.

A third bottleneck is *spatial permanence*. A useful game world model must preserve more than visual continuity; it must preserve the identity of the world itself across revisits, viewpoint changes, occlusions, and partially observed interactions. In practice, however, this remains highly unstable. Many current models can continue a scene smoothly, yet fail to maintain geometric consistency when the agent loops back to a previously visited region, re-encounters an object from a new angle, or expects the world state to remain anchored in a persistent map. Recent efforts therefore move beyond plain next-frame prediction toward explicit spatial memory, compressed world-state representations, and revisit-oriented finetuning, all of which are attempts to restore loop closure and object permanence (Oshima et al., 2025; Sun et al., 2025b; Wu et al., 2026; Zhou et al., 2025). This challenge

is structurally central for gaming AI. Navigation, combat, resource collection, puzzle solving, and multi-room planning all presuppose that the environment remains the *same* environment as time unfolds. Without robust spatial permanence, a world model may still serve as a visually rich generator, but it cannot yet serve as a reliable substrate for training agents that must act in coherent, revisitable worlds.

A fourth and perhaps most consequential challenge concerns *reward design and verifiability*. Code-based engines expose deterministic rules, externally checkable task conditions, and a well-defined notion of success. Once training is moved into a learned simulator, this guarantee weakens substantially. The reward must then be inherited from model predictions, attached through a learned verifier, or approximated by heuristic judgments over imagined trajectories, any of which may deviate from genuine task completion. Recent theory on learned rewards shows that low prediction error on a training distribution does not in itself prevent large downstream regret once optimization shifts the policy into new regions of behavior (Fluri et al., 2024). Empirically, recent world-model RL systems likewise show that stable gains require explicit mechanisms to control rollout error, maintain policy-simulator alignment, and reduce opportunities for exploiting simulator flaws (Jiang et al., 2026; Sharma et al., 2026; Zhu et al., 2026). This makes reward specification inside world models fundamentally different from ordinary offline data augmentation. The goal is no longer just to synthesize more trajectories, but to synthesize trajectories whose causal structure, spatial persistence, and success signals are trustworthy enough to support policy improvement. Until that problem is solved, the world model will remain a promising data engine, but not yet a dependable replacement for the rule-grounded certainty of traditional game environments.